AT Command Set User Manual

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1 Introduction

1.1 Scope

This document presents the AT Command Set for the AICXTEK LTE mobile terminal, including the command syntax, the command lists with possible setting values and the responses of these commands.

1.2 Definitions

For the purposes of the present document, the following syntactical definitions apply (refer also clause 4):

<CR> Carriage return character, which value is specified with command S3.

<LF> Linefeed character, which value is specified with command S4.

<...> Name enclosed in angle brackets is a syntactical element. Brackets themselves do not appear in the command line.

[...] Optional subparameter of a command or an optional part of TA information response is enclosed in square brackets. Brackets themselves do not appear in the command line. When subparameter is not given in parameter type commands, new value equals to its previous value. In action type commands, action should be done on the basis of the recommended default setting of the subparameter.

underline Underlined defined subparameter value is the recommended default setting of this subparameter. In parameter type commands, this value should be used in factory settings which are configured by ITU-T Recommendation V.250 [1] command &F0. In action type commands, this value should be used when subparameter is not given.

1.3 Reference

[1] ITU-T Recommendation V.250: “Serial asynchronous automatic dialling and control”.

[2] 3GPP TS 27.007: “AT command set for User Equipment (UE)”.

[3] 3GPP TS 27.005: “Use of Data Terminal Equipment - Data Circuit terminating Equipment(DTE - DCE) interface for Short Message Service (SMS) and Cell Broadcast Service (CBS)”.

1.4 Abbreviations

For the purposes of the present document, the following abbreviations apply:

AT ATtention; this two-character abbreviation is always used to start a

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	command line to be sent from TE to TA
BCD	Binary Coded Decimal
CTM	Cell Test Mode
EPS	Evolved Packet System
GSM	Global System for Mobile communications
IMEI	International Mobile station Equipment Identity
JS	Joint Solution
LTE	Long Term Evolution
ME	Mobile Equipment
MT	Mobile Termination
SIM	Subscriber Identity Module
TA	Terminal Adaptor, e.g. a GSM data card (equal to DCE; Data Circuit terminating Equipment)
TE	Terminal Equipment, e.g. a computer (equal to DTE; Data Terminal Equipment)
UE	User Equipment
UICC	Universal Integrated Circuit Card
URC	Unsolicited Result Code
USIM	Universal Subscriber Identity Module
DM	Dual Mode
DM1	Dual Mode Phase1
DM2	Dual Mode Phase2
TD-SCDMA	Time Division- Synchronous Code Division Multiple
Access MBMS	Multimedia Broadcast/Multicast Service
MRBT	Multimedia ring back tone

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2 AT command syntax

This clause summarizes general aspects on AT commands and issues related to them.

TE software implementers must take account that future versions of this specification may include additional parameters beyond what is expected in any response to an AT Set Command, Read Command, Test Command, or unsolicited result code. Implementations must therefore analyses all parameters provided from the TA and discard (ignore) any parameters received following the parameters expected by the TE software.

For further information refer ITU-T Recommendation V.250

2.1 Command line

See figure 1 for general structure of a command line. Standardized basic commands are found only in V.250. The commands use syntax rules of extended commands. Every extended command has a test command

(trailing =?) to test the existence of the command and to give information about the type of its subparameters. Parameter type commands also have a read command (trailing ?) to check the current values of subparameters. Action type commands do not store the values of any of their possible subparameters, and therefore do not have a read command.

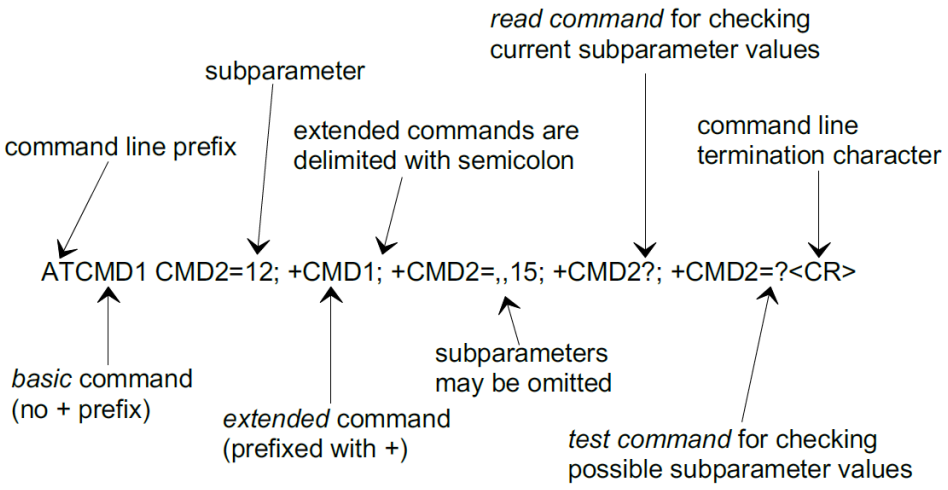


Figure 2-1 Basic structure of a command line(Reference 3GPP TS 27.007, Figure 2)

If verbose responses are enabled with command V1 and all commands in a command line has been performed successfully, result code <CR><LF>OK<CR><LF> is sent from the TA to the TE. If numeric responses are enabled with command V0, result code 0<CR> is sent instead.

If verbose responses are enabled with command V1 and subparameter values of a command are not

accepted by the TA (or command itself is invalid, or command cannot be performed for some reason), result code <CR><LF>ERROR<CR><LF> is sent to the TE and no subsequent commands in the command line are processed. If numeric responses are enabled with command V0, result code 4<CR> is sent instead. ERROR (or 4) response may be replaced by +CME ERROR: <err> (refer clause 9) when command was not processed due to an error related to MT operation.

2.2 Information responses and result codes

The TA response for the example command line of figure 1 could be as shown in figure 2. Here, verbose response format is enabled with command V1. If the numeric format V0 would have been used, the <CR><LF> headers of information responses would have been left out and the final result code should be changed to 0<CR>.

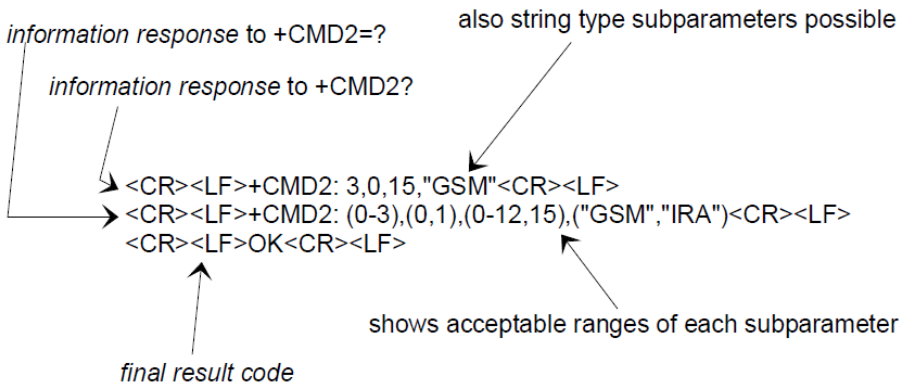


Figure 2-2: Response to a command line(Reference 3GPP TS 27.007, Figure 3)

So called intermediate result codes inform about progress of TA operation (e.g. connection establishment CONNECT), and so called unsolicited result codes indicate occurrence of an event not directly associated with issuance of a command from TE (e.g. ring indication RING).

It's strongly suggested that concatenating commands on the same line should be avoided since if one command fails to be executed, all the commands after it on the same line will be discarded silently and you have to send AT+CEER command to check the execution. Refer the command in Section 4.

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However, you may enter several AT commands on the same line if you will. This eliminates the need to type the "AT" prefix before each command. Instead, it is only needed once at the beginning of the command line. Use a semicolon as command delimiter.

The command line buffer accepts a maximum of 250 characters. If this number is exceeded, none of the commands will be executed and ME returns ERROR directly.

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3 General control and status AT Commands

These AT commands are related to general control and status reporting.

3.1 ATE Command echo

Set command ATE[<n>]	The setting of this parameter determines whether or not the MT echoes characters received from the TE during command state and online command state. Response: **OK** If parameter is not supported: **ERROR** Parameter **<n>** 0 Echo mode off 1 Echo mode on
Reference	V.250
Note	

3.2 AT+CEER Extended error report

Set command AT+CEER	TA returns one or more lines of information text <report> Response **+CEER: <report>** **OK** Parameter **<report>:** the total number of characters, including line terminators, in the information text shall not exceed

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	2041 characters.
Test command AT+CEER=?	Response **OK**
Reference	3GPP TS 27.007, section 6.10
Note	

3.3 AT+CMEE Report mobile termination error

Set command AT+CMEE=[<n>]	Set command disables or enables the use of result code +CME ERROR: <err> and +CMS ERROR: <err> as an indication of an error relating to the functionality of the MT. Response **OK** If <n> is not recognized or supported **ERROR** Parameter **<n>** 0 disable +CME ERROR: <err> result code and use ERROR instead 1 enable +CME ERROR: <err> result code and use numeric <err> values (refer next subclause) 2 enable +CME ERROR: <err> result code and use verbose <err> values (refer next subclause) Example: AT+CMEE=1 OK

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Read command: AT+CMEE?	Response **+CMEE: <n>** **OK** Parameter See set command
Test command AT+CMEE=?	Response **+CMEE:** (list of supported **<n>s**) **OK** Parameter See set command
Reference	3GPP TS 27.007, section 9
Note	

3.4 AT+CSCS Set TE character set

Set command AT+CSCS=[<chset>]	Set command informs TA which character set <chset> is used by the TE. TA is then able to convert character strings correctly between TE and MT character sets. Response **OK** If <chset> is not recognized or supported **ERROR** Parameter **<chset>** "HEX" character strings consist only of hexadecimal numbers from 00 to FF; e.g. "032FE6" equals three 8-bit characters with decimal values 3, 47 and 230; no conversions to the original MT character set shall be done. "GSM" GSM 7 bits default alphabet (3GPP TS 23.038) "UCS2" 16 bits universal multiple-octet coded character set (ISO/IEC10646); UCS2 character strings are converted to hexadecimal number from 0000 to FFFF; eg "004100620063" equals three 16-bit characters with decimal value 65,98,99

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	"IRA" international reference alphabet (ITU-T T.50 [13]) Example: **AT+CSCS=”HEX”** **OK**
Read command AT+CSCS?	Response **+CSCS: <chset>** **OK** Parameter See set command
Test command AT+CSCS=?	Response **+CSCS:** (list of supported **<chset>s**) **OK** Parameter See set command
Reference	3GPP TS 27.007, section 5.5
Note	when input string is null, handling as AT+CSCS="IRA"

3.5 ATV Set result code format mode

Set command ATV[<n>]	The setting of this parameter determines the contents of the header and trailer transmitted with result codes and information responses. It also determines whether result codes are transmitted in a numeric form or an alphabetic (or “verbose”) form. The text portion of information responses is not affected by this setting. Response: **OK** If parameter is not supported: **ERROR** Parameter **<n>:** integer type, default is 1 and only support value 1. 1 Information response: <CR><LF><text><CR><LF>

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	Result codes: <CR><LF><verbose code><CR><LF>
Reference	ITU-T V.250
Note	The parameter <n> is only support 1.

3.6 ATS0 Automatic answer

Set command ATS0=[<n>]	This S-parameter controls the automatic answering feature of the MT. If set to 0, automatic answering is disabled. If set to a non-zero value, the MT shall cause the MT to answer when the incoming call indication (ring) has occurred the number of times indicated by the value. Response: **OK** If <n> is not recognized or supported **ERROR** Parameter **<n>** 0 Automatic answering is disabled. 1-255 Automatic answering after <n> rings
Read command ATS0?	Response **<n>** **OK** Parameter See set command
Reference	ITU-T V.250
Note	

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3.7 A/ Repeat last command line

Execute command A/	Repeats the last command line. No terminating character is required. If no commands input before, **OK** will be returned immediately.
Reference	V.250 Note After beginning with the character “a” or “A”, a second character “t”, “T” or “/” has to follow. In case of using a wrong second character, it is necessary to start again with character “a” or “A”.
Note	

3.8 ATO Return to online data state

Set command ATO[<n>]	ATO is the corresponding command to the +++ escape sequence. When a CS data call or a GPRS connection is changed as command mode, ATO causes the TA to resume the data or GPRS connection and takes back to data mode or PPP online mode. Response If connection is not successfully resumed **NO CARRIER** If TA returns to data or PPP online mode from command mode **CONNECT** If <n> is not recognized or supported **ERROR** Parameter <n> 0 switch from command mode to data or PPP online mode
Reference	ITU-T V.250
Note	

3.9 ATZ Reset to default configuration

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Set command ATZ[<n>]	This command instructs the MT to set all TA related parameters to their factory defaults, the parameters include error report setting, echo setting, response format setting, character set setting, auto answer setting, etc. Response: **OK** If <n> is not recognized or supported **ERROR** <n> 0 Reset to factory defaults
Reference	ITU-T V.250
Note	

3.10 AT&F Set to factory-defined configuration

Set command AT&F[<n>]	This command instructs the MT to set all TA related parameters to their factory default values, the parameters include error report setting, echo setting, response format setting, character set setting, auto answer setting, etc. Response: **OK** If <n> is not recognized or supported **ERROR** <n> 0 Reset to factory defaults
Reference	ITU-T V.250
Note	

3.11 ATQ Set result code presentation

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Set command ATQ[<n>]	This command determines whether the TA transmits any result code to the TE. Response: If succeed **OK** If fail +CME ERROR: <err> Parameter <n>: integer type, default is 0 and only support value 0. 0 TA transmits result code
Reference	ITU-T V.250
Note	The parameter <n> is only support 0.

3.12 AT&D Circuit108 (Data terminal ready) behavior

Set command AT&D[<value>]	This command control how the DCE responds when circuit 108/2 is changed from the ON to the OFF condition during online data state. Response: **OK** If <value> is not recognized or supported **ERROR** <value> 0 DCE ignores circuit 108/2 1 DCE enters online command state 2 DCE perform to disconnect all CS call
Reference	ITU-T V.250
Note	

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3.13 AT+CGSN Request product serial number identification

Execute command AT+CGSN[=<snt>]	Execute command causes the TA to return IMEI and related information to identify the MT that the TE is connect to. Response When <snt>=0 (or omitted) and command successful: **<sn>** When <snt>=1 and command successful: **+CGSN: <imei>** When <snt>=2 and command successful: **+CGSN: <imeisv>** When <snt>=3 and command successful: **+CGSN: <svn>** **ERROR/+CME ERROR** Parameter <snt>: integer type indicating the serial number type that has been requested. 0 returns <sn> 1 returns the IMEI 2 returns the IMEISV 3 returns the SVN <sn>: one or more lines of information text determined by the MT manufacturer.

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	<imei>: string type in decimal format indicating the IMEI. <imeisv>: string type in decimal format indicating the IMEISV. <svn>: string type in decimal format indicating the current SVN which is a part of IMEISV.
Test command AT+CGSN=?	Response +CGSN: (list of supported <snt>s)
Reference	3GPP TS 27.007, section 5.4
Note	The default value <snt>=0 returns the information text <sn> with no command name prefixed.

3.14 AT+GSN Request product serial number identification

Execute command AT+GSN[=<snt>]	Execute command causes the TA to return IMEI and related information to identify the MT that the TE is connect to. Response When <snt>=0 (or omitted) and command successful: **<sn>** When <snt>=1 and command successful: **+CGSN: <imei>** **OK** **ERROR/+CME ERROR** Parameter <snt>: integer type indicating the serial number type that has been requested. 0 returns <sn> 1 returns the IMEI

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	<sn>: one or more lines of information text determined by the MT manufacturer. <imei>: string type in decimal format indicating the IMEI.
Test command AT+CSN=?	Response +GSN: (list of supported <snt>s)
Reference	ITU-T V.250
Note	

3.15 AT+CIMI Request International Mobile Subscriber Identity

Execute command AT+CIMI	Execution command causes the TA to return <IMSI>, which is intended to permit the TE to identify the individual SIM card or active application in the UICC (GSM or USIM) which is attached to MT. Response **<IMSI>** **OK** If error is related to ME functionality: **ERROR/+CME ERROR** Parameter **<IMSI>**: International Mobile Subscriber Identity (string without double quotes)
Test command AT+CIMI=?	Response **OK**
Reference	3GPP TS 27.007, section 5.6
Note	

3.16 AT+GCAP Request complete TA capabilities list

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Execute command AT+GCAP	TA reports a list of additional capabilities. Response **+GCAP: <name>** **OK** Parameter <name>: e.g.: +CGSM
Test command AT+GCAP=?	Response **OK**
Reference	V.250
Note	

3.17 ATS3 Command line termination character

Set command ATS3=<n>	This S-parameter represents the decimal IA5 value of the character recognized by the DCE from the DTE to terminate an incoming command line. It is also generated by the DCE as part of the header, trailer, and terminator for result codes and information text, along with the **S4** parameter (see the description of the **V** parameter for usage). Response: **OK** If <n> is not recognized or supported **ERROR** Parameter **<n>** 1 - 127, Set command line termination character to this value. Now only support Mandatory default setting: 13 Carriage return character (CR, IA5 0/13).
Read command ATS3?	Response <**n**> OK

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	Parameter <**n**> See set command.
Reference	ITU-T V.250

3.18 ATS4 Response formatting character

Set command ATS4=<n>	This S-parameter represents the decimal IA5 value of the character generated by the DCE as part of the header, trailer, and terminator for result codes and information text, along with the **S3** parameter (see the description of the **V** parameter for usage). Response: **OK** If <n> is not recognized or supported **ERROR** Parameter **<n>** 1 - 127, Set response formatting character to this value. Now only support Recommended default setting: 10 Line feed character (LF, IA5 0/10).
Read command ATS4?	Response <**n**> OK Parameter <**n**> See set command.
Reference	ITU-T V.250

3.19 ATS5 Command line editing character

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Set command ATS5=<n>	This S-parameter represents the decimal IA5 value of the character recognized by the DCE as a request to delete from the command line the immediately preceding character. Response: **OK** If <n> is not recognized or supported **ERROR** **<n>** 1 - 127, Set command line editing character to this value. Now only support Recommended default setting: 8 Backspace character (BS, IA5 0/8).
Read command ATS4?	Response: **OK** If <n> is not recognized or supported Parameter <**n**> See set command.
Reference	ITU-T V.250

3.20 ATS7 Connection completion timeout

Set command ATS7=<n>	This S-parameter specifies the amount of time, in seconds, that the DCE shall allow between either answering a call (automatically or by the A command) or completion of signalling of call addressing information to network (dialling), and establishment of a connection with the remote DCE. If no connection is established during this time, the DCE disconnects from the line and returns a result code indicating the cause of the disconnection (see the descriptions of the **A** and **D** commands and related dial modifiers for more information). Response: **OK** If <n> is not recognized or supported

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	ERROR Parameter **<n>** Number of seconds in which connection must be established or call will be disconnected. <n> is greater than 0.
Read command ATS7?	Response <**n**> OK Parameter <**n**> See set command.
Reference	ITU-T V.250

3.21 AT+CGMI Request manufacturer identification

Execute command AT+CGMI or AT+GMI	Execution command causes the TA to return one or more lines of information about the manufacturer, determined by the MT manufacturer. Response: **<manufacturer>** **OK** If error: **ERROR** Parameters **<manufacturer>** the manufacturer information, the information text shall not exceed 2048 characters.
Test command AT+CGMI=? or	Response **OK**

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AT+GMI=?	
Reference	ITU-T V.250

3.22 AT+CGMM Request model identification

Execute command AT+CGMM or AT+GMM	Execution command causes the TA to return one or more lines of information about the model, determined by the MT manufacturer. Response: **<model>** **OK** If error: **ERROR** Parameters **<model>** the model information, the information text shall not exceed 2048 characters.
Test command AT+CGMM=? or AT+GMM=?	Response **OK**
Reference	ITU-T V.250

3.23 AT+CGMR Request revision identification

Execute command AT+CGMR or AT+GMR	Execution command causes the TA to return one or more lines of information about the revision, determined by the MT manufacturer. Response: **<revision>** **OK** If error: **ERROR**

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	Parameters **<revision>** the revision information, the information text shall not exceed 2048 characters.
Test command AT+CGMR=? or AT+GMR=?	Response **OK**
Reference	ITU-T V.250

3.24 ATI Request product identification

Execute command ATI	Execution command causes the TA to return one or more lines of information about the product. Response: **<product>** **OK** If error: **ERROR** Parameters **<product>** the product information, including the manufacturer, the model and the revision information. The information text shall not exceed 2048 characters.
Test command ATI=?	Response **OK**
Reference	ITU-T V.250

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3.25 AT+IFC TE-TA local flow control

Execute command AT+IFC=[<by_te> [,<by_ta>]]	Execution command enable or disable the TE-TA local flow control for the serial port. Response: **OK** If error: **ERROR** Parameters **<by_te>** the flow control from TE to TA. 0 disabled, default value. 2 enabled **<by_ta>** the flow control from TA to TE. 0 disabled, default value. 2 enabled
Read command AT+IFC?	**+IFC: <by_te>,<by_ta>** **OK**
Test command AT+IFC=?	**+IFC: (0,2),(0,2)** **OK**
Reference	ITU-T V.250

3.26 AT+IPR Set baud rate

Set command AT+IPR=<baud_rate>	Set the baud rate for the serial port. Response: **OK** If error: **ERROR**

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	Parameters **<baud_rate>** the baud rate of the serial port. Refer to the response of AT+IPR=? for the value range in bps.
Read command AT+IPR?	**+IPR: <baud_rate>** **OK**
Test command AT+IPR=?	**+IPR: (300,600,1200,2400,4800,9600,19200,38400,43000,56000, 57600,115200,921600,2000000,2500000, 3000000,3500000,4000000)** **OK**
Reference	ITU-T V.250

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4 Termination control and status AT Commands

These AT Commands are related to MT control and status reporting.

4.1 AT+CFUN Set phone functionality

Set command AT+CFUN=[<fun>[,<rst>]]	The set command selects the level of functionality <fun> in the MT. Level "full functionality" is where the highest level of power is drawn. "Minimum functionality" is where minimum power is drawn. This AT command is used during the power up sequence and the shutdown sequence. It is used to trigger the detection of the SIM and activation of the protocol stack (depending on <fun>). Note, that CFUN=1 also triggers network registration or CFUN=0 network deregistration. Response **OK/ERROR/+CME ERROR** Parameter **< fun >:** 0: minimum functionality (power off and deactivate USIM) 1: full functionality (activate USIM and power on) 4: switch off PS. 5: deactivate USIM **< rst >:** 0: do not reset the MT before setting it to <**fun**> power level
Read command AT+CFUN?	Response +**CFUN**: <**fun**> **OK**

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	Parameter See above
Test command AT+CFUN=?	Response +**CFUN**: (list of supported <**fun**>s), (list of supported <**rst**>s) **OK** Parameter See above
Example	AT+CFUN=1,0 OK AT+CFUN? +CFUN: 1 OK AT+CFUN=? +CFUN: (0,1,4,5),(0,1) OK
Reference	
Note	

4.2 AT+CMER Mobile Termination event reporting

Set command AT+CMER=[<mode>[,<keyp>[,<disp>[,<ind>[,<bfr>]]]]]	The set command enables and disables the presentation of Unsolicited Result Codes for event reporting. MC35 supports only the type +**CIEV** (indicator event reporting). If enabled the +CIEV URCs are sent whenever the value of an indicator changes. Response **OK/ERROR/+CME ERROR**

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	Parameter <**mode**> 2 buffer unsolicited result codes in the TA when TA-TE link is reserved (e.g. in on line data mode) and flush them to the TE after reservation; otherwise forward them directly to the TE <**keyp**> 0 No keypad event reporting <**disp**> 0 No display event reporting <**ind**> 0 No indicator event reporting 2 Indicator event reporting using result code +CIEV: <ind>,<value> <desc> indicates the indicator's name and <value> is the new value of this indicator. All indicator events shall be directed from TA to TE. <**bfr**> 0 TA buffer of Unsolicited Result Codes defined within this command; is cleared when <mode> 3 is entered
	Unsolicited Result Code: +**CIEV**: <**ind**>,<**value**> <**ind**>: values and their <value> ranges. 1 Battery charge level (0-5), where (0-5) means 0, 20, 40, 60, 80 or 100 per cent of the remaining

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	battery capacity. If no battery is connected to the ME, then the value will be always equal to 5. 2 Signal quality (0-5) 0 is the least field strength, the greater the number is, the stronger the field strength is.5 is the greatest field strength (If customized by user, greatest field strength value may greater than 5). 3 Service level, Service availability (0-1). 4 Sounder activity (0-1). The indicator provides information about tones generated in the ME. The value 1 means for example: Incoming call - ME is ringing. Waiting call. ME generates waiting call tone (if call waiting is enabled). Outgoing call. ME generates Call Progress tone. Outgoing call. ME generates BUSY tone. The value changes to 0, when the tone stops. 5 Unread short message(s) (0-1). 6 Call in progress (0-1). Voice and data calls only. The indicator changes its value as soon as a call has been established, for example when both interlocutors are connected or when the call ends. 7 Roaming status, roaming indicator (0-1). 8 A short message memory storage in the MT has become full (1) or memory locations are available (0); a short message memory storage in the USIM/SIM and NVRAM has become full and a new short message is coming (255); i.e. the range is (0,1,255). <**value**>: integer type value, in the range stated above for the corresponding <ind>.
Read command AT+CMER?	Response **+CMER: <mode>,<keyp>,<disp>,<ind>,<bfr>** **OK**

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	Parameter See above
Test command AT+CMER=?	Response +**CMER**: (list of supported <**mode**>s),(list of supported <**keyp**>s),(list of supported <**disp**>s),(list of supported <**ind**>s),(list of supported <**bfr**>s) **OK** Parameter See above
Remark	1.Only the indicators which are registered (see AT+CIND command), will be reported if CMER is enabled. 2.After CMER has been switched on, +CIEV URCs for all registered indicators with their default value will be presented to the TE. 3.While the ME is in online data mode, no URC will be displayed. Each +CIEV URC is replaced with a break (100ms), and is stored in a buffer. Once the ME goes into command mode (after +++ was entered), all URCs stored in the buffer will be output. Exception: For message indicator, please note that each incoming short message will cause a break.
Example	AT+CMER=2,0,0,2,0 OK AT+CMER? +CMER: 2,0,0,2,0 OK AT+CMER=? +CMER: (2),(0),(0),(0,2),(0) OK
Reference	3GPP TS 27.007, section 8.10
Note	

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4.3 AT+CIND Indicator control

Set command AT+CIND=[<ind>[,<ind>[,...]]]	The set command is used to set the value of MT indicators. Response **OK/ERROR/+CME ERROR** Parameter See Test command
Read command AT+CIND?	Response TA returns the status of the MT indicators. +**CIND**: <**ind**>[,<**ind**>[,...]] Parameter See Test command

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Test command AT+CIND=?	Response +**CIND**: (<**descr**>,(list of supported <**ind**>s))[,(<**descr**>,(list of supported <**ind**>s))[,...]] OK Parameters <**descr**>: values and their <ind> ranges. 1 "battchg" Battery charge level (0-5), where (0-5) means 0, 20, 40, 60, 80 or 100 per cent of the remaining battery capacity. If no battery is connected to the ME, then the value will be always equal to 5. 2 "signal" Signal quality (0-5) 0 is the least field strength, the greater the number is, the stronger the field strength is.5 is the greatest field strength 3 "service" Service availability (0-1). 4 "sounder" Sounder activity (0-1). The indicator provides information about tones generated in the ME. The value 1 means for example: Incoming call - ME is ringing. Waiting call - ME generates waiting call tone (if call waiting is enabled). Outgoing call - ME generates Call Progress tone. Outgoing call - ME generates BUSY tone. The value changes to 0, when the tone stops. 5 "message" Unread short message(s) (0-1). 6 "call" Call in progress (0-1). 7 "roam" Roaming indicator (0-1). 8 "smsfull" A short message memory storage in the MT has become full (1) or memory locations are available (0); i.e. the range is (0-1). <**ind**>: integer type value, in the range stated above for the

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	corresponding <descr>. In the list above, all parameters are presented by the sequence used in the Test, Read and Write commands.
Example	AT+CIND? +CIND: 5,5,1,0,0,0,0,0 The battery is either full or an external supply source is used. The signal quality level is 5, and the ME is registered to its home network. OK AT+CMER=2,0,0,2 Activate Indicator Event Report with AT+CMER OK +CIEV: 1,0 +CIEV: 2,2 +CIEV: 3,1 +CIEV: 4,1 +CIEV: 5,0 +CIEV: 6,0 AT+CIND? +CIND: 1,1,1,0,0,0,0,0 +CIEV:1,X +CIEV:2,X and +CIEV:3,X are enabled OK AT+CMER=2,0,0,2 Activate Indicator Event Report with AT+CMER OK

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	ATD"0123456"; OK You make a call. +CIEV: 5,1 A set of +CIEV URCs reports is presented +CIEV: 6,1 +CIEV: 5,0 +CIEV: 6,0 The receiver hangs up. NO CARRIER ATD"0123456"; OK You make a call. NO CARRIER This time, no +CIEV URCs are displayed.
Reference	3GPP TS 27.007, section 8.9
Note	In current version, only support report signal quality and smsfull. (+CIEV: 2,<signal_strength> ; +CIEV : 8, <smsfull_flag>) Others parameter shall be not supported.

4.4 AT+CSQ Signal quality

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Execute command AT+CSQ	Execution command returns received signal strength indication <rssi> and channel bit error rate <ber> from the MT. Response **+CSQ: <rssi>,<ber>** **OK** If error is related to ME functionality: **+CME ERROR: <err>** Parameter <rssi> 0 -113 dBm or less 1 -111 dBm 2...30 -109... -53 dBm 31 -51 dBm or greater 99 not known or not detectable <ber> (in percent): 0...7 as RXQUAL values in the table in TS 45.008 [20] subclause 8.2.4 99 not known or not detectable
Test command AT+CSQ=?	Response +CSQ: (list of supported <rssi>s),(list of supported <ber>s) **OK**
Example	AT+CSQ +CSQ: 26,99 OK
Reference	3GPP TS 27.007, section 8.5
Note	

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4.5 AT+CTZR Time Zone Reporting

Set command AT+CTZR=[<onoff>]	The set command enables and disables the time zone change event reporting. If the reporting is enabled the MT returns the unsolicited result code +**CTZV**: <**tz**>, +**CTZE**: <**tz**>,<**dst**>,[<**time**>] whenever the time zone is changed. Response **OK/ERROR/+CME ERROR** Parameter **<onoff>:** integer type value indicating 0 disable time zone change event reporting (default). 1 Enable time zone change event reporting by unsolicited result code +**CTZV**: <**tz**> 2 Enable extended time zone and local time reporting by unsolicited result code +**CTZE**: <**tz**>,<**dst**>,[<**time**>] 3 Enable extended time zone and local time reporting by unsolicited result code **+CTZEU: <tz>,<dst>,[<utime>]** <**tz**>**:** string type value representing the sum of the local time zone plus daylight saving time. <**dst**>: integer type value indicating whether <tz> includes daylight savings adjustment. 0 <tz> includes no adjustment for daylight saving time. 1 <tz> includes +1 hour adjustment for daylight saving time. 2 <tz> includes +2 hours adjustment for daylight saving time. <**time**>: string type value representing the local time.
Read command AT+CTZR?	Response +**CTZR**: < **onoff** > **OK** Parameter

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	See above
Test command AT+CTZR=?	Response +**CTZR**: (list of supported <**onoff**>s)) **OK** Parameter See above
Example	AT+CTZR=1 OK AT+CTZR? +CTZR: 1 OK AT+CTZR=? +CTZR: (0-3) OK
Reference	3GPP TS 27.007, section 8.40
Note	

4.6 AT+CPAS Phone activity status

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Set command AT+CPAS	Execution command returns the activity status <pas> of the MT. It can be used to interrogate the MT before requesting action from the phone. Refer subclause 9.2 for possible <err>values. Response **OK** +CME ERROR: <err> Parameter <pas>: 0 ready (MT allows commands from TA/TE) 1 unavailable (MT does not allow commands from TA/TE) 2 unknown (MT is not guaranteed to respond to instructions) 3 ringing (MT is ready for commands from TA/TE, but the ringer is active) 4 call in progress (MT is ready for commands from TA/TE, but a call is in progress) 5 asleep (MT is unable to process commands from TA/TE because it is in a low functionality state)
Test command AT+CPAS=?	Response +CPAS: (list of supported <pas>s) **OK** Parameter See Set command
Reference	3GPP 27.007, section 8.1

4.7 AT+CESQ Extended Signal Quality

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Execution Command AT+CESQ	Execution command returns received signal quality parameters. Response **+CESQ:** **<rxlev>,<ber>,<rscp>,<ecno>,<rsrq>,<rsrp>,<ss_rsrq>,<ss_rsrp>** **,<ss_sinr>** **OK** **+CME ERROR: <err>** Parameter <rxlev>: Integer type. Received signal strength level 0 <rssi>< -110dBm 1 -110dBm ≤<rssi>< -109dBm 2 -109dBm ≤<rssi>< -108dBm ... 61 -50dBm ≤<rssi>< -49dBm 62 -49dBm ≤<rssi>< -48dBm 63 -48dBm ≤<rssi> 99 Not known or not detectable <ber>: Integer type. Channel bit error rate (in percent) 0 <0.01% 1 0.01% --- 0.1% 2 0.1% --- 0.5% 3 0.5% --- 1.0% 4 1.0% --- 2.0% 5 2.0% --- 4.0% 6 4.0% --- 8.0% 7 >=8.0% 99 not known or not detectable <rscp>: Integer type, received signal code power 0 -120 dBm or less 1 -120 dBm ≤<rscp>< -119 dBm 2 -119 dBm ≤<rscp>< -118 dBm ... 94 -27 dBm ≤<rscp>< -26 dBm 95 -26 dBm ≤<rscp>< -25 dBm 96 -25 dBm ≤<rscp> 255 Not known or not detectable <ecno>: Integer type, Ec/No (see 3GPP 25.133) 0 -24 dBm or less 1 -24dBm ≤<ecno>< -23.5 dBm 2 -23.5dBm ≤<ecno>< -23 dBm ... 47 -1dBm ≤<ecno>< -0.5 dBm

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	48 -0.5dBm ≤<ecno>< 0 dBm 49 0 dBm ≤<ecno> 255 Not known or not detectable <rsrq>: Integer type. Reference signal received quality 0 -19.5dB or less 1 -19.5dB ≤<rsrq>< -19dB 2 -19dB ≤<rsrq>< -18.5dB ... 32 -4dB ≤<rsrq>< -3.5dB 33 -3.5dB ≤<rsrq>< -3dB 34 -3dB ≤<rsrq> 255 Not known or not detectable <rsrp>: Integer type. Reference signal received power 0 -140 dBm or less 1 -140dBm ≤<rsrp>< -139 dBm 2 -139dBm ≤<rsrp>< -138 dBm ... 95 -46dBm ≤<rsrp>< -45 dBm 96 -45dBm ≤<rsrp>< -44 dBm 97 -44dBm ≤<rsrp> 255 Not known or not detectable <ss_rsrq>: Integer type. Synchronization signal based reference signal received quality. 0 -43dB or less 1 -43dB ≤<ss_rsrq>< -42.5dB 2 -42.5dB ≤<ss_rsrq>< -42dB ... 124 18.5dB ≤<ss_rsrq>< 19dB 125 19dB ≤<ss_rsrq>< 19.5dB 126 19.5dB ≤<ss_rsrq>< 20dB 255 Not known or not detectable <ss_rsrp>: Integer type. Synchronization signal based reference signal received power. 0 -156 dBm or less 1 -156dBm ≤<ss_rsrp>< -155 dBm 2 -155dBm ≤<ss_rsrp>< -154 dBm ... 124 -33dBm ≤<ss_rsrp>< -32 dBm 125 -32dBm ≤<ss_rsrp>< -31 dBm 126 -31dBm ≤<ss_rsrp> 255 Not known or not detectable <ss_sinr>: Integer type. Synchronization signal based signal to noise and interference ratio. 0 -23 dB or less 1 -23 dB ≤<ss_sinr>< -22.5 dB 2 -22.5 dB ≤<ss_sinr>< -22 dB ...

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	125 39dB ≤<ss_sinr>< 39.5 dB 125 39.5dB ≤<ss_sinr>< 40 dB 126 40dB ≤<ss_sinr> 255 Not known or not detectable
Test Command AT+CESQ=?	Response **+CESQ:** (list of supported <**rxlev**>s,99),(list of supported <**ber**>s,99),(list of supported <**rscp**>s,255),(list of supported <**ecno**>s,255),(list of supported <**rsrq**>s,255),(list of supported <**rsrp**>s,255), (list of supported <**ss_rsrq**>s,255), (list of supported <**ss_rsrp**>s,255), (list of supported <**ss_sinr**>s,255) **OK**
Reference	3GPP 27.007, section 8.69
Note	This command only supports to get <rsrq> and <rsrp>.

4.8 AT+CGPIAF Print IP address format

Set command AT+CGPIAF=[<IPv6_AddressFormat>[,<IPv6_SubnetNotation>[,<IPv6LeadingZeros>[,<IPv6_CompressZeros>]]]]	The set command will decide what format to print IPV6 address parameters of other AT commands. Response **OK/ERROR/+CME ERROR** Parameter **<IPv6_AddressFormat>:** integer type, decides the IPv6 address format. Relevant for all AT command parameters that can hold an IPv6 address. 0 Use IPv4-like dot-notation. IP address, and subnetwork mask if applicable, are dot-separated. 1 Use IPv6-like colon-notation. IP address, and

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	subnetwork mask if applicable and when given explicitly, are separated by a space. <**IPv6_SubnetNotation**>**:** integer type, decides the subnet-notation for remote address and subnet mask. Setting does not apply if <IPv6_AddressFormat>=0. 0 Both IP Address and subnet mask are stated explicitly, separated by a space. 1 The printout format is applying /(forward slash) subnet-prefix Classless Inter-Domain Routing notation. <**IPv6_LeadingZeros**>**:** integer type, decides whether leading zeros are omitted or not. Setting does not apply if <IPv6_AddressFormat>=0. 0 Leading zeros are omitted. 1 Leading zeros are included. <**IPv6_CompressZeros**>**:** integer type, decides whether 1-n instances of 16-bit zero-values are replaced by only '::'. This applies only once. Setting does not apply if <IPv6_AddressFormat>=0. 0 No zero compression. 1 Use zero compression.
Read command AT+CGPIAF?	Response **+CGPIAF**: <IPv6_AddressFormat>, <IPv6_SubnetNotation>, <IPv6LeadingZeros>, <IPv6_CompressZeros> **OK** Parameter See above
Test command AT+CGPIAF =?	Response **+CGPIAF**: (0-1),(0-1),(0-1),(0-1) **OK**
Reference	3GPP TS 27.007, section 8.62
Note	

4.9 AT+CAVIMS Availability for voice calls with IMS

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Set command AT+CAVIMS=[<state>]	The set command will inform MT whether the UEs is currently available for voice calls with IMS. Response **OK/ERROR/+CME ERROR** Parameter **<state>:** integer type. The UEs IMS voice call availability status stored in the MT. 0 voice calls with the IMS are not available 1 voice calls with the IMS are available
Read command AT+CAVIMS?	Read command returns the UEs IMS voice call availability status stored in the MT. Response +**CAVIMS**: <state> **OK** Parameter See above
Test command AT+CAVIMS=?	Test command returns supported values as a compound value. Response **+CAVIMS**: (0-1) **OK**
Reference	3GPP TS 27.007, section 8.68
Note	

4.10 AT+CIREP IMS network reporting

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Set command AT+CIREP=[<reporting>]	Set command enables or disables reporting of IMS network. Response **OK/ERROR/+CME ERROR** Parameter **<reporting>:** integer type. Enables or disables reporting of IMS network. 0 Disable reporting 1 Enable reporting <**ue_report_nw_IMSVOPS_flag**>: integer type; shows whether the UE reports IMS Voice Over PS session supported indication received from network. 0 UE does not report IMS Voice Over PS session supported indication received from network. 1 UE reports IMS Voice Over PS session supported indication received from network. <**nw_support_ims_vops**>: integer type. Gives the last IMS Voice Over PS session supported indication received from network. 0 IMSVOPS support indication is not received from network, or is negative. 1 IMSVOPS support indication as received from network is positive.
Read command AT+CIREP?	Read command returns the status of result code presentation and the IMSVOPS supported indication. Response +**CIREP**: <ue_report_nw_IMSVOPS_flag>, <nw_support_ims_vops> **OK** Parameter See above

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Test command AT+CIREP=?	Test command returns supported values as a compound value. Response **+CIREP**: (0-1) **OK**
Reference	3GPP TS 27.007, section 8.64
Note	

4.11 AT+CCLK Set clock

Set command AT+CCLK=<time>	Set the real-time clock of the MT. Response: **OK** If error: **ERROR** Parameters **<time>** the real-time clock, string type, format is "yy/mm/dd, hh:mm:ss±zz". zz is time zone, range -96…+96.
Read command AT+CCLK?	**+CCLK: <time>** **OK**
Reference	Refer 27.007 section 8.15

4.12 AT+CLAC List all available AT commands

Execute command AT+CLAC	Execution command causes the MT to return one or more lines of AT Commands. Response: <AT Command1>

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	[<CR><LF><AT Command2> […]] **OK** If error: **ERROR** Parameters **<AT Command>** Defines the AT command including the prefix AT. Text shall not contain the sequence 0<CR> or OK<CR>
Test command AT+CLAC=?	Response **OK**
Reference	Refer 27.007 section 8.37

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5 Network service related AT Commands

This clause describes network related commands. Commands include network registration information query, access technology switch, and basic supplementary service handing.

5.1 AT+CREG Network registration

Set command AT+CREG=[<n>]	Set command controls the presentation of an unsolicited result code +CREG: <stat> when <n>=1 and there is a change in the MT network registration status, or code +CREG: <stat>[,<lac>,<ci>[,<AcT>]] when <n>=2 and there is a change of the network cell. when <n>=0 report nothing if the MT network registration status is change. When <n>=3, the unsolicited result code format (+CREG: <stat>) is the same as <n>=1, but <stat>=5 and <stat>=1 is regards as the same status: registered success. In other words, <stat>=5 does not exist when <n>=3. Response **OK/ERROR** Parameter <**n**>: 0 disable network registration unsolicited result code 1 enable network registration unsolicited result code +CREG: <stat> 2 enable network registration unsolicited result code +CREG: <stat>[,<lac>,<ci>[,<AcT>]]

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Read command AT+CREG?	Read command returns the status of result code presentation and an integer <stat> which shows whether the network has currently indicated the registration of the MT. Response +CREG: <n>,<stat>[,<lac>,<ci>[,<AcT>]] **OK** If error is related to ME functionality: **+CME ERROR: <err>** Parameter <**stat**>: 0 not registered, MT is not currently searching a new operator to register to 1 registered, home network 2 not registered, but MT is currently searching a new operator to register to 3 registration denied 4 unknown 5 registered, roaming <lac>: string type, two bytes location area code in hexadecimal format (e.g. "00C3" equal 195 in decimal) <ci>: string type, two bytes (four bytes in LTE version) cell ID in hexadecimal format **<AcT>**: refer +CEREG
Test command AT+CREG=?	Response +**CREG**: (list of supported <**n**>s) **OK**
Reference	3GPP 27.007, section 7.2
Note	

5.2 AT+COPS PLMN Selection

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Set command AT+COPS=[<mode>[,<format>[,<oper>[,<AcT>]]]]	This command allows you to choose or register to the GSM/UMTS/EPS network operator in automatically or manually. When using the manual mode, the <**oper**> must be entered, no matter whether you want to search for the home operator or another one. Note: If <mode>=1/4 and <AcT> is ignored, <AcT> shall be NOT specified. Response **OK** If error is related to ME functionality: **+CME ERROR: <err>** Parameter: <**mode**> 0 automatic mode; <**oper**> field is ignored 1 manual operator selection. <**oper**> field must be present, 2 manually deregister from network and remain unregistered until mode 0,1,4 is selected 3 set <**format**> for read command +COPS? 4 combination of manual/automatic mode; if manual selection fails, ME switches to automatic mode (<**mode**>=0). (<**oper**> field must be present) <**format**> 0 long format alphanumeric <**oper**> 1 short format alphanumeric <**oper**> 2 numeric <**oper**> <**stat**>: 0 unknown 1 available 2 current 3 forbidden <**oper**> string type; <**format**> indicates if the format is alphanumeric or

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	numeric; long alphanumeric format can be up to 16 characters long and short format up to 8 characters; numeric format is the Location Area Identification number which consists of a three BCD digit country code, plus a two BCD digit network code, which is administration specific; returned <**oper**> shall not be in BCD format, but in IRA characters converted from BCD; hence the number has structure: (country code digit 3)(country code digit 2)(country code digit 1)(network code digit 3)(network code digit 2)(network code digit 1) **<AcT>** Access technology selected 0 GSM 1 GSM compact 2 UTRAN 7 EUTRAN
Read command AT+COPS?	TA returns the current mode and, if registered, the currently used operator. If the ME is unregistered, <**format**> and <**oper**> are omitted. Response **+COPS: <mode>[, <format>, <oper>[, <ACT>]]** **OK** If error is related to ME functionality: **+CME ERROR: <err>** Parameter Only the following one is different with the set command. **<AcT>** Access technology selected 0 GSM 1 GSM compact 2 UTRAN 7 EUTRAN Other parameters refer to set command

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	Example To query the status of the ME's network registration using the read command: AT+COPS? +COPS: 0,2,"46000",7 OK
Test command AT+COPS=?	TA returns a list of quadruplets, each representing an operator present in the network. The list of operators is presented in the following order: Home network, networks referenced in SIM, and other networks. Response **+COPS:** (list of supported (**<stat>**, long alphanumeric **<oper>**, short alphanumeric <**oper**>, numeric<**oper**>s,<AcT>s)) [,(list of supported <**mode**>s), (list of supported <**format**>s)] **OK** If error is related to ME functionality: **+CME ERROR: <err>** Parameter See Set command
Example	Attempt to manually select a forbidden operator: AT+COPS=1,2,"46000",7 OK To query the present status of ME's network registration using the test command: AT+COPS=? +COPS: (2,"D2","XX","26202"),(3,"E-Plus","YY","26203"),(3,"TD1",,"26201"),(3,"Interkom","ZZ","26207"),,(0-4),(0,2) OK Registered operator is D2. The other operators are present in the network, but not allowed to be used with the current SIM card.
Reference	3GPP 27.007, section 7.3

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Note	AT+COPS? report the read values according to the current settings.

5.3 AT+CPLS Selection of preferred PLMN list

Set command AT+CPLS=<list>	This command is used to select one PLMN selector with Access Technology list in the SIM card or active application in the UICC (GSM or USIM), that is used by +CPOL command. Execute command selects a list in the SIM/USIM. Response **OK** If error is related to ME functionality: **+CME ERROR: <err>** Parameter: **<list>** 0 User controlled PLMN selector with Access Technology EFPLMNwAcT, if not found in the SIM/UICC then PLMN preferred list EFPLMNsel (this file is only available in SIM card or GSM application selected in UICC) 1 Operator controlled PLMN selector with Access Technology EFOPLMNwAcT 2 HPLMN selector with Access Technology EFHPLMNwAcT 3 EHPLMN selector with Access Technology EFEHPLMNwAcT
Read command AT+CPLS?	TA returns the selected PLMN selector list from the SIM/USIM. Response **+CPLS: <list>** **OK**

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	Parameter See Set command
Test command AT+CPLS=?	TA returns the whole index range supported lists by the SIM/USIM. Response **+CPLS:** (list of supported (**<list>**s) **OK** Parameter See Set command
Reference	3GPP 27.007, section 7.20
Note	

5.4 AT+CPOL Preferred PLMN list

Set command AT+CPOL=[<index>][,<format>[,<oper>[,<GSM_AcT>,<GSM_Compact_AcT>,<UTRAN_AcT>,<EUTRAN_AcT>]]]	This command is used to edit the PLMN selector with Access Technology lists in the SIM card or active application in the UICC (GSM or USIM). Execute command writes an entry in the SIM/USIM list of preferred PLMNs, previously selected by the command +**CPLS**. If no list has been previously selected, the User controlled PLMN selector with Access Technology, $EF_{PLMNwAcT}$, is the one accessed by default. If <**index**> is given but <**oper**> is left out, entry is deleted. If <**oper**> is given but <**index**> is left out, <**oper**> is put in the next free location. If only <**format**> is given, the format of the <**oper**> in the read command is changed. The Access Technology selection parameters, <**E-UTRAN_AcT**>are required when writing User controlled PLMN selector with Access Technology, **$EF_{PLMNwAcT}$**, Operator controlled PLMN selector with Access Technology $EF_{OPLMNwAcT}$ and HPLMN selector with Access Technology $EF_{HPLMNwAcT}$. Response **OK**

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	If error is related to ME functionality: **+CME ERROR: <err>** Parameter: <**index**> 1-8 integer type; the order number of operator in the SIM/USIM preferred operator list <**format**> 0 long format alphanumeric <**oper**> 1 short format alphanumeric <**oper**> 2 numeric <**oper**> <**oper**n> string type; <**format**> indicates if the format is alphanumeric or numeric (see +**COPS**) <**GSM_AcT**n>: integer type; GSM access technology 0 access technology not selected 1 access technology selected <**GSM_Compact_AcT**n>: integer type; GSM compact access technology 0 access technology not selected 1 access technology selected <**UTRAN_AcTn**>: integer type; UTRAN access technology 0 access technology not selected 1 access technology selected <**E-UTRAN_AcT**n>: integer type; E-UTRAN access technology 0 access technology not selected 1 access technology selected
Read command AT+CPOL?	TA returns all used entries from the SIM/USIM list of preferred PLMNs, previously selected by the command +**CPLS**, with the

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	Access Technologies for each PLMN in the list. Response **+CPOL:<index1>,<format>,<oper1>[,<GSM_AcT1>,<GSM_Compact_AcT1>,<UTRAN_AcT1>,<E-UTRAN_AcT1>]** [<CR><LF>+**CPOL**:<**index2**>,<**format**>,<**oper2**>[,<**GSM_AcT2**>,<**GSM_Compact_AcT2**>,<**UTRAN_AcT2**>,<**E-UTRAN_AcT2**>] [...]] **OK** If error is related to ME functionality: **+CME ERROR: <err>** Parameter See Set command
Test command AT+CPOL=?	TA returns the whole index range supported by the USIM. Response **+CPOL:** (list of supported <**index**>s),(list of supported <**format**>s) **OK** If error is related to ME functionality: **+CME ERROR: <err>** Parameter See Set command
Reference	3GPP 27.007, section 7.19
Note	

5.5 AT+CLCK Facility lock

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Execute command AT+CLCK=<fac>, <mode>[,<passwd >[,<class>]]	Execute command is used to lock, unlock or interrogate a MT or a network facility <fac>. Password is normally needed to do such actions. When querying the status of a network service (<mode>=2) the response line for 'not active' case (<status>=0) should be returned only if service is not active for any <class>. This command should be abortable when network facilities are set or interrogated. Calls barring facilities are based on supplementary services. The interaction of these with other commands based on other supplementary services is described in the specification of the relevant supplementary service. Response **when <mode>=2 and command successful:** **+CLCK**: <**status**>[,<**class1**> [<**CR**><**LF**>**+CLCK: <status>,<class2>** **[…]]** **OK** If error is related to ME functionality: **+CME ERROR: <err>** Parameter <**fac**> values reserved by the present document: "SC" SIM (lock SIM/UICC card) (SIM/UICC asks password in MT power-up and when this lock command issued) "AO" BAOC (Barr All Outgoing Calls) "OI" BOIC (Barr Outgoing International Calls) "OX" BOIC-exHC (Barr Outgoing International Calls except to Home Country) (refer 3GPP TS 22.088 [6] clause 1) "IR" BIC-Roam (Barr Incoming Calls when Roaming outside the home country) (refer 3GPP TS 22.088 [6] clause 2)"AB" All Barring services (applicable only for <mode>=0)"OX" BOIC-exHC (Barr Outgoing

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	International Calls except to Home Country) "AI" BAIC (Barr All Incoming Calls) "AB" All Barring services (applicable only for <mode>=0) "AG" All outgoing barring services (applicable only for <mode>=0) "AC" All incoming barring services (applicable only for <mode>=0) <**mode**>: 0 unlock 1 lock 2 query status <**status**>: 0 not active 1 active <**passwd**>: string type; shall be the same as password specified for the facility from the MT user interface or with command Change Password +CPWD <**classx**> is a sum of integers each representing a class of information (default 7): 1 voice (telephony) 2 data (refers to all bearer services; with <mode>=2 this may refer only to some bearer service if TA does not support values 16, 32, 64 and 128) 4 fax (facsimile services) 5 all Teleservices-ExeptSMS 7 voice, data and fax 8 short message service 12 all Data Teleservices 13 all Teleservices 16 data circuit sync 32 data circuit async 247 all Synchronous Services Others reserved all Asynchronous Services

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Test command AT+CLCK=?	Response +**CLCK**: (list of supported <**fac**>s) **OK**
Reference	3GPP 27.007, section 7.4 NOTE: The supported values of the parameters are not the same as the 3GPP 27.007.
Note	

5.6 AT+CPWD Change password

Execute command AT+CPWD=<fac>, <oldpwd>,<newpwd>	Action command sets a new password for the facility lock function defined by command Facility Lock +CLCK. Response **OK / ERROR** If error is related to ME functionality: **+CME ERROR: <err>** Parameter <**fac**> values reserved by the present document: "P2" SIM PIN2 Refer Facility Lock +CLCK for other values <**oldpwd**>, <**newpwd**>: String type; <oldpwd> shall be the same as password specified for the facility from the MT user interface or with command Change Password +CPWD and <newpwd> is the new password; maximum length of password can be determined with <pwdlength>
Test command AT+CPWD=?	Response +**CPWD**: (list of supported <**fac**>,<**pwdlength**>)s **OK** Parameter <fac>: see above. <**pwdlength**>: integer type maximum length of the password for the facility
Reference	3GPP 27.007, section 7.5 NOTE: The supported values of the parameters are not the same as the 3GPP 27.007
Note	This command only supports the following <fac>:

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	“AB”,”AG”,”AI”,”OI”,”OX”,”IR””SC”,”P2”

5.7 AT+CEREG EPS network registration status

Set command AT+CEREG=[<n>]	The set command controls the presentation of an unsolicited result code **+CEREG: <stat>** when <**n**>=1 and there is a change in the MT's EPS network registration status, or code **+CEREG: <stat>[,<tac>,<ci>[,<AcT>]]** when <**n**>=2 and there is a change of the network cell. when <n>=4 the unsolicited result code will provide the UE with additional information for the Active Time value and the extended periodic TAU value if there is a change of the network cell in E-UTRAN. Response **OK/ERROR/CME ERROR** Parameter <**n**>: 0 disable network registration unsolicited result code 1 enable network registration unsolicited result code +CEREG: <stat> 2 enable network registration and location information unsolicited result code +CEREG: <stat>[,<tac>,<ci>[,<AcT>]] 4 For a UE that wants to apply PSM, enable network registration and location information unsolicited result code + CEREG: <stat>[,[<tac>],[<ci>],[<AcT>][,,[,[<Active-Time>],[<Periodic-TAU>]]]] <**stat**>: EPS registration status 0 not registered, MT is not currently searching an operator to register to 1 registered, home network 2 not registered, but MT is currently trying to attach or searching an operator to register to 3 registration denied 4 unknown 5 registered, roaming <**tac**>:

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	string type; two bytes tracking area code in hexadecimal format (e.g. "00C3" equals 195 in decimal) <**ci**>: string type; four byte E-UTRAN cell ID in hexadecimal format <**AcT**>: access technology of the registered network 0 GSM 1 GSM Compact 2 UTRAN 7 E-UTRAN <**Active-Time**>: string type; one byte in an 8 bit format. Indicates the Active Time value (T3324) allocated to the UE in E-UTRAN. <**Periodic-TAU**>: string type; one byte in an 8 bit format. Indicates the extended periodic TAU value (T3412) allocated to the UE in E-UTRAN.
Read command AT+CEREG?	The read command returns the status of result code presentation and an integer <**stat**> which shows whether the network has currently indicated the registration of the MT. Response When <n> = 0,1 or 2 and command successful: **+CEREG: <n>,<stat>[,<tac>,<ci>[,<AcT>]]** **OK** When <n> = 4 and command successful: **+CEREG: <n>,<stat>[,[<tac>],[<ci>],[<AcT>][,,[,[<Active-Time>],[<Periodic-TAU>]]]]** **OK** Parameter see set command
Test command AT+CEREG=?	Test command returns the modes and buffer settings supported by the MT as compound values. Response **+CEREG: (**list of supported**<n>**s**)** **OK** Parameter

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	see set command
Example	AT+CEREG =1 OK AT+CEREG? +CEREG: 1,1 OK AT+CEREG =? +CEREG: (0-2,4) OK
Reference	Refer 27.007 section 10.1.22
Note	1. The list of supported <n> is (0-2,4) 2. The default value of <n> is 0. <n> is usually controlled by user/AP based on requirement.

5.8 AT+CPSMS Power saving mode setting

Set command AT+CPSMS=[<mode>[,<Requested_Periodic-RAU>[,<Requested_GPRS-READY-timer>[,<Requested_ Periodic_TAU>[,<Requested_Active-Time>]]]]	The set command controls the setting of the UE's power saving mode (PSMS) parameters. Response **OK/ERROR** Parameter <**mode**>: integer type. Indication to disable or enable the use of PSM in the UE. disable network registration unsolicited result code 0 Disable the use of PSM 1 Enable the use of PSM 2 Disable the use of PSM and discard all parameters for PSM or, if available, reset to the manufacturer specific default values. <**Requested_Periodic-RAU**>: string type; one byte in an 8 bit format. Requested extended periodic RAU value (T3312) to be allocated to the UE in GERAN/UTRAN. The requested extended periodic RAU value is coded as one byte (octet3) of the GPRS Timer 3 information element coded as bit format (e.g. "01000111" equals 70 hours). For the coding and value range, see the GPRS Timer 3 IE in 3GPP TS 24.008 [8] Table

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	10.5.163a/3GPP TS 24.008. See also 3GPP TS 23.682 [149] and 3GPP TS 23.060 [47]. The default value, if available, is manufacturer specific. <**Requested_GPRS-READY-timer**>: string type; one byte in an 8 bit format. Requested GPRS READY timer value (T3314) to be allocated to the UE in GERAN/UTRAN. The requested GPRS READY timer value is coded as one byte (octet 2) of the GPRS Timer information element coded as bit format. For the coding and value range, see the GPRS Timer IE in 3GPP TS 24.008[8] Table 10.5.172/3GPP TS 24.008. See also 3GPP TS 23.060 [47]. The default value, if available, is manufacturer specific. <**Requested_Periodic-TAU**>: string type; one byte in an 8 bit format. Requested extended periodic TAU value (T3412) to be allocated to the UE in E-UTRAN. The requested extended periodic TAU value is coded as one byte (octet3) of the GPRS Timer 3 information element coded as bit format (e.g. “01000111” equals 70 hours). For the coding and value range, see the GPRS Timer 3 IE in 3GPP TS 24.008 [8] Table 10.5.163a/3GPP TS 24.008. See also 3GPP TS 23.682 [149] and 3GPP TS 23.401 [82]. The default value, if available, is manufacturer specific. <**Requested_Active-Time**>: string type; one byte in an 8 bit format. Requested Active Time value (T3324) to be allocated to the UE. The requested Active Time value is coded as one byte (octet 3) of the GPRS Timer 2 information element coded as bit format (e.g. “00100100” equals 4 minutes).For the coding and value range, see the GPRS Timer 2 IE in 3GPP TS 24.008[8] Table 10.5.163/3GPP TS 24.008. See also 3GPP TS 23.682 [149], 3GPP TS 23.060 [47] and 3GPP TS 23.401 [82]. The default value, if available, is manufacturer specific.

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Read command AT+CPSMS?	Read command returns the current parameters values. Response +CPSMS: <mode>, , ,[<Requested_ Periodic_ TAU>],[<Requested _Active-Time>] **OK** If error: **+CME ERROR: <err>** Parameter See Set command
Test command AT+CPSMS=?	Response +**CPSMS**: (list of supported <**mode**>s), , , (list of supported <**Requested_ Periodic_ TAU**>s),(list of supported <**Requested_Active-Time**>s) **OK** Parameter See Set command
Reference	3GPP 27.007, section 7.38
Note	The <**Requested_Periodic-RAU** > and <**Requested_GPRS-READY-timer**> is null in set command.

5.9 AT+CEDRXS eDRX setting

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Set command AT+CEDRXS=[<mode>[,<AcT-type>[,<Requested_eDRX_value>]]]	The set command controls the setting of the UEs eDRX parameters. Response **OK/ERROR** Parameter <**mode**>: integer type, indicates to disable or enable the use of eDRX in the UE. This parameter is applicable to all specified types of access technology, i.e. the most recent setting of <mode> will take effect for all specified values of <AcT>. 0 Disable the use of eDRX 1 Enable the use of eDRX 2 Enable the use of eDRX and report URC +CEDRXP 3 Disable the use of eDRX and discard all parameters for eDRX or, if available, reset to the manufacturer specific default values. <**AcT-type**>: integer type, indicates the type of access technology. This AT-command is used to specify the relationship between the type of access technology and the requested eDRX value. 0 Access technology is not using eDRXS. This parameter value is only used in the unsolicited result code. 1 EC-GSM-IoT (A/Gb mode) 2 GSM (A/Gb mode) 3 UTRAN (lu mode) 4 E-UTRAN (WB-S1 mode) 5 E-UTRAN (NB-S1 mode) <**Requested_eDRX_value**>: string type; half a byte in a 4 bits format. The eDRX value refers to bit 4 to 1 of octet 3 of the Extended DRX parameters information element (see clause 10.5.5.32 of 3GPP TS 24.008 [8]). For the coding and the value range, see Extended DRX parameters information element in 3GPP TS 24.008 [8] Table 10.5.5.32/3GPP TS 24.008. The default value, if available, is manufacturer specific.

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Read command AT+CEDRXS?	Read command returns the current settings for each defined value of <AcT-type>. Response [+CEDRXS: <AcT-type>, <Requested_eDRX_value> [<CR><LF>+CEDRXS: <AcT-type>, <Requested_eDRX_value> […]]] If error: **+CME ERROR: <err>** Parameter See Set command
Test command AT+CEDRXS=?	Response +**CEDRXS**: (list of supported <**mode**>s), (list of supported <**AcT-type**>s),(list of supported <**Requested_eDRX_value**>s) **OK** Parameter See Set command
Reference	3GPP 27.007, section 7.40
Note	

5.10 AT+CNEM Network emergency (bearer) services support

Set command AT+CNEM=[<reporting>]	The set command enables reporting of changes in the emergency (bearer) services support indicators. Response **OK/ERROR** Parameter <**reporting**>: integer type 0 Reporting not enabled 1 Reporting enabled

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	<**emb_Iu_supp**>: integer type. Emergency bearer services support indicator for Iu mode. 0 Emergency bearer services in Iu mode and A/Gb mode not supported 1 Emergency bearer services supported in Iu mode, but not supported in A/Gb mode <**emb_S1_supp**>: integer type. Emergency bearer services support indicator for S1 mode. 0 Emergency bearer services in S1 mode not supported 1 Emergency bearer services in S1 mode supported
Read command AT+CNEM?	Response **+CNEM**: <reporting>[,<emb_Iu_supp>[,<emb_S1_supp>]] **OK** Parameter See Set command
Test command AT+CNEM=?	Response **+CNEM**: (0-1) **OK**
Reference	3GPP 27.007, section 7.33
Note	The <emb_Iu_supp> is always 0 in read command.

5.11 AT+COLP Connected line identification presentation

Set command AT+COLP=[<n>]	Set command to enable/disable +COLP result code. When <n>=1, the presentation of the connected line identity at the TE enabled and when the called subscriber allows the intermediate result code, **+COLP: <number>,<type>[,<subaddr>,<satype> [,<alpha>]]**. If error is related to ME functionality: **+CME ERROR: <err>** Parameter **<n>:** integer type (parameter sets/shows the result code presentation status to the TE).

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	0 disable 1 enable **<m>:** integer type (parameter shows the subscriber COLP service status in the network). 0 COLP / TIP not provisioned 1 COLP / TIP provisioned 2 unknown (e.g. no network, etc.) **<number>**: string type phone number in format specified by <type>. **<type>:** type of address octet in integer format (refer 3GPP TS 24.008 [8] clause 10.5.4.7)
Read command AT+COLP?	Response: **+COLP: <n>,<m>** Parameter See set command
Test command AT+COLP=?	Response: **+COLP: (list of supported <n>s)** Parameter See set command
Reference	Refer 27.007 section 7.8
Note	

5.12 AT+COPN Read operator names

Execute command AT+COPN	Execute command returns the list of operator names from the MT. Response **+COPN: <numeric1>,<alpha1>** [<CR><LF>+COPN: <numeric2>,<alpha2> […]] If error is related to ME functionality: **+CME ERROR: <err>** Parameter **<numericn>:** string type; operator in numeric format. **<alphan>:** string type; operator in long alphanumeric format.

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Test command AT+COPN=?	Response **OK**
Reference	Refer 27.007 section 7.21
Note	

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6 SMS related AT Commands

6.1 AT+CSMS Select Message Service

Set command AT+CSMS=<service>	Set command selects messaging service <service>. It returns the types of messages supported by the ME: <mt> for mobile terminated messages, <mo> for mobile originated messages and <bm> for broadcast type messages. Response **+CSMS: <mt>,<mo>,<bm>** **OK** If error is related to ME functionality: **+CMS ERROR: <err>** Parameter **<service>** 0 3GPP TS 23.040 [3] and 3GPP TS 23.041 [4] 1 3GPP TS 23.040 [3] and 3GPP TS 23.041 [4] the requirement of <service> setting 1 is mentioned under corresponding command descriptions). **<mt>** Mobile Terminated Messages: 0 Type not supported 1 Type supported **<mo>** Mobile Originated Messages: 0 Type not supported 1 Type supported **<bm>** Broadcast Type Messages: 0 Type not supported 1 Type supported Example: AT+CSMS=0 +CSMS: 1,1,0

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	OK
Read command AT+CSMS?	Response **+CSMS: <service>,<mt>,<mo>,<bm>** **OK** Parameter See set command
Test command AT+CSMS=?	Response **+CSMS**: (list of supported **<service>**s) **OK** Parameter See set command
Reference	3GPP TS 27.005, section 3.2.1
Note	

6.2 AT+CGSMS Select service for MO SMS messages

Set command AT+CGSMS=[<service>]	The set command is used to specify the service or service preference that the MT will use to send MO SMS messages. Response **OK/ERROR/CME ERROR** Parameter <**service**>: a numeric parameter which indicates the service or service preference to be used 0 Packet Domain 1 circuit switched 2 Packet Domain Preferred (use circuit switched if GPRS not available) 3 Circuit switched Preferred (use Packet Domain if circuit switched not available)
Read command AT+CGSMS?	The read command returns the currently selected service or service preference.

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	Response **+CGSMS: <service>** **OK** Parameter **<service >** see set command
Test command AT+CGSMS =?	The test command is used for requesting information on the currently available services and service preferences. Response **+CGSMS: (**list of currently available **<service>**s**)** **OK** Parameter **<service>** see set command
Example	AT+CGSMS =1 OK AT+CGSMS? +CGSMS: 1 OK AT+CGSMS =? +CGSMS: (0-3) OK
Reference	Refer 27.007 section 10.1.21
Note	

6.3 AT+CMGF Select SMS message format

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Set command AT+CMGF=[<mode>]	TA sets parameter which specifies the input and output format of messages to be used. Response **OK** Parameter **<mode>** 0 PDU mode 1 text mode Example AT+CMGF=0 OK
Read command AT+CMGF?	Response **+CMGF: <mode>** **OK** Parameter See set command
Test command AT+CMGF=?	Response **+CMGF:** (list of supported **<mode>s**) **OK** Parameter See set command
Reference	3GPP TS 27.005, section 3.2.3

6.4 AT+CSCA SMS service center address

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Set command AT+CSCA=<sca>[,<tosca>]	TA updates the SMSC address, through which mobile originated SMs are transmitted. In test mode, setting is used by send and write commands. In PDU mode, setting is used by the same commands, but only when the length of the SMSC address coded into <pdu> parameter equals zero. Note: This command writes the service center address to non-volatile memory. Response **OK** Parameter **<sca>** 3GPP TS 24.011 RP SC address value field in string format; BCD numbers (or GSM default alphabet characters) are converted into characters of the currently selected TE character set (refer to command +CSCS in 3GPP TS 27.007 [9]); type of address is given by **<tosca>.** Maximum length of address: 20 characters (excluding '+' mark) **<tosca>** Service center address format 3GPP TS 24.011 RP SC address Type-of-Address octet in integer format Example: AT+CSCA="+8613800210500",145 OK
Read command AT+CSCA?	Response **+CSCA: <sca>,<tosca>** **OK** Parameter See set command
Test command AT+CSCA=?	Response **OK**
Reference	3GPP TS 27.005, section 3.3.1
Note	

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6.5 AT+CSMP Set Text Mode Parameters

Set command AT+CSMP=[<fo>[,<vp>[,<pid>[,<dcs>]]]]	Set command is used to select values for additional parameters needed when SM is sent to the network or placed in a storage when text format message mode is selected. It is possible to set the validity period starting from when the SM is received by the SMSC (<vp> is in range 0... 255) or define the absolute time of the validity period termination (<vp> is a string). The format of <vp> is given by <fo>. If TA supports the EVPF, see 3GPP TS 23.040 [3], it shall be given as a hexadecimal coded string (refer e.g. <pdu>) with double quotes. Response **OK** Parameter **<fo>** depending on the command or result code: first octet of 3GPP TS 23.040 [3] SMS-DELIVER, SMS-SUBMIT (default 17), SMS-STATUS-REPORT, or SMS-COMMAND (default 2) in integer format <**vp**> depending on SMS-SUBMIT <fo> setting: 3GPP TS 23.040 [3] TP-Validity-Period either in integer format (default 167), in time-string format (refer <dt>), or if EVPF is supported, in enhanced format (hexadecimal coded string with double quotes) **<pid>** 3GPP TS 23.040 [3] TP-Protocol-Identifier in integer format (default 0) <**dcs**> depending on the command or result code: 3GPP TS 23.038 [2] SMS Data Coding Scheme (default 0), or Cell Broadcast Data Coding Scheme in integer format
Read command AT+CSMP?	Response **+CSMP: <fo>,<vp>,<pid>,<dcs>** **OK** Parameter See set command
Test command AT+CSMP=?	Response **OK**

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Reference	3GPP TS 27.005, section 3.3.2
Note	

6.6 AT+CSDH Show Text Mode Parameters

Set command AT+CSDH=[<show>]	Set command controls whether detailed header information is shown in text mode result codes. Response **OK** Parameter **<show>** 0 do not show header values defined in commands +CSCA and +CSMP (<sca>, <tosca>, <fo>, <vp>, <pid> and <dcs>) nor <length>, <toda> or <tooa> in +CMT, +CMGL, +CMGR result codes for SMS-DELIVERs and SMS-SUBMITs in text mode; for SMS-COMMANDs in +CMGR result code, do not show<pid>, <mn>, <da>, <toda>, <length> or <cdata> 1 show the values in result codes
Read command AT+CSDH?	Response **+CSDH: <show>** **OK** Parameter See set command
Test command AT+CSDH=?	Response **+CSDH: (list of supported <show>s)** **OK**
Reference	3GPP TS 27.005, section 3.3.3
Note	

6.7 AT+CSAS Save Settings

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Set command AT+CSAS[=<profile>]	Execution command saves active message service settings to a non-volatile memory. A TA can contain several profiles of settings. Settings specified in commands Service Centre Address **+CSCA**, Set Message Parameters **+CSMP** are saved. Response **OK** If error is related to ME functionality: **+CMS ERROR:<err>** Parameter **<profile>** 1-4 profile number Example: AT+CSAS=1 OK
Test command AT+CSAS=?	Response **+CSAS:** (list of supported **<profile>s**) **OK** Parameter See set command
Reference	3GPP TS 27.005, section 3.3.5
Note	

6.8 AT+CRES Restore Settings

Set command AT+CRES[=<profile>]	Execution command restores message service settings from non-volatile memory to active memory. A TA can contain several profiles of settings. Settings specified in commands Service Centre Address **+CSCA**, Set Message Parameters **+CSMP** are restored. Response **OK** If error is related to ME functionality:

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	+CMS ERROR:<err> Parameter **<profile>** 1-4 profile number
Test command AT+CRES=?	Response **+CRES:** (list of supported **<profile>s**) **OK** Parameter See set command
Reference	3GPP TS 27.005, section 3.3.6
Note	

6.9 AT+CNMI New Message Indications to TE

Set command AT+CNMI=[<mode>[, <mt>[,<bm>[,<ds>[, <bfr>]]]]]	Set command selects the procedure, how receiving of new messages from the network is indicated to the TE when TE is active, e.g. DTR signal is ON. If TE is inactive (e.g. DTR signal is OFF), message receiving should be done as specified in 3GPP TS 23.038 [2]. NOTE 1: When DTR signal is not available or the state of the signal is ignored (V.250 command &D0), reliable message transfer can be assured by using **+CNMA** acknowledgement procedure. Response **OK** If error is related to ME functionality: **+CMS ERROR:<err>**

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Parameter

<mode>

0 Buffer unsolicited result codes in the TA. If TA result code buffer is full, indications can be buffered in some other place or the oldest indications may be discarded and replaced with the new received indications.

1 Discard indication and reject new received message unsolicited result codes when TA-TE link is reserved (e.g. in on-line data mode). Otherwise forward them directly to the TE.

2 Buffer unsolicited result codes in the TA when TA-TE link is reserved (e.g. in on-line data mode) and flush them to the TE after reservation. Otherwise forward them directly to the TE.

<mt>

0 No SMS-DELIVER indications are routed to the TE.

1 If SMS-DELIVER is stored into ME/TA, indication of the memory location is routed to the TE using unsolicited result code:

+CMTI: <mem>,<index>

2 SMS-DELIVERs (except class 2 messages and messages in the message waiting indication group (store message)) are routed directly to the TE using unsolicited result code:

+CMT: [<alpha>],<length><CR><LF><pdu> (PDU mode enabled); or +CMT: <oa>,[<alpha>],<scts>*[,<tooa>,<fo>,<pid>,<dcs>,<sca>,<tosca>, <length>]*<CR><LF><data> (text mode enabled; about parameters in italics, refer command Show Text Mode Parameters +CSDH)

If ME has its own display device then class 0 messages and messages in the message waiting indication group (discard message) may be copied to both ME display and to TE. In this case, ME shall send the acknowledgement to the network.

Class 2 messages and messages in the message waiting indication group (store message) result in indication as defined in <mt>=1

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	3 Class 3 SMS-DELIVERs are routed directly to TE using unsolicited result codes defined in <mt>=2. Messages of other data coding schemes result in indication as defined in <mt>=1. <**bm**> 0 No CBM indications are routed to the TE. 1 If CBM is stored into ME/TA, indication of the memory location is routed to the TE using unsolicited result code: +CBMI: <mem>,<index> 2 New CBMs are routed directly to the TE using unsolicited result code: +CBM: <length><CR><LF><pdu>(PDU mode enabled); or +CBM: <sn>,<mid>,<dcs>,<page>,<pages><CR><LF><data>(text mode enabled) **<ds>:** 0 No SMS-STATUS-REPORTs are routed to the TE. 1 SMS-STATUS-REPORTs are routed to the TE using unsolicited result code: +CDS: <length><CR><LF><pdu> (PDU mode enabled); or +CDS: <fo>,<mr>,[<ra>],[<tora>],<scts>,<dt>,<st> (text mode enabled) 2 If SMS-STATUS-REPORT is stored into ME/TA, indication of the memory location is routed to the TE using unsolicited result code: +CDSI: <mem>,<index> **<bfr>:** 0 TA buffer of unsolicited result codes defined within this command is flushed to the TE when **<mode>** 1...3 is entered (**OK** response shall be given before flushing the codes). 1 TA buffer of unsolicited result codes defined within this command

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	is cleared when <mode> 1...3 is entered. Example: AT+CNMI=2,2,0,0,0 OK
Read command AT+CNMI?	Response **+CNMI: <mode>,<mt>,<bm>,<ds>,<bfr>** **OK** Parameter See set command
Test command AT+CNMI=?	Response **+CNMI**: (list of supported **<mode>s**), (list of supported **<mt>s**), (list of supported **<bm>s**), (list of supported **<ds>s**), (list of supported**<bfr>s**) **OK** Parameter See set command
Reference	3GPP TS 27.005, section 3.4.1
Note	

6.10 AT+CMGS Send SMS message

Execute command if PDU mode (+CMGF=0) AT+CMGS=<length><CR>**PDU is given**<ctrl-Z/ESC>	TA transmits SMS message from TE to network (SMS-SUBMIT). Message reference value **<mr>** is returned to TE on successful message delivery. Value can be used to identify message upon unsolicited delivery status report result code. Response If PDU mode (+CMGF=0) and sending successful: **+CMGS: <mr>[,<ackpdu>]** **OK** If error is related to ME functionality: **+CMS ERROR: <err>** Parameter

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	<length>: integer type value indicating in PDU mode (+CMGF=0), the length of the actual TP data unit in octets (i.e. the RP layer SMSC address octets are not counted in the length) **<mr>:** 3GPP 23.040 TP-Message-Reference in integer format **<ackpdu>:** 3GPP 23.040 RP-User-Data element of RP-ACK PDU; format is same as for <pdu> in case of SMS, but without 3GPP 24.011 SC address field and parameter shall be enclosed in double quote characters like a normal string type parameter.
Test command AT+CMGS=?	Response **OK**
Reference	3GPP TS 27.005, section 4.3 - **<length>** must indicate the number of octets coded in the TP layer data unit to be given (i.e. SMSC address octets are excluded). - the TA shall send a four characters sequence **<CR><LF><greater_than><space>** (IRA 13, 10, 62, 32) after command line is terminated with <CR>; after that PDU can be given from TE to ME/TA. - the DCD signal shall be in ON state while PDU is given. - the echoing of given characters back from the TA is controlled by V.250 echo command E. - the PDU shall be hexadecimal format (similarly as specified for **<pdu>**) and given in one line; ME/TA converts this coding into the actual octets of PDU. - when the length octet of the SMSC address (given in the PDU) equals zero, the SMSC address set with command Service Centre Address +CSCA is used; in this case the SMSC Type-of-Address octet shall not be present in the PDU, i.e. TPDU starts right after SMSC length octet. - sending can be cancelled by giving <ESC> character (IRA 27). <ctrl-Z> (IRA 26) must be used to indicate the ending of PDU.
Note	

If text mode

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Execution command if text mode (+CMGF=1): AT+CMGS=<da>[,<toda>]<CR>**text is entered**<ctrl-Z/ESC>	Execution command sends message from a TE to the network (SMS-SUBMIT). Message reference value <mr> is returned to the TE on successful message delivery. Value can be used to identify message upon unsolicited delivery status report result code. If sending fails in a network or an ME error, final result code +CMS ERROR: <err> is returned. See chapter Message Service Failure Result Code for a list of <err> values. This command should be abortable. Response if text mode (+CMGF=1) and sending successful: **+CMGS:<mr>[,<scts>]** **OK** If error is related to ME functionality: +CMS ERROR: <err> Parameter **<mr>** 3GPP 23.040 TP-Message-Reference in integer format **<da>** 3GPP TS 23.040 [3] TP-Destination-Address Address- Value field in string format; BCD numbers (or GSM 7 bits default alphabet characters) are converted to characters of the currently selected TE character set (refer command +CSCS in 3GPP TS 27.007 [9]); type of address given by <toda> **<toda>** 3GPP TS 24.011 [6] TP-Destination-Address Type-of-Address octet in integer format (when first character of <da> is + (IRA 43) default is 145, otherwise default is 129) **<scts>** 3GPP TS 23.040 [3] TP-Service-Centre-Time-Stamp in time-string format
Test command AT+CMGS=?	Response **OK**
Reference	entered text (3GPP TS 23.040 [3] TP-Data-Unit) is sent to address <da> and all current settings (refer Set Text Mode Parameters +CSMP and Service Centre Address +CSCA) are used to construct the actual PDU in ME/TA. -the TA shall send a four characters sequence

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<CR><LF><greater_than><space> (IRA 13, 10, 62, 32) after command line is terminated with <CR>; after that text can be entered from TE to ME/TA.

- the DCD signal shall be in ON state while text is entered.
- the echoing of entered characters back from the TA is controlled by V.250 echo command E.
- the entered text should be formatted as follows:
  - if <dcs> (set with +CSMP) indicates that 3GPP TS 23.038 [2] GSM 7 bits default alphabet is used and <fo> indicates that 3GPP TS 23.040 [3] TP-User-Data-Header-Indication is not set:
    - if TE character set other than "HEX" (refer command Select TE Character Set +CSCS in 3GPP TS 27.007 [9]): ME/TA converts the entered text into the GSM 7 bits default alphabet according to rules of Annex A; backspace can be used to delete last character and carriage returns can be used (previously mentioned four characters sequence shall be sent to the TE after every carriage return entered by the user);
    - if TE character set is "HEX": the entered text should consist of two IRA character long hexadecimal numbers which ME/TA converts into the GSM 7 bits default alphabet characters. (e.g. 17 (IRA 49 and 55) will be converted to character Π (GSM 7 bit default alphabet 23)).
  - if <dcs> indicates that 8-bit or UCS2 data coding scheme is used or <fo> indicates that 3GPP TS 23.040 [3] TP-User- Data-Header-Indication is set: the entered text should consist of two IRA character long hexadecimal numbers which ME/TA converts into 8-bit octet (e.g. two characters 2A (IRA 50 and 65) will be converted to an octet with integer value 42).

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	- sending can be cancelled by giving <ESC> character (IRA 27). - <ctrl-Z> (IRA 26) must be used to indicate the ending of the message body
Note	

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6.11 AT+CMGC Send command

Execution command **If PDU mode (+CMGF=0)** AT+CMGC=<length><CR>**PDU is given**<ctrl-Z/ESC>	TA transmits SMS message from TE to network (SMS-SUBMIT). Message reference value <mr> is returned to TE on successful message delivery. Value can be used to identify message upon unsolicited delivery status report result code. Response If PDU mode (+CMGF=0) and sending successful: **+CMGC: <mr>[,<ackpdu>]** **OK** If error is related to ME functionality: **+CMS ERROR: <err>** Parameter **<length>** integer type value indicating in PDU mode (+CMGF=0), the length of the actual TP data unit in octets (i.e. the RP layer SMSC address octets are not counted in the length). **<mr>** 3GPP 23.040 TP-Message-Reference in integer format **<ackpdu>** 3GPP 23.040 RP-User-Data element of RP-ACK PDU; format is same as for <pdu> in case of SMS, but without 3GPP 24.011 SC address field and parameter shall be enclosed in double quote characters like a normal string type parameter
Test command AT+CMGC=?	Response **OK**
Reference	3GPP TS 27.005, section 4.5 - <length> must indicate the number of octets coded in the TP layer data unit to be given (i.e. SMSC address octets are excluded). - the TA shall send a four characters sequence

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	<CR><LF><greater_than><space> (IRA 13, 10, 62, 32) after command line is terminated with <CR>; after that PDU can be given from TE to ME/TA. - the DCD signal shall be in ON state while PDU is given. - the echoing of given characters back from the TA is controlled by V.250 echo command E. - the PDU shall be hexadecimal format (similarly as specified for <pdu>) and given in one line; ME/TA converts this coding into the actual octets of PDU. when the length octet of the SMSC address (given in the PDU) equals zero, the SMSC address set with command Service Centre Address +CSCA is used; in this case the SMSC Type-of-Address octet shall not be present in the PDU, i.e. TPDU starts right after SMSC length octet. - sending can be cancelled by giving <ESC> character (IRA 27). - <ctrl-Z> (IRA 26) must be used to indicate the ending of PDU.
Note	

If text mode

If text mode Execution command **If text mode (+CMGF=1):** AT+CMGC=<fo>,<ct>[,<pid>[,<mn>[,<da>[,<toda>]]]]<CR> ***text is entered***<ctrl-Z/ESC>	Execution command sends a command message from a TE to the network (SMS-COMMAND). The entering of text (3GPP TS 23.040[3] TP-Command-Data) is done similarly as specified in command Send Message +CMGS, but the format is fixed to be a sequence of two IRA character long hexadecimal numbers which ME/TA converts into 8-bit octets (refer +CMGS). Message reference value <mr> is returned to the TE on successful message delivery. Optionally (when +CSMS <service> value is 1 and network supports) <scts> is returned. Values can be used to identify message upon unsolicited delivery status report result code. If sending fails in a network or an ME error, final result code +CMS ERROR: <err> is returned. See chapter Message Service Failure Result Code for a list of <err> values. This command should be abortable.

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	Response if text mode (+CMGF=1) and sending successful: **+CMGC: <mr>[,<scts>]** **OK** If error is related to ME functionality: **+CMS ERROR: <err>** Parameter **<fo>** depending on the command or result code: first octet of 3GPP TS 23.040 [3] SMS-DELIVER, SMS-SUBMIT (default 17), SMS-STATUS-REPORT, or SMS-COMMAND (default 2) in integer format **<ct>** 3GPP TS 23.040 [3] TP-Command-Type in integer format (default 0) **<pid>** 3GPP TS 23.040 [3] TP-Protocol-Identifier in integer format (default 0) **<mn>** 3GPP TS 23.040 [3] TP-Message-Number in integer format **<da>** 3GPP TS 23.040 [3] TP-Destination-Address Address-Value field in string format; BCD numbers (or GSM 7 bit default alphabet characters) are converted to characters of the currently selected TE character set (refer command +CSCS in3GPP TS 27.007 [9]); type of address given by <toda> **<toda>** 3GPP TS 24.011 [6] TP-Destination-Address Type-of-Address octet in integer format (when first character of <da> is + (IRA 43) default is 145, otherwise default is 129) **<mr>** 3GPP 23.040 TP-Message-Reference in integer format **<scts>** 3GPP TS 23.040 [3] TP-Service-Centre-Time-Stamp in time-string format
Test command AT+CMGC=?	Response **OK**
Reference	The entering of text (3GPP TS 23.040 [3] TP-Command-Data) is

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	similar as specified in command Send Message +CMGS
Note	

6.12 AT+CMMS More Messages to Send

Set command AT+CMMS=[<n>]	Set command controls the continuity of SMS relay protocol link. When feature is enabled (and supported by network) multiple messages can be sent much faster as link is kept open. Response **OK** If error is related to ME functionality: **+CMS ERROR: <err>** Parameter **<n>**: 0 Disable 1 keep enabled until the time between the response of the latest message send command (+CMGS, +CMSS, etc.) and the next send command exceeds 1-5 seconds (the exact value is up to ME implementation), then ME shall close the link and TA switches <n> automatically back to 0 2 enable (if the time between the response of the latest message send command and the next send command exceeds 1-5 seconds (the exact value is up to ME implementation), ME shall close the link but TA shall not switch automatically back to <n>=0)
Read command AT+CMMS?	Response **+CMMS: <n>** **OK** Parameter See above
Test command AT+CMMS =?	Response +**CMMS**: (list of supported <**n**>s)

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	OK
Example	AT+CMMS=1 OK AT+CMMS? +CMMS: 1 OK AT+CMMS=? +CMMS: (0-2) OK
Reference	3GPP TS 27.005, section 3.5.6

6.13 AT+CNMA New Message Acknowledgement to ME/TA

PDU mode:

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Execute command **If PDU mode** **(+CMGF=0):** AT+CNMA [=<n>[,<length>[<CR> ***PDU is given*** <ctrl- Z/ESC>]]]	Execution command confirms reception of a new message (SMS-DELIVER or SMS-STATUS-REPORT) which is routed directly to the TE (refer command +CNMI). This acknowledgement command shall be used when +CSMS parameter <service> equals 1. In PDU mode, it is possible to send either positive (RP-ACK) or negative (RP-ERROR) acknowledgement to the network. Parameter <n> defines which one will be sent. Optionally (when <length> is greater than zero) an acknowledgement TPDU (SMS-DELIVER-REPORT for RP-ACK or RP-ERROR) may be sent to the network. The entering of PDU is done similarly as specified in command Send Message +CMGS, except that the format of <ackpdu> is used instead of <pdu> (i.e. SMSC address field is not present). PDU shall not be bounded by double quotes. TA shall not send another +CMT or +CDS result code to TE before previous one is acknowledged. If ME does not get acknowledgement within required time (network timeout), ME should respond as specified in 3GPP TS 24.011 [6] to the network. ME/TA shall automatically disable routing to TE by setting both <mt> and <ds> values of +CNMI to zero. If command is executed, but no acknowledgement is expected, or some other ME related error occurs, final result code +CMS ERROR: <err> is returned. See chapter Message Service Failure Result Code for a list of <err> values. Response **OK** If error is related to ME functionality: **+CMS ERROR: <err>** Parameter **<n>**: 0 command operates similarly as defined for the text mode 1 send RP-ACK (or buffered result code received correctly) 2 send RP-ERROR (if PDU is not given, ME/TA shall send SMS-DELIVER-REPORT with 3GPP TS 23.040 [3] TP-FCS value set to 'FF' (unspecified error cause))

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	<length>: integer type value indicating in in PDU mode (+CMGF=0), the length of the actual TP data unit in octets (i.e. the RP layer SMSC address octets are not counted in the length) **<pdu>**: In the case of SMS: TPDU in hexadecimal format: ME/TA converts each octet of TP data unit into two IRA character long hexadecimal number (e.g. octet with integer value 42 is presented to TE as two characters 2A (IRA 50 and 65)). NOTE: In case that a directly routed message must be buffered in ME/TA (possible when +CNMI parameter <mode> equals 0 or 2) or AT interpreter remains too long in a state where result codes cannot be sent to TE (e.g. user is entering a message using +CMGS), acknowledgement (RP-ACK) must be sent to the network without waiting +CNMA command from TE. Later, when buffered result codes are flushed to TE, TE must send +CNMA[=0] acknowledgement for each result code. In this way, ME/TA can determine if message should be placed in non-volatile memory and routing to TE disabled (+CNMA [=0] not received). Refer command +CNMI for more details how to use <mode> parameter reliably.
Test command AT+CNMA=?	Test command returns a list of supported <n> values. If the only value supported is 0, the device does not support sending of TPDU. Response if PDU mode (+CMGF=0): **+CNMA: (list of supported <n>s)** **OK**
Example	
Reference	3GPP TS 27.005, section 4.6
Note	

6.14 AT+CMSS Send Message from Storage

Set command AT+CMSS=<index>[,	The set command sends message with location value <index> from preferred message storage <mem2> to the network. shall be used instead of the one stored with the message. Reference value <mr> is

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<da>[,<toda>]]	returned to the TE on successful message delivery. Response If text mode (+CMGF=1) and sending successful: **+CMSS: <mr> [,<scts>]** If PDU mode (+CMGF=0) and sending successful **+CMSS: <mr> [,<ackpdu>]** If error (sending fails) is related to ME functionality: **+CMS ERROR: <err>** Parameter **<index>** integer type, value in the range of location numbers supported by the associated memory **<da>** 3GPP TS 23.040 [3] TP-Destination-Address Address-Value field in string format; BCD numbers (or GSM 7 bits default alphabet characters) are converted to characters of the currently selected TE character set (refer command +CSCS in3GPP TS 27.007 [9]); type of address given by <toda> **<toda>** 3GPP TS 24.011 [6] TP-Destination-Address Type-of-Address octet in integer format (when first character of <da> is + (IRA 43) default is 145, otherwise default is 129) **<mr>** integer type,3GPP 23.040 TP-Message-Reference **<scts>** string type,3GPP TS 23.040 [3] TP-Service-Centre-Time-Stamp in time-string format. **<ackpdu>** 3GPP 23.040 RP-User-Data element of RP-ACK PDU
Test command AT+CMSS=?	Response **OK**
Example	AT+CMSS=14 +CMSS: 186 OK AT+CMSS=? OK
Reference	3GPP TS 27.005, section 3.5.2 and 4.7
Note	

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6.15 AT+CPMS Preferred message storage

Set command AT+CPMS=<mem1>[,<mem2>[,<mem3>]]	The set command is used to select memory storages <mem1>, <mem2> and <mem3> to be used for reading, writing, etc. Response +CPMS: <uesed1>, <total1>, <used2>, <total2>, <used3>, <total3> **OK** If error is related to ME functionality: **ERROR** Parameter <**mem1**>: string type, memory from which messages are read and deleted. "ME" FLASH message storage "SM" SIM message storage <**mem2**>: string type, memory to which writing and sending operations are made. "ME" FLASH message storage "SM" SIM message storage <**mem3**>: string type, memory to which received SMS is preferred to be stored. "ME" FLASH message storage "SM" SIM message storage <**usedX**>: integer type, the number of messages used in <memX> <**totalX**>: integer type, total number of message locations in <memX>
Read command AT+CPMS?	Response **+CPMS:**

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	<mem1>,<used1>,<total1>,<mem2>,<used2>,<total2>,<mem3>,<used3>,<total3> **OK** Parameter See set command
Test command AT+CPMS =?	Test command returns lists of memory storages supported by the TA. Response +**CPMS**: (list of supported <**mem1**>s), (list of supported <**mem2**>s), (list of supported <**mem3**>s) **OK**
Example	AT+CPMS=? +CPMS: (“SM”,“ME”), (“SM”,“ME”), (“SM”,“ME”) OK AT+CPMS? +CPMS: “ME”,0,180, “ME”,0,180, “ME”,0,180 OK AT+CMPS= “SM”, “SM”, “SM” +CPMS: 3,50,3,50,3,50 OK
Reference	3GPP TS 27.005, section 3.2.2

6.16 AT+CMGR Read message

Execute command AT+CMGR=<index>	Execution command is used to return message with location value <index> from message storage <mem1> to the TE. Response 1) If text mode (AT+CMGF=1), command successful and SMS-DELIVER: +CMGR: <stat>,<oa>,[<alpha>],<scts>[,<tooa>,<fo>,<pid>,<dcs>,<sca>,<tosca>,<length>] <CR><LF><data> **OK** 2) If text mode (AT+CMGF=1), command successful and SMS-SUBMIT: +CMGR:

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<stat>,<da>,[<alpha>][,<toda>,<fo>,<pid>,<dcs>,[<vp>],<sca>,<tosca>,<length>]<CR><LF><data>

OK

3) If text mode (AT+CMGF=1), command successful and SMS-STATUS-REPORT:

+CMGR: <stat>,<fo>,<mr>,[<ra>],[<tora>],<scts>,<dt>,<st>

OK

4) If text mode (AT+CMGF=1), command successful and SMS-COMMAND:

+CMGR:

<stat>,<fo>,<ct>[,<pid>,[<mn>],[<da>],[<toda>],<length>]<CR><LF><data>

OK

5) If text mode (AT+CMGF=1), command successful and CBM storage:

+CMGR:

<stat>,<sn>,<mid>,<dcs>,<page>,<pages><CR><LF><data>

OK

6) If PDU mode (AT+CMGF=0) and Command successful:

+CMGR: <stat>,[<alpha>],<length><CR><LF><pdu>

OK

7) If error is related to ME functionality:

+CMS ERROR: <err>

Parameter

<**index**>: Integer type; value in the range of location numbers supported by the associated memory and start with one.

<**stat**>

1. Text mode:

"REC UNREAD"	received unread message
"REC READ"	received read message
"STO UNSENT"	stored unsent message
"STO SENT"	stored sent message

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2. PDU mode:

0	received unread message
1	received read message
2	stored unsent message
3	stored sent message

<**oa**>: Originating-Address, Address-Value field in string format; BCD numbers (or GSM 7 bits default alphabet characters) are converted to characters of the currently selected TE character set, type of address given by <tooa>.

<**alpha**>: string type alphanumeric representation of <da> or <oa> corresponding to the entry found in MT phonebook; implementation of this feature is manufacturer specific; used character set should be the one selected with command Select TE Character Set AT+CSCS.

<**scts**>: TP-Service-Centre-Time-Stamp in time-string format (refer <dt>).

<**tooa**>: TP-Originating-Address, Type-of-Address octet in integer format.

<**fo**>: Depending on the command or result code: first octet of GSM 03.40 SMS-DELIVER, SMS-SUBMIT (default 17), SMS-STATUS-REPORT, or SMS-COMMAND (default2) in integer format. SMS status report is supported under text mode if <fo> is set to 49.

<**pid**>: Protocol Identifier
GSM 03.40 TP-Protocol-Identifier in integer format
0…255

<**dcs**>: Depending on the command or result code: SMS Data Coding Scheme (default 0), or Cell Broadcast Data Coding Scheme in integer format.

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	<**sca**>: RP SC address Address-Value field in string format; BCD numbers (or GSM 7 bits default alphabet characters) are converted to characters of the currently selected TE character set, type of address given by <tosca>. <**tosca**>: RP SC address Type-of-Address octet in integer format. <**length**>: Integer type value indicating in the text mode (AT+CMGF=1) the length of message body <data> in characters; or in PDU mode (AT+CMGF=0), the length of the actual TP data unit in octets. <**data**>: In the case of SMS: TP-User-Data in text mode responses; <**da**>: TP-Destination-Address Address- Value field in string format; BCD numbers (or GSM 7 bits default alphabet characters) are converted to characters of the currently selected TE character set; type of address given by <toda>. <**toda**>: TP-Destination-Address Type-of- Address octet in integer format (when first character of <da> is + (IRA 43) default is 145, otherwise default is 129). The range of value is from 128 to 255. <**vp**>: depending on SMS-SUBMIT <fo> setting: TP-Validity-Period either in integer format (default 167), in time-string format (refer <dt>), or if EVPF is supported, in enhanced format (hexadecimal coded string with double quotes) <**mr**>: Message reference GSM 03.40 TP-Message-Reference in integer format <**ra**>: Recipient Address GSM 03.40 TP-Recipient-Address Address-Value field in string format; BCD numbers (or GSM default alphabet characters) are converted to characters of the currently selected TE character set; type of address given by <tora>

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	<**tora**>: Type of Recipient Address GSM 04.11 TP-Recipient-Address Type-of-address octet in integer format (default refer <toda>) <**dt**>: Discharge Time GSM 03.40 TP-Discharge-Time in time string format: "yy/MM/dd,hh:mm:ss+zz", where characters indicate year (two last digits), month, day, hour, minutes, seconds, and time zone. <**st**>: Status GSM 03.40 TP-Status in integer format 0…255 <**ct**>: Command Type GSM 03.40 TP-Command-Type in integer format 0…255 <**mn**>: Message Number GSM 03.40 TP-Message-Number in integer format <**sn**>: Serial Number GSM 03.41 CBM Serial Number in integer format <**mid**>: Message Identifier GSM 03.41 CBM Message Identifier in integer format <**page**>: Page Parameter GSM 03.41 CBM Page Parameter bits 4-7 in integer format <**pages**>: Page Parameter GSM 03.41 CBM Page Parameter bits 0-3 in integer format <**pdu**>: In the case of SMS: TPDU in hexadecimal format: ME/TA converts each octet of TP data unit into two IRA character long hexadecimal number (e.g. octet with integer value 42 is presented to TE as two characters 2A (IRA 50 and 65)).
Test command	Response

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AT+CMGR =?	**OK**
Example	AT+CMGR =? OK AT+CMGR=1 +CMGR: “STO UNSENT”, “+10011”,,145,17,0,0,167, “+8613800100500”,145,11 Hello World OK
Reference	3GPP TS 27.005, section 3.4.3

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6.17 AT+CMGW Write message to memory

Execution command If text mode (AT+CMGF=1) AT+CMGW[=<oa>/<da>[,<tooa>/<toda>[,<stat>]]]<CR>**text is entered**<ctrl-Z/ESC> If PDU mode (AT+CMGF=0) AT+CMGW=<length>[,<stat>]<CR> **PDU is given**<ctrl-Z/ESC>	Execution command is used to store message (either SMS-DELIVER or SMS-SUBMIT) to memory storage <mem2>. Response 1) If write successfully: +CMGW: <index> **OK** 2) If write fails: **ERROR** 3) If write fails: **+CMS ERROR: <err>** Parameter <**index**>: Integer type; value in the range of location numbers supported by the associated memory and start with one. <**stat**> 1. Text mode: "STO UNSENT" stored unsent message "STO SENT" stored sent message 2. PDU mode: 0 stored unsent message 1 stored sent message <**oa**>: Originating-Address, Address-Value field in string format; BCD numbers (or GSM 7 bits default alphabet characters) are converted to characters of the currently selected TE character set, type of address given by <tooa>. <**tooa**>: TP-Originating-Address, Type-of-Address octet in integer format. <**length**>: Integer type value indicating in the text mode (AT+CMGF=1) the length of message body <data> in characters; or in

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	PDU mode (AT+CMGF=0), the length of the actual TP data unit in octets. <**da**>: TP-Destination-Address Address- Value field in string format; BCD numbers (or GSM 7 bits default alphabet characters) are converted to characters of the currently selected TE character set; type of address given by <toda>. <**toda**>: TP-Destination-Address Type-of- Address octet in integer format (when first character of <da> is + (IRA 43) default is 145, otherwise default is 129). The range of value is from 128 to 255.
Test command AT+CMGW =?	Response **OK**
Example	AT+CMGW =? OK AT+CMGW= “13012832788”>ABCD<ctrl-Z/ESC> +CMGW: 1 OK
Reference	3GPP TS 27.005, section 3.5.3

6.18 AT+CMGL List SMS messages from preferred store

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Execute command AT+CMGL=<stat>	Execution command is used to return messages with status value <stat> from message storage <mem1> to the TE. Response 1) If text mode (AT+CMGF=1), command successful and SMS-SUBMITs or SMS-DELIVERs: +CMGL: <index>,<stat>,<oa>/<da>,[<alpha>],[<scts>][,<tooa>/<toda>,<fo>,<pid>,<dcs>,<sca>,<tosca>,<length>]<CR><LF><data> [+CMGL: <index>,<stat>,<oa>/<da>,[<alpha>],[<scts>][,<tooa>/<toda>,<fo>,<pid>,<dcs>,<sca>,<tosca>,<length>]<CR><LF><data> […]] **OK** 2) If text mode (AT+CMGF=1), command successful and SMS-STATUS-REPORTs: +CMGL: <index>,<stat>,<fo>,<mr>,[<ra>],[<tora>],<scts>,<dt>,<st> [+CMGL: <index>,<stat>,<fo>,<mr>,[<ra>],[<tora>],<scts>,<dt>,<st> […]]] **OK** 3) If text mode (AT+CMGF=1), command successful and SMS-COMMANDs: +CMGL: <index>,<stat>,<fo>,<ct> [+CMGL: <index>,<stat>,<fo>,<ct> […]] **OK** 4) If text mode (AT+CMGF=1), command successful and CBM storage: +CMGL: <index>,<stat>,<sn>,<mid>,<page>,<pages>,<data> [+CMGL: <index>,<stat>,<sn>,<mid>,<page>,<pages>,<data> […]] **OK** 5) If PDU mode (AT+CMGF=0) and Command successful: +CMGL: <index><stat>,[<alpha>],<length><pdu> [+CMGL: <index><stat>,[<alpha>],<length><pdu> […]] **OK**

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6) If error is related to ME functionality:

+CMS ERROR: <err>

Parameter

<**index**>: Integer type; value in the range of location numbers supported by the associated memory and start with one.

<**stat**>

1. Text mode:

"REC UNREAD"	received unread message
"REC READ"	received read message
"STO UNSENT"	stored unsent message
"STO SENT"	stored sent message
"ALL"	all messages

2. PDU mode:

0	received unread message
1	received read message
2	stored unsent message
3	stored sent message
4	all messages

<**oa**>: Originating-Address, Address-Value field in string format; BCD numbers (or GSM 7 bits default alphabet characters) are converted to characters of the currently selected TE character set, type of address given by <tooa>.

<**alpha**>: string type alphanumeric representation of <da> or <oa> corresponding to the entry found in MT phonebook; implementation of this feature is manufacturer specific; used character set should be the one selected with command Select TE Character Set AT+CSCS.

<**scts**>: TP-Service-Centre-Time-Stamp in time-string format (refer <dt>).

<**tooa**>: TP-Originating-Address, Type-of-Address octet in integer format.

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<fo>: Depending on the command or result code: first octet of GSM 03.40 SMS-DELIVER, SMS-SUBMIT (default 17), SMS-STATUS-REPORT, or SMS-COMMAND (default2) in integer format. SMS status report is supported under text mode if <fo> is set to 49.

<pid>: Protocol Identifier
GSM 03.40 TP-Protocol-Identifier in integer format
0…255

<dcs>: Depending on the command or result code: SMS Data Coding Scheme (default 0), or Cell Broadcast Data Coding Scheme in integer format.

<sca>: RP SC address Address-Value field in string format; BCD numbers (or GSM 7 bits default alphabet characters) are converted to characters of the currently selected TE character set, type of address given by <tosca>.

<tosca>: RP SC address Type-of-Address octet in integer format.

<length>: Integer type value indicating in the text mode (AT+CMGF=1) the length of message body <data> in characters; or in PDU mode (AT+CMGF=0), the length of the actual TP data unit in octets.

<data>: In the case of SMS: TP-User-Data in text mode responses;

<da>: TP-Destination-Address Address- Value field in string format; BCD numbers (or GSM 7 bits default alphabet characters) are converted to characters of the currently selected TE character set; type of address given by <toda>.

<toda>: TP-Destination-Address Type-of- Address octet in integer format (when first character of
<da> is + (IRA 43) default is 145, otherwise default is 129). The range of value is from 128 to 255.

<mr>: Message reference

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	GSM 03.40 TP-Message-Reference in integer format <**ra**>: Recipient Address GSM 03.40 TP-Recipient-Address Address-Value field in string format; BCD numbers (or GSM default alphabet characters) are converted to characters of the currently selected TE character set; type of address given by <tora> <**tora**>: Type of Recipient Address GSM 04.11 TP-Recipient-Address Type-of-address octet in integer format (default refer <toda>) <**dt**>: Discharge Time GSM 03.40 TP-Discharge-Time in time string format: "yy/MM/dd,hh:mm:ss+zz", where characters indicate year (two last digits), month, day, hour, minutes, seconds, and time zone. <**st**>: Status GSM 03.40 TP-Status in integer format 0…255 <**ct**>: Command Type GSM 03.40 TP-Command-Type in integer format 0…255 <**sn**>: Serial Number GSM 03.41 CBM Serial Number in integer format <**mid**>: Message Identifier GSM 03.41 CBM Message Identifier in integer format <**page**>: Page Parameter GSM 03.41 CBM Page Parameter bits 4-7 in integer format <**pages**>: Page Parameter GSM 03.41 CBM Page Parameter bits 0-3 in integer format

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	<**pdu**>: In the case of SMS: TPDU in hexadecimal format: ME/TA converts each octet of TP data unit into two IRA character long hexadecimal number (e.g. octet with integer value 42 is presented to TE as two characters 2A (IRA 50 and 65)).
Test command AT+CMGL =?	Response **+CMGL:** (list of supported <stat>s) **OK**
Example	AT+CMGL=? +CMGL: ("REC UNREAD","REC READ","STO UNSENT","STO SENT","ALL") OK AT+CMGL="ALL" +CMGL: 0,“STO UNSENT”, “+10011”,,167 Hello World OK
Reference	3GPP TS 27.005, section 3.4.2

6.19 AT+CMGD Delete message

Execute command AT+CMGD=<index>[,<delflag>]	Execution command is used to delete message from preferred message storage <mem1> location <index>. If <delflag> is present and not set to 0 then the ME shall ignore <index> and follow the rules for <delflag> shown below. Response 1) OK 2) ERROR 3) +CMS ERROR: <err> Parameter <**index**>: Integer type; value in the range of location numbers supported by

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	the associated memory and start with one. <**delflag**> 0 Delete the message specified in <index> 1 Delete all read messages from preferred message storage, leaving unread messages and stored mobile originated messages (whether sent or not) untouched. 2 Delete all read messages from preferred message storage and sent mobile originated messages, leaving unread messages and unsent mobile originated messages untouched. 3 Delete all read messages from preferred message storage, sent and unsent mobile originated messages leaving unread messages untouched. 4 Delete all messages from preferred message storage including unread messages.
Test command AT+CMGD =?	Response **+CMGD:** (list of supported <index>s)[, (list of supported <delflag>s)] **OK**
Example	AT+CMGD=? +CMGD: (1), (0-4) OK AT+CMGD=1 OK
Reference	3GPP TS 27.005, section 3.5.4

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7 Packet Domain Commands

7.1 AT+CGDCONT Define PDP Context

Set command AT+CGDCONT= <cid>[,<PDP_type>[, <APN>[,<PDP_addr>, [<d_comp>[,<h_comp> [, <IPv4AddrAlloc>[,< <requset_type>[,< P- CSCF_discovery>[,<I M_CN_ Signalling_ Flag_Ind>[,<NSLPI>]]]]]]]]]	This command specifies PDP context parameter values for a PDP context identified by the (local) context identification parameter, **<cid>**. A special form of the set command, **+CGDCONT= <cid>** causes the values for context number **<cid>** to become undefined. Response **OK/ERROR/+CME ERROR** Parameter **<cid>** (PDP Context Identifier) a numeric parameter which specifies a particular PDP context definition. The parameter is local to the TE-MT interface and is used in other PDP context-related commands. The range of permitted values (minimum value = 1) is returned by the test form of the command. If no cid is given, nothing will be changed (neither context definition nor undefinition). 1-5 Valid range of cid **<PDP_type>** (Packet Data Protocol type) a string parameter which specifies the type of packet data protocol IP Internet Protocol version 4(default) PPP Point to Point Protocol IPV6Internet Protocol, version 6 (IETFRFC2460) **<APN>** Access Point Name is a string parameter (framed by quotation marks) which is a logical name that is used to select the GGSN or the external packet data network. If the value is null or omitted, then the subscription value will be requested. **<PDP_addr>** a string parameter that identifies the MT in the address space applicable to the PDP. If the value is null or omitted, then a value may be provided by the TE during the PDP startup procedure or, failing that, a dynamic address will be requested. The read form of the command will continue to return the null string even if an address has been allocated during the PDP startup procedure. The allocated address may be read using the **+CGPADDR** command. This parameter supports both ipv4 and ipv6 address.

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<d_comp>a numeric parameter that controls PDP data compression (0 is only supported mode in this version, any other value is omitted and take no effect)

0 -off (default if value is omitted)

1 -on (manufacturer preferred compression)

2 -V.42bis

3 -V.44

<h_comp>a numeric parameter that controls PDP header Compression (0 is only supported mode in this version, any other value is omitted and take no effect)

0 -off (default if value is omitted)

1 -on (manufacturer preferred compression)

2 -RFC1144 (applicable for SNDCP only)

3 -RFC2507

4 -RFC3095 (applicable for PDCP only)

<**IPv4AddrAlloc**>: integer type; controls how the MT/TA requests to get the IPv4 address information.

0 IPv4 address allocation through NAS signalling.

<**request_type**>: integer type; indicates the type of PDP context activation request for the PDP context.

0 PDP context is for new PDP context establishment or for handover from a non-3GPP access network.

1 PDP context is for emergency (bearer) service.

<**P-CSCF_discovery**>: integer type; influences how the MT/TA requests to get the P-CSCF address.

0 Preference of P-CSCF address discovery not influenced by +CGDCONT

1 Preference of P-CSCF address discovery through NAS signalling

<**IM_CN_Signalling_Flag_Ind**>: integer type; indicates to the network whether the PDP context is for IM CN subsystem-related signalling only or not.

0 UE indicates that the PDP context is not for IM CN subsystem-related signalling only.

1 UE indicates that the PDP context is for IM CN subsystem-related signalling only.

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	<**NSLPI**>: integer type; indicates the NAS signalling priority requested for this PDP context: 0 Indicates that this PDP context is to be activated with the value for the low priority indicator configured in the MT. 1 Indicates that this PDP context is to be activated with the value for the low priority indicator configured in the MT. **Remark:** 1. If **<cid>**has been defined by **CGDSCONT** or **<cid>**has been activated, response **ERROR/+CME ERROR** 2. If undefine a **<cid>** which is used as <**p_cid**> for a exist Secondary PDP cid, response ERROR/+CME ERROR
Read command AT+CGDCONT?	The read command returns the current settings for each defined context. If there is no context define simply OK will be returned. Response **+CGDCONT:** **<cid>,<PDP_type>,<APN>,<PDP_addr>,<d_comp>,<h_comp>,**<**IPv4AddrAlloc**>,<**request_type**>,<**P-CSCF_discovery**>,<**IM_CN_Signalling_Flag_Ind**>, <**NSLPI**> **[<CR><LF>+CGDCONT: ...]** **OK** Parameter see set command
Test command AT+CGDCONT=?	The test command returns values supported as a compound value. If the MT supports several PDP types, **<PDP_type>**, the parameter value range for each **<PDP_type>** are returned on a separate line. Response **+CGDCONT:** (range of supported **<cid>**s), **<PDP_type>,,,**(list of supported **<d_comp>**s), (list of supported **<h_comp>s),**(list of supported **<IPv4AddrAlloc>s),**(list of supported **<request_type>s),**(list of supported **<P-CSCF_discovery>s),**(list of supported <**IM_CN_Signalling_Flag_Ind>s),**(list of supported <**NSLPI**>**s),**

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	[<CR><LF>+CGDCONT: ...] **OK** Parameter see set command
Reference	Refer 27.007 section 10.1.1
Note	

7.2 AT+CGDSCONT Define Secondary PDP Context

Set command AT+CGDSCONT=<cid>, <p_cid>[,<d_comp>[, <h_comp>[,<IM_CN_ Signalling_Flag_Ind>]]]	The set command specifies PDP context parameter values for a Secondary PDP context identified by the (local) context identification parameter, **<cid>**. The number of PDP contexts that may be in a defined state at the same time is given by the range returned by the test command. A special form of the set command, **+CGDSCONT= <cid>**causes the values for context number <cid> to become undefined. Response **OK/ERROR/+CME ERROR** Parameter **<cid>** (PDP Context Identifier) a numeric parameter which specifies a particular PDP context definition. The parameter is local to the TE-MT interface and is used in other PDP context-related commands. The range of permitted values (minimum value = 1) is returned by the test form of the command. If no cid is given, nothing will be changed (neither context definition nor undefinition). 1-5 Valid range of cid **<p_cid>** (Primary PDP Context Identifier) a numeric parameter which specifies a particular PDP context definition which has been defined by use of the +CGDCONT command. The parameter is local to the TE-MT interface. The list of permitted values is returned by the test form of the command. 1-5 Valid range of p_cid **<d_comp>**a numeric parameter that controls PDP data compression (0 is only supported mode in this version, any other value is omitted and take no effect) 0 -off (default if value is omitted)

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	1 -on (manufacturer preferred compression) 2 -V.42bis 3 -V.44 **<h_comp>**a numeric parameter that controls PDP header compression (0 is only supported mode in this version, any other value is omitted and take no effect) 0 -off (default if value is omitted) 1 -on (manufacturer preferred compression) 2 -RFC1144 (applicable for SNDCP only) 3 -RFC2507 4 -RFC3095 (applicable for PDCP only) <**IM_CN_Signalling_Flag_Ind**>: integer type; indicates to the network whether the PDP context is for IM CN subsystem-related signalling only or not. 0 UE indicates that the PDP context is not for IM CN subsystem-related signalling only. 1 UE indicates that the PDP context is for IM CN subsystem-related signalling only. Remark: 1. If **<cid>**is defined by **CGDCONT** or **<cid>**has been activated, response **ERROR/+CME ERROR** 2. If **<p_cid>** is not defined by **CGDCONT**, response **ERROR/+CME ERROR** 3. **<cid>** must not be equal to **<p_cid>**
Read command AT+CGDSCONT?	The read command returns the current settings for each defined context. If there is no context define simply OK will be returned. Response **+CGDSCONT:** **<cid>,<p_cid>,<d_comp>,<h_comp>,<IM_CN_Signalling_Flag_Ind>** **[<CR><LF>+CGDSCONT: …]** **OK** Parameter see set command

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Test command AT+CGDSCONT=?	Response +**CGDSCONT**: (range of supported **<cid>**s), (list of **<p_cid>**s for active primary contexts),(list of supported **<d_comp>**s),(list of supported **<h_comp>**s),(list of supported **<IM_CN_Signalling_Flag_Ind** >s) **OK** Parameter see set command
Reference	Refer 27.007 section 10.1.2
Note	

7.3 AT+CGTFT Traffic Flow Template

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Set command AT+CGTFT=<cid>,[<packet filter identifier>,<evaluation precedence index> [,<source address and subnet mask>[,<protocol number (ipv4) / next header (ipv6)> [,<destination port range> [,<source port range> [,<ipsec security parameter index (spi)> [,<type of service (tos) (ipv4) and mask / traffic class (ipv6) and mask> [,<flow label(ipv6)>[,<direction>]]]]]]]]	The set command specifies a Packet Filters that is to be added to the TFT stored in the MT and used for the context identified by the (local) context identification parameter, <cid>. The specified TFT will be stored in the GGSN only at activation or MS-initiated modification of the related context. Since this is the same parameter that is used in the **+CGDCONT** and **+CGDSCONT** commands, the **+CGTFT** command is effectively an extension to these commands. The Packet Filters consist of a number of parameters, each of which may be set to a separate value. A special form of the set command, **+CGTFT**= <**cid**> causes all of the Packet Filters in the TFT for context number <**cid**> to become undefined. At any time there may exist only one PDP context with no associated TFT amongst all PDP contexts associated to one PDP address. At an attempt to delete a TFT, which would violate this rule, an **ERROR** or **+CME ERROR** response is returned (This version, this rule is processed by network). Extended error responses are enabled by the **+CMEE** command. Response **OK/ERROR/+CME ERROR** Parameter <**cid**>: a numeric parameter which specifies a particular PDP context definition (see the +CGDCONT and +CGDSCONT commands). The following parameters are defined in 3GPP TS 23.060[47] - <**packet filter identifier**>: Numeric parameter, value range from 1 to 8(TD dual

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mode

version) or 1 to 16(LTE version).

<**source address and subnet mask**>:

Consists of dot-separated numeric (0-255) parameters on the form 'a1.a2.a3.a4.m1.m2.m3.m4', for IPv4 and 'a1.a2.a3.a4.a5.a6.a7.a8.a9.a10.a11.a12.a13.a14.a15.a16.m1.m2.m3.m4.m5.m6.m7.m8.m9.m10.m11.m12.m13.m14.m15.m16', for IPv6.

<**protocol number (ipv4) / next header (ipv6)**>:

Numeric parameter, value range from 0 to 255.

<**destination port range**>:

Consists of dot-separated numeric (0-65535) parameters on the form 'f.t'.

<**source port range**>:

Consists of dot-separated numeric (0-65535) parameters on the form 'f.t'.

<**ipsec security parameter index (spi)**>:

Hexadecimal parameter, value range from 0x00000000 to 0xFFFFFFFF.

<**type of service (tos) (ipv4) and mask / traffic class (ipv6) and mask**>:

Dot-separated numeric (0-255) parameters on the form 't.m'.

<**flow label (ipv6)**>:

Hexadecimal parameter, value range from 0x00000 to 0xFFFFF.

Valid for IPv6 only.

<**evaluation precedence index**>:

Numeric parameter, value range from 0 to 255.

<direction>:

a numeric parameter which specifies the transmission direction in which the packet filter shall be applied.

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	0 - Pre-Release 7 TFT filter (see 3GPP TS 24.008 [8], table 10.5.162) 1 - Uplink 2 - Downlink - Directional (Up & Downlink) (default if omitted) NOTE: this field is added in LTE.
Read command AT+CGTFT?	The read command returns the current settings for all Packet Filters for each defined context. Response **+CGTFT:** <cid>,<packet filter identifier>,<evaluation precedence index>,<source address and subnet mask>,<protocol number (ipv4) / next header (ipv6)>,<destination port range>,<source port range>,<ipsec security parameter index (spi)>,<type of service (tos) (ipv4) and mask / traffic class (ipv6) and mask>,<flow label (ipv6)>,<direction> [<CR><LF>+CGTFT: …] OK Parameter <**cid**>: see set command <**packet filter identifier**>: see set command <**source address and subnet mask**>: see set command <**protocol number (ipv4) / next header (ipv6)**>: see set command <**destination port range**>: see set command <**source port range**>: see set command <**ipsec security parameter index (spi)**>: see set command <**type of service (tos) (ipv4) and mask / traffic class (ipv6) and mask**>: see set command

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	<**flow label (ipv6)**>: see set command <**evaluation precedence index**>: see set command **<direction>:** see set command
Test command AT+CGTFT =?	The test command returns values supported as a compound value. If the MT supports several PDP types, the parameter value ranges for each PDP type are returned on a separate line. TFTs shall be used for PDP- type IP and PPP only. For PDP-type PPP a TFT is applicable only when IP traffic is carried over PPP. If PPP carries header-compressed IP packets, then a TFT cannot be used Response **+CGTFT:** <PDP_type>, (list of supported <packet filter identifier>s), (list of supported <evaluation precedence index>s), (list of supported <source address and subnet mask>s), (list of supported <protocol number (ipv4) / next header (ipv6)>s), (list of supported <destination port range>s), (list of supported <source port range>s), (list of supported <ipsec security parameter index (spi)>s), (list of supported <type of service (tos) (ipv4) and mask / traffic class (ipv6) and mask>s), (list of supported <flow label (ipv6)>s) [<CR><LF>+CGTFT: …] OK Parameter **<PDP_type>** see the +CGDCONT <**cid**>: see set command <**packet filter identifier**>: see set command <**source address and subnet mask**>: see set command <**protocol number (ipv4) / next header (ipv6)**>: see set command

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	<destination port range>: see set command **<source port range>**: see set command **<ipsec security parameter index (spi)>**: see set command **<type of service (tos) (ipv4) and mask / traffic class (ipv6) and mask>**: see set command **<flow label (ipv6)>**: see set command **<evaluation precedence index>**: see set command **<direction>:** see set command
Reference	Refer 27.007 section 10.1.3 Refer 23.060
Note	

7.4 AT+CGATT PS attach or detach

Execute command AT+CGATT=<state>	The execution command is used to attach the MT to, or detach the MT from, the Packet Domain service. After the command has completed, the MT remains in V.250 command state. If the MT is already in the requested state, the command is ignored and the **OK** response is returned. Any active PDP contexts will be automatically deactivated when the attachment state changes to detached. Response **OK/ERROR/+CME ERROR** Parameter If no state is specified nothing will be changed. But OK will be sent to TE.

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	<state> indicates the state of PS attachment 0 detached 1 attached
Read command AT+CGATT?	The read command returns the current Packet Domain service state. Response **+CGATT: <state>** **OK** Parameter **<state>** see execute command
Test command AT+CGATT=?	The test command is used for requesting information on the supported Packet Domain service states. Response **+CGATT:** (list of supported **<state>**s) **OK** Parameter **<state>** see set command
Example	AT+CGATT=1 OK AT+CGATT=? +CGATT: (0,1) OK
Reference	Refer 27.007 chapter 10.1.9
Note	

7.5 AT+CGACT PDP context activate or deactivate

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Execute command AT+CGACT=[<state>[,<cid>]]	The execution command is used to activate or deactivate the specified PDP context. After the command has completed, the MT remains in V.250 command state. If any PDP context is already in the requested state, the state for that context remains unchanged. If <state> exists and <cid> is left out, when <state> = 1, MT will find an undefined cid and activate it; when <state> = 0, MT will find a cid in the state of ACTIVE/ ACT_REQ and deactivate it, but if there exists 2 or more cids which are in the state of ACTIVE / ACT_REQ at current time, MT will send "+CME ERROR: 3" to TE. Response **OK/ERROR/+CME ERROR** Parameter If no parameter is specified nothing will be changed. But **OK** will be sent to TE。 **<cid>** a numeric parameter which specifies a particular PDP context definition (see the +CGDCONT and +CGDSCONT commands). **<state>** indicates the state of PDP context activation 0 - deactivated 1 - activated If <state> equals to 0 and the PDP was deactivated successfully, there will be a "+CGDAC" URC report such as **+CGDAC: <cid>**, it means that this deactivate request has been accepted by network.
Read command AT+CGACT?	The read command returns the current activation states for all the defined PDP contexts. Response **+CGACT: <cid>, <state>[<CR><LF>+CGACT: <cid>, <state> [...]]** **OK** Parameter **<state>** see execute command **<cid>** see execute command
Test command AT+CGACT=?	The test command is used for requesting information on the supported PDP context activation states.

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	Response **+CGACT:** (list of supported **<state>**s) **OK** Parameter **<state>** see set command
Example	AT+CGACT=1,1 OK AT+CGACT? +CGACT: 1, 1 OK AT+CGACT=? +CGACT: (0,1) OK
Reference	Refer 27.007 section 10.1.10
Note	

7.6 AT+CGCMOD PDP Context Modify

Execute command AT+CGCMOD[=<cid>[,<cid>[,…]]]	The execution command is used to modify the specified PDP context (s) with respect to QoS profiles and TFTs. After the command has completed, the MT returns to V.250 online data state. If the requested modification for any specified context cannot be achieved, an ERROR or **+CME ERROR** response is returned. Extended error responses are enabled by the **+CMEE** command. If no <**cid**>s are specified the activation form of the command modifies all active contexts. Response **OK/ERROR/+CME ERROR** Parameter

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	<cid> a numeric parameter which specifies a particular PDP context definition (see the +CGDCONT and +CGDSCONT)
Test command AT+CGCMOD =?	The test command returns a list of <cid>s associated with active contexts Response **+CGCMOD: (**list of **<cid>**s associated with active contexts**)** **OK** Parameter **<cid>** see set command
Reference	Refer 27.007 section 10.1.11
Note	

7.7 AT+CGDATA Enter data state

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Execute command AT+CGDATA[=<L2P>, [<cid>]]	The execution command causes the MT to perform whatever actions are necessary to establish communication between the TE and the network. This may include performing a GPRS attach and one PDP context activation. Commands following **+CGDATA** command in the AT command line shall not be processed by the MT. Response **OK/ERROR/+CME ERROR** If successful, the MT issues the intermediate result code CONNECT and enters V.250 online data state: **CONNECT** If **<L2P>** set as M-0000, and the intermediate result code is enabled, the MT issues the intermediate result code **^DPPPI**: <cid>,<pdp_address>,<gateway>,<primary_dns>,<secondary_dns> before **CONNECT** After data transfer is complete, and the layer 2 protocol termination procedure has completed successfully, the command state is re-entered and the MT re-turns the final result code **OK** If the <L2P> parameter value is unacceptable to the MT, the MT returns an ERROR or +CME ERROR response: **ERROR/+CME ERROR** In the event of erroneous termination or a failure to start up, the command state is re-entered and the MT returns **NO CARRIER** or, if enabled, **+CME ERROR** Parameter

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	If no parameter is specified nothing will be changed). But **OK** will be sent to TE. **<cid>** a numeric parameter which specifies a particular PDP context definition (see the +CGDCONT and +CGDSCONT commands) **<L2P>** a string parameter that indicates the layer 2 protocol to be used between the TE and MT PPP Point-to-point protocol for a PDP such (default value) M-0000 Manufacturer-specific protocol intermediate result code is enabled, the MT issues the intermediate result code **^DPPPI**: <cid>,<pdp_address>,<gatewa y>,<primary_dns>,<secondary_ dns> before **CONNECT** <pdp_address>,<gateway>,<primary_dns>,<secondary_dns>: a string type parameter, which identifies the MT in the address space applicable to the PDP (e.g. IP V4 address for PDP type IP) These parameters support both ipv4 and ipv6 address. Remark Only one <**cid**> is supported this version
Test command AT+CGDATA=?	The test command is used for requesting information on the supported layer 2 protocols Response **+CGDATA**: (list of supported **<L2P>**s) **OK** Parameter **<L2P>** see execute command
Reference	Refer 27.007 section 10.1.12
Note	

7.8 AT+CGPADDR Show PDP address

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Execute command AT+CGPADDR[=<cid> [,<cid> [,…]]]	The execution command returns a list of PDP addresses for the specified context identifiers If no <**cid**> is specified, then a list of PDP address for all defined context will be returned. Response **+CGPADDR: <cid>,<PDP_addr> [<CR><LF>+CGPADDR: …]** **OK/ERROR/+CME ERROR** Parameter <**cid**>: a numeric parameter which specifies a particular PDP context definition (see the +CGDCONT and +CGDSCONT commands). If no <cid> is specified, the addresses for all defined contexts are returned. If none of the <cid>s is defined according to the test command, reports error. <**PDP_address**>: a string that identifies the MT in the address space applicable to the PDP. The address may be static or dynamic. For a static address, it will be the one set by the +CGDCONT and +CGDSCONT commands when the context was defined. For a dynamic address it will be the one assigned during the last PDP context activation that used the context definition referred to by <cid>. <PDP_address> is omitted if none is available. This parameter supports both ipv4 and ipv6 address.
Test command AT+CGPADDR=?	The test command returns a list of defined **<cid>**s Response **+CGPADDR:** (list of defined **<cid>**) **OK** Parameter **<cid>** see execute command
Example	AT+CGPADDR=1

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	+CGPADDR: 1,"107.210.5.4" OK AT+CGPADDR=? +CGAPDDR: (1,2,3,4) OK
Reference	Refer 27.007 chapter 10.1.14
Note	

7.9 AT+CGEREP Packet Domain event reporting

Set command AT+CGEREP=[<mode>[,<bfr>]]	Set command enables or disables sending of unsolicited result codes, **+CGEV: XXX** from MT to TE in the case of certain events occurring in the Packet Domain MT or the network. **<mode>** controls the processing of unsolicited result codes specified within this command. **<bfr>** controls the effect on buffered codes when **<mode>** 1 or 2 is entered Response The following unsolicited result codes and the corresponding events are defined – **+CGEV: REJECT <PDP_type>, <PDP_addr>** A network request for PDP context activation occurred when the MT was unable to report it to the TE with a +CRING unsolicited result code and was automatically rejected. NOTE1: This event is not applicable for EPS. **+CGEV: NW REACT<PDP_type>, <PDP_addr>, [<cid>]** The network has requested a context reactivation. The <cid> that was used to reactivate the context is provided if known to the MT. NOTE2: This event is not applicable for EPS. **+CGEV: NW DEACT <PDP_type>, <PDP_addr>, [<cid>]** The network has forced a context deactivation. The <cid> that was used to activate the context is provided if known to the MT. **+CGEV: ME DEACT <PDP_type>, <PDP_addr>, [<cid>]** The mobile termination has forced a context deactivation. The **<cid>**that was

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used to activate the context is provided if known to the MT.

+CGEV: ME ACT <PDP_type>, <PDP_addr>, [<cid>]

The mobile termination has forced a context activation. The <cid> that was used to activate the context is provided if known to the MT.

+CGEV: NW DETACH

The network has forced a PS detach. This implies that all active contexts have been deactivated. These are not reported separately.

+CGEV: ME DETACH

The mobile termination has forced a PS detach. This implies that all active contexts have been deactivated. These are not reported separately.

+CGEV: NW CLASS <class>

The network has forced a change of MT class. The highest available class is reported (see +CGCLASS). The format of the parameter <class> is found in command +CGCLASS.

+CGEV: ME CLASS <class>

The mobile termination has forced a change of MT class. The highest available class is reported (see +CGCLASS). The format of the parameter <class> is found in command +CGCLASS.

+CGEV: NWPDNACT<cid>

The network has activated a context. The context represents a Primary PDP context. The <cid> for this context is provided to the TE. The format of the parameter <cid> is found in command +CGDCONT.

+CGEV: NWDEACT <p_cid>, <cid>, <event_type>

The network has deactivated a context.The <cid> for this context is provided to the TE in addition to the associated primary <p_cid>. The format of the parameters <p_cid> and <cid> are found in command +CGDSCONT. The format of the parameter <event_type> is defined above.

NOTE: Occurrence of this event replaces usage of the event +CGEV: NWDEACT<PDP_type>,<PDP_addr>,[<cid>].

+CGEV: MEDEACT<p_cid>, <cid>, <event_type>

The network has responded to an ME initiated context deactivation request. The associated <cid> is provided to the TE in addition to the associated primary <p_cid>. The format of the parameters <p_cid> and <cid> are found in command +CGDSCONT. The format of the parameter <event_type> is defined above.

NOTE: Occurrence of this event replaces usage of the event

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+CGEV: MEDEACT<PDP_type>,<PDP_addr>,[<cid>].

+CGEV: NW MODIFY <cid>, <change_reason>, <event_type>

The network has modified EPS bearer context parameter(s). The associated <cid> is provided to the TE in addition of the change reason: TFT and/or QoS modification. The format of the parameter <cid> is found in command +CGDCONT. The format of the parameter <event_type> is defined above.

<change_reason> integer type parameter indicates what kind of change occurred.

1: TFT only changed

2: Qos only changed3: Both TFT andQoS changed

+CGEV: ME MODIFY <cid>, <change_reason>, <event_type>

The network has responded to an ME initiated Traffic Flow modification request with an EPS bearer modification request. The associated <cid> is provided to the TE in addition of the change reason: TFT and/or QoS modification. The format of the parameter <cid> is found in command +CGDCONT. The format of the parameter <event_type> is defined above.

OK/ERROR/+CME ERROR

If a setting is not supported by the MT, ERROR or **+CME ERROR** is returned.

Parameter

<PDP_type>see command +CGDCONT

<PDP_addr>see command +CGDCONT

<cid>see command +CGDCONT

<mode>

0 buffer unsolicited result codes in the MT, and discard them when memory is full, the oldest ones can be discarded. No codes are forwarded to the TE.

1 Discard unsolicited Commands when MT-TE link is reserved (e.g. in on-line data mode); otherwise forward them directly to the TE

2 buffer unsolicited result codes in the MT when MT-TE link is reserved (e.g. in on-line data mode) and flush them to the TE when MT-TE link becomes available; otherwise forward them directly to the TE

<bfr>

0 MT buffer of unsolicited result codes defined within this command is cleared when <mode>1 or 2 is entered

1 MT buffer of unsolicited result codes defined within this command is flushed

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	to the TE when **<mode>**1 or 2 is entered (**OK** response shall be given before flushing the codes) Remark: If no parameter is specified nothing will be changed, but OK will be sent.
Read command AT+CGEREP?	Read command returns the current mode and buffer settings Response **+CGEREP: <mode>,<bfr>** **OK** Parameter **<mode>** see set command **<bfr>** see set command
Test command AT+CGEREP=?	Test command returns the modes and buffer settings supported by the MT as compound values. Response **+CGEREP:**(list of supported **<mode>**s),(list of supported **<bfr>**s) **OK** Parameter **<mode>** see set command **<bfr>** see set command
Example	AT+CGEREP=1,1 OK AT+CGEREP? +CGEREP: 1,1 OK AT+CGEREP=? +CGEREP: (0-2),(0-1) OK
Reference	Refer 27.007 section 10.1.18
Note	

7.10 AT+CGTFTRDP Traffic Flow Template Read Dynamic Parameters

Execute command AT+CGTFTRDP[=<cid>]	The execution command returns the relevant information about Traffic Flow Template of **<cid>** together with the additional network assigned values when established by the network. If the context cannot be found an ERROR response is

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returned.

If the parameter <**cid**> is omitted, the Traffic Flow Templates for all established PDP contexts are returned.

Response

+CGTFTRDP: <cid>,<packet filter identifier>,<evaluation precedence index>, <source address and subnet mask>,<protocol number (ipv4) / next header (ipv6)>,<destination port range>,<source port range>,<ipsec security parameter index (spi)>,<type of service (tos) (ipv4) and mask / traffic class (ipv6) and mask>,<flow label (ipv6)>,<direction>,<NW packet filter Identifier>

OK

If error is related to ME functionality:

ERROR/+CME ERROR: <err>

Parameter

<**cid**>:

a numeric parameter which specifies a particular PDP context definition or Traffic Flows definition (see +CGDCONT and +CGDSCONT commands). The following parameters are defined in 3GPP TS 23.060

<**packet filter identifier**>:

a numeric parameter. The value range is from 1 to 16.

<**evaluation precedence index**>:

a numeric parameter. The value range is from 0 to 255.

<**source address and subnet mask**>:

string type. The string is given as dot-separated numeric (0-255) parameters on the form:"a1.a2.a3.a4.m1.m2.m3.m4" for IPv4 or "a1.a2.a3.a4.a5.a6.a7.a8.a9.a10.a11.a12.a13.a14.a15.a16.m1.m2.m3 .m4.m5.m6.m7.m8.m9.m10.m11.m12.m13.m14.m15.m16", for IPv6.

<**protocol number (ipv4) / next header (ipv6)**>:

a numeric parameter, value range from 0 to 255.

<**destination port range**>:

string type. The string is given as dot-separated numeric (0-65535) parameters on the form "f.t".

<**source port range**>:

string type. The string is given as dot-separated numeric (0-65535) parameters on the form "f.t".

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	<**ipsec security parameter index (spi)**>: numeric value in hexadecimal format. The value range is from 00000000 to FFFFFFFF. <**type of service (tos) (ipv4) and mask / traffic class (ipv6) and mask**>: string type. The string is given as dot-separated numeric (0-255) parameters on the form "t.m". <**flow label (ipv6)**>: numeric value in hexadecimal format. The value range is from 00000 to FFFFF. Valid for IPv6 only. <**direction**>: a numeric parameter which specifies the transmission direction in which the Packet Filter shall be applied. 0 Pre Release 7 TFT Filter (see 3GPP TS 24.008 [8], table 10.5.162) 1 Uplink 2 Downlink 3 Bidirectional (Used for Uplink and Downlink) <**NW packet filter Identifier**>: a numeric parameter. The value range is from 1 to 16. In EPS the value is assigned by the network when established NOTE: Some of the above listed attributes can coexist in a Packet Filter while others mutually exclude each other. The possible combinations are shown in 3GPP TS 23.060 [47]
Test command AT+CGTFTRDP =?	Test command returns the modes and buffer settings supported by the MT as compound values. Response +CGTFTRDP**:** (list of <**cid**>s associated with active contexts) **OK** Parameter see set command
Example	
Reference	Refer 27.007 section 10.1.25
Note	

7.11 AT+CGEQOS Define EPS Quality Of Service

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Set command AT+CGEQOS=<cid> [,<QCI> [,<DL_GBR>, <UL_GBR> [,<DL_MBR>,<UL_M BR]]]	The set command allows the TE to specify the EPS Quality of Service parameters **<cid>, <QCI>, [<DL_GBR>** and **<UL_GBR>]** and **[<DL_MBR>** and **<UL_MBR>]** for a PDP context or Traffic Flows. When in UMTS/GPRS the MT applies a mapping function to UTMS/GPRS Quality of Service. A special form of the set command, **+CGEQOS= <cid>** causes the values for context number **<cid>** to become undefined. Response **OK/ERROR/CME ERROR** Parameter **<cid>**: a numeric parameter which specifies a particular EPS Traffic Flows definition in EPS and a PDP Context definition in UMTS/GPRS. **<QCI>**: a numeric parameter that specifies a class of EPS QoS. (see 3GPP TS 23.203) 0 QCI is selected by network [1 – 4] value range for guaranteed bit rate Traffic Flows [5 – 9] value range for non-guaranteed bit rate Traffic Flows **<DL_GBR>**: a numeric parameter which indicates DL GBR in case of GBR QCI. The value is in kbit/s. This parameter is omitted for a non-GBR QCI. (see 3GPP TS 24.301) **<UL_GBR>**: a numeric parameter which indicates UL GBR in case of GBR QCI. The value is in kbit/s. This parameter is omitted for a non-GBR QCI. (see 3GPP TS 24.301) **<DL_MBR>**: a numeric parameter which indicates DL MBR in case of GBR QCI. The value is in kbit/s. This parameter is omitted for a non-GBR QCI. (see 3GPP TS 24.301) **<UL_MBR>**:

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	a numeric parameter which indicates UL MBR in case of GBR QCI. The value is in kbit/s. This parameter is omitted for a non-GBR QCI. (see 3GPP TS 24.301)
Read command AT+CGEQOS?	The read command returns the current settings for each defined QoS. Response **[+CGEQOS: <cid>, <QCI>, [<DL_GBR>,<UL_GBR>],** **[<DL_MBR>,<UL_MBR>]]** **[<CR><LF>+CGEQOS: <cid>, <QCI>, [<DL_GBR>,<UL_GBR>],** **[<DL_MBR>,<UL_MBR>]** **[…]]** **OK** Parameter see set command
Test command AT+CGEQOS=?	Test command returns the modes and buffer settings supported by the MT as compound values. Response **+CGEQOS: (range of supported <cid>s),(list of supported<QCI>s),(list of supported <DL_GBR>s),(list of supported<UL_GBR>s), (list of supported <DL_MBR>s) ,(list of supported<UL_MBR>s)** **OK**
Example	
Reference	Refer 27.007 section 10.1.26
Note	

7.12 AT+CGEQOSRDP EPS Quality Of Service Read Dynamic Parameters

Execute command AT+CGEQOSRDP[=<cid>]	The execution command returns the Quality of Service parameters **<QCI>, [<DL_GBR>** and **<UL_GBR>]** and **[<DL_MBR>** and **<UL_MBR>]** of the established PDP Context associated to the provided context identifier **<cid>**. If the context cannot be found an ERROR response is returned. If the parameter **<cid>** is omitted, the Quality of Service parameters for all established PDP contexts are returned. Response

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	+CGEQOSRDP: <cid>, <QCI>, [<DL_GBR>,<UL_GBR>], [<DL_MBR>,<UL_MBR>] **OK** If error is related to ME functionality: **ERROR/+CME ERROR: <err>** Parameter **<cid>**: a numeric parameter which specifies a particular Traffic Flows definition in EPS and a PDP Context definition in UMTS/GPRS. **<QCI>**: a numeric parameter that specifies a class of EPS QoS. (see 3GPP TS 23.203) 0 QCI is selected by network [1 – 4] value range for guaranteed bit rate Traffic Flows [5 – 9] value range for non-guaranteed bit rate Traffic Flows **<DL_GBR>**: a numeric parameter which indicates DL GBR in case of GBR QCI. The value is in kbit/s. This parameter is omitted for a non-GBR QCI. (see 3GPP TS 24.301) **<UL_GBR>**: a numeric parameter which indicates UL GBR in case of GBR QCI. The value is in kbit/s. This parameter is omitted for a non-GBR QCI. (see 3GPP TS 24.301) **<DL_MBR>**: a numeric parameter which indicates DL MBR in case of GBR QCI. The value is in kbit/s. This parameter is omitted for a non-GBR QCI. (see 3GPP TS 24.301) **<UL_MBR>**: a numeric parameter which indicates UL MBR in case of GBR QCI. The value is in kbit/s. This parameter is omitted for a non-GBR QCI. (see 3GPP TS

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	24.301)
Test command AT+CGEQOSRDP=?	Response **+CGEQOSRDP:** (list of <**cid**>s associated with active contexts) **OK** Parameter see set command
Example	
Reference	Refer 27.007 section 10.1.27
Note	

7.13 AT+CEMODE UE modes of operation for EPS

Set command AT+CEMODE=[<mode>]	The set command is used to set the MT to operate according to the specified mode of operation for EPS, see 3GPP TS 24.301. If the requested mode of operation is not supported, an ERROR or +CME ERROR response is returned. Extended error responses are enabled by the +CMEE command. Response **OK/ERROR/CME ERROR** Parameter **<mode>**: a numeric parameter which indicates the mode of operation 0 PS mode 2 of operation 1 CS/PS mode 1 of operation 2 CS/PS mode 2 of operation 3 PS mode 1 of operation NOTE: <mode> PS mode means that the MT would only operate EPS services
Read command AT+CEMODE?	The read command returns the mode of operation set by the TE, independent of the current serving cell capability and independent of the current serving cell Access Technology.

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	+CEMODE: <mode> **OK** Parameter see set command
Test command AT+CEMODE =?	Test command returns the modes and buffer settings supported by the MT as compound values. Response **+CEMODE: (list of supported <mode>s)** **OK** Parameter see set command
Example	
Reference	Refer 27.007 section 10.1.28
Note	

7.14 AT^DDPDN UE set EPS Default PDN Parameters

Set command AT^DDPDN=<PDN type>,<APN>[,<username>,<password>,<auth_type>]	The set command is used to configure parameters PDN type and APN which are used by NAS during ATTACH REQUEST procedure, so it should be invoked before MS triggers network registration. Response **OK/ERROR/CME ERROR** Parameter **<PDN type>**: 1 -IPv4 2 - IPv6 3 - IPv4IPv6 **<APN>**: Access Point Name a string parameter which is a logical name that is used to select the GGSN or the external packet data network. If the value is null or omitted, it is decided by operator.

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	<**username**>: string type; User name for access to the IP network. <**password**>: string type; Password for access to the IP network. <**auth_type**>: integer type; Authentication protocol used for this PDP context. 0 None. Used to indicate that no authentication protocol is used for this PDP context. Username and password are removed if previously specified. 1 PAP 2 CHAP
Read command AT^DDPDN?	The read command returns the current configuration. **^DDPDN: <PDN type>,<APN>[,<username>,<password>,<auth_type>]** OK Parameter see set command
Test command AT^DDPDN=?	Test command returns the modes and buffer settings supported by the MT as compound values. Response **^DDPDN: (list of supported < PDN type >s)** **OK** Parameter see set command
Reference	
Note	

7.15 AT+CGCONTRDP PDP Context read dynamic parameters

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Execute command AT+CGCONTRDP [=<cid>]	The execution command returns the relevant information <bearer_id>, <apn>, <PDP_addr_mask >, <gw_addr>, <DNS_prim_addr>, <DNS_sec_addr>, <P-CSCF_prim_addr>, <P-CSCF_sec_addr>for an active non secondary PDP context established by the network with <cid>. Response [+CGCONTRDP: <cid>,<bearer_id>, <apn>[,<PDP_addr_mask>[, <gw_addr>[,<DNS_prim_addr>[,<DNS_sec_addr>[, <P-CSCF_prim_addr>[,<P-CSCF_sec_addr>[,<IM_CN_Signalling_Flag>]]]]]]]] [<CR><LF>+CGCONTRDP: <cid>,<bearer_id>, <apn>[,<PDP_addr_mask>[, <gw_addr>[,<DNS_prim_addr>[,<DNS_sec_addr>[, <P-CSCF_prim_addr>[,<P-CSCF_sec_addr>[,<IM_CN_Signalling_Flag>]]]]]]] […]] **OK** If error is related to ME functionality: **+CME ERROR: <err>** Parameter **<cid>** (PDP Context Identifier) a numeric parameter which specifies a particular PDP context definition. The parameter is local to the TE-MT interface and is used in other PDP context-related commands. The range of permitted values (minimum value = 1) is returned by the test form of the command. If no cid is given, nothing will be changed (neither con-text definition nor undefinition). 1-5 Valid range of cid **<bearer_id>**: integer type; identifies the bearer, i.e. the EPS bearer and the NSAPI. **<APN>** Access Point Name is a string parameter (framed by quotation marks) which is a logical name that is used to select the GGSN or the external packet data network. If the value is null or omitted, then the subscription value will be requested. **<PDP_addr_mask>**: string type; shows the PDP address mask of the MT. **<gw_addr>**: string type; shows the Gateway Address of the MT. The string is given as dot-separated numeric (0-255) parameters. When +CGPIAF is supported, its settings can influence the format of this parameter returned with the execute form of +CGCONTRDP.

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	<DNS_prim_addr>: string type; shows the IP address of the primary DNS server. When +CGPIAF is supported, its settings can influence the format of this parameter returned with the execute form of +CGCONTRDP. **<DNS_sec_addr>**: string type; shows the IP address of the secondary DNS server. When +CGPIAF is supported, its settings can influence the format of this parameter returned with the execute form of +CGCONTRDP. **<P-CSCF_prim_addr>**: string type; shows the IP address of the primary P-CSCF server. When +CGPIAF is supported, its settings can influence the format of this parameter returned with the execute form of +CGCONTRDP. **<P-CSCF_sec_addr>**: string type; shows the IP address of the secondary P-CSCF server. When +CGPIAF is supported, its settings can influence the format of this parameter returned with the execute form of +CGCONTRDP. <**IM_CN_Signalling_Flag**>: integer type; shows whether the PDP context is for IM CN subsystem-related signalling only or not. 0 PDP context is not for IM CN subsystem-related signalling only 1 PDP context is for IM CN subsystem-related signalling only
Test command AT+CGCONTRDP=?	The test command returns a list of supported **<cid>**s associated with active contexts Response **+CGCONTRDP:** (list of supported **<cid>**s associated with active contexts) **OK**
Reference	Refer 27.007 section 10.1.23
Note	The list of supported **<cid>**s is (1-5).

7.16 AT+CGSCONTRDP Secondary PDP Context read dynamic parameters

Execute command AT+CGSCONTRDP	The execution command returns <p_cid>,<bearer_id> and <IM_CN_Signalling_Flag> for a given <cid>

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[=<cid>]	Response [+CGSCONTRDP: <cid>,<p_cid>, <bearer_id >[,< IM_CN_Signalling_Flag >]] [<CR><LF>+CGSCONTRDP: <cid>,<p_cid>, <bearer_id >[, IM_CN_Signalling_Flag >] […]] **OK** If error is related to ME functionality: **+CME ERROR: <err>** Parameter **<cid>** (PDP Context Identifier) a numeric parameter which specifies a particular PDP context definition. The parameter is local to the TE-MT interface and is used in other PDP context-related commands. The range of permitted values (minimum value = 1) is returned by the test form of the command. If no cid is given, nothing will be changed (neither con-text definition nor undefinition). 1-5 Valid range of cid **<p_cid>**: integer type; specifies a particular PDP context definition or default EPS context Identifier which has been specified by use of the +CGDCONT command. The parameter is local to the TE-MT interface. **<bearer_id>**: integer type; identifies the bearer, i.e. the EPS bearer and the NSAPI. <**IM_CN_Signalling_Flag**>: integer type; shows whether the PDP context is for IM CN subsystem-related signalling only or not. 0 PDP context is not for IM CN subsystem-related signalling only 1 PDP context is for IM CN subsystem-related signalling only
Test command AT+CGSCONTRDP=?	The test command returns a list of **<cid>**s associated with active secondary PDP contexts Response **+CGSCONTRDP:** (list of **<cid>**s associated with active contexts) **OK**
Reference	Refer 27.007 section 10.1.24

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7.17 AT+CGAUTH Define PDP context authentication parameters

Set command AT+CGAUTH=<cid>[,<auth_prot>[,<userid>[,<password>]]]	The set command allows the TE to specify authentication parameters for a PDP context identified by the (local) context identification parameters <cid> used during the PDP context activation and the PDP context modification procedures. Response **OK/ERROR/CME ERROR** Parameter **<cid>**: integer type; specifies a particular PDP context definition. **<auth_prot>:** integer type; Authentication protocol used for this PDP context. 0 None. Used to indicate that no authentication protocol is used for this PDP context. Username and password are removed if previously specified. 1 PAP 2 CHAP **<userid>**: string type; User name for access to the IP network. **<password>**: string type; Password for access to the IP network.
Read command AT+CGAUTH?	The read command returns the current settings for each defined context. [**+CGAUTH**: <cid>,<auth_prot>[,<userid>,<password>] [<CR><LF>+CGAUTH: <cid>,<auth_prot>,<userid>,<password>] […]]] **OK** Parameter see set command
Test command AT+CGAUTH=?	The test command returns values supported as compound values. Response **+CGAUTH:** (1-5),(0-2), , **OK**
Reference	Refer 27.007 section 10.1.31

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7.18 AT+CGDEL Delete non-active PDP contexts

Set command AT+CGDEL [=<cid>]	The set command +CGDEL=<cid> removes the indicated PDP context and removes all associated data related to the indicated PDP context that are not activated. Response [+CGDEL: <cid>[,<cid>[,…]]] **OK** **+CME ERROR: <err>** Parameter **<cid>**: integer type; specifies a particular PDP context definition.
Test command AT+CGDEL=?	The test command gets the supported cid for set operation. Response **+CGDEL: (list of supported <cid>s)** **OK** Parameter see set command
Example	
Reference	Refer 27.007 section 10.1.29
Note	

7.19 AT+CSCON Signaling connection status

Set command AT+CSCON= [<n>]	The set command controls the presentation of an unsolicited result code +CSCON. If <n>=1, +CSCON: <mode> is sent from the MT when the connection mode of the MT is changed. If <n>=2 and there is a state within the current mode, +CSCON: <mode>[,<state>] is sent from the MT when the connection mode or state information of the MT is changed. If <n>=3, +CSCON: <mode>[,<state>[,<access>]] is sent from the MT when the connection mode, state or access information of the MT is changed. If <n>=4, +CSCON: <mode>[,<state>[,<access>[,<coreNetwork>]]] is sent from the MT. Response **OK/ERROR/CME ERROR** Parameter **<n>**: integer type

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	0 Disable unsolicited result code 1 Enable unsolicited result code +CSCON: <mode> 2 Enable unsolicited result code +CSCON: <mode>[,<state>] 3 Enable unsolicited result code +CSCON: <mode>[,<state>[,<access>]] 4 Enable unsolicited result code +CSCON: <mode>[,<state>[,<access>[,<coreNetwork>]]] <mode>: integer type; indicates the signaling connection status. 0 Idle 1 Connected <state>: integer type; indicates the CS or PS state while in GERAN and the RRC state information if the MT is connected mode while in UTRAN, EUTRAN and NG-RAN. 7 E-UTRAN connected state <access>: integer type; indicates the current radio access type. 3 Indicates usage of radio access of type E-UTRA TDD. 4 Indicates usage of radio access of type E-UTRA FDD. <coreNetwork>: integer type; indicates the core network type the UE is connected to. 0 Indicates MT is connected to EPC.
Read command AT+CSCON?	The read command returns the status of result code presentation and an integer <mode> which shows whether the MT is currently in idle mode or connected mode. State information <state> is returned only when <n>=2. Radio access type information <access> is returned only when <n>=3. Core network type information <coreNetwork> is returned only when <n>=4. Response **+CSCON: <n>,<mode>[,<state>[,<access>]]** **[<CR><LF>+CSCON: <n>,<mode>[,<state>[,<access>[,<coreNetwork>]]]** **[…]]** **OK**

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	Parameter see set command
Test command AT+CSCON=?	Test command returns supported values as a compound value. Response +CSCON: (list of supported <n>s) OK Parameter see set command
Reference	Refer 27.007 section 10.1.30
Note	

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8 IMS related AT Commands

8.1 AT+IMSREG Registration with IMS

Set command AT+IMSREG=<command>	Set command informs the UE registers or deregisters with IMS. Response **OK** If error is related to ME functionality: **+CME ERROR: <err>** Parameter **<command>** 0 IMS de-registration. 1 IMS registration. 10 Disable IMS automatic registration. 11 Enable IMS automatic registration.
Read command AT+IMSREG?	Read command returns the UEs IMS registration behavior. Response **+IMSREG: <command>** **OK** Parameter See set command
Test command AT+IMSREG =?	Response **+IMSREG**: (list of supported **<command>**s) **OK** Parameter See set command
Reference	
Note	The list of supported **<command>**s is (0-1,10-11). AT+IMSREG=10 and AT+IMSREG=11, the settings do not modify the configuration file and are lost after reboot.

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8.2 AT+CIREG IMS registration information

Set command AT+CIREG=[<n>]	Set command controls the presentation of an unsolicited result code **+CIREGU: <reg_info>[,<ext_info>]** when there is a change in the MT's IMS registration information. Refer clause 9.2 for possible <err> values. Response **OK** If error is related to ME functionality: **+CME ERROR: <err>** Parameter **<n>** integer type. Enables or disables reporting of changes in the MT's IMS registration information. 0 disable reporting. 1 enable reporting (parameter <reg_info>). 2 enable extended reporting (parameters <reg_info> and <ext_info>).
Read command AT+CIREG?	Read command returns the UEs IMS registration information. Response **+CIREG: <n>,<reg_info>[,<ext_info>]** **OK** Parameter **<n>**: see set command **<reg_info>**: integer type. Indicates the IMS registration status. The UE is seen as registered as long as one or more of its public user identities are registered with any of its contact addresses, see 3GPP TS 24.229 [89]. 0 not registered. 1 registered. **<ext_info>**: numeric value in hexadecimal format. The value range is from 1 to FFFFFFFF. It is a sum of hexadecimal values, each representing a particular IMS capability of the MT. The MT can have IMS capability not covered by the below list. This parameter is not present if the IMS registration status is "not registered". 1 RTP-based transfer of voice according to MMTEL, see 3GPP TS 24.173 [87]. This functionality can not be indicated if the UE is not available for voice over PS, see 3GPP TS 24.229 [89]. 2 RTP-based transfer of text according to MMTEL, see 3GPP TS 24.173 [87].

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	4 SMS using IMS functionality, see 3GPP TS 24.341 [101]. 8 RTP-based transfer of video according to MMTEL, see 3GPP TS 24.173 [87]. The hexadecimal values 10, 20, 40 … 80000 are reserved by the present document.
Test command AT+CIREG=?	Response **+CIREG**: (list of supported **<n>**s) **OK** Parameter See set command
Reference	Refer 27.007 section 8.71
Note	The parameter <ext_info> is not implemented

8.3 AT+CLCCS List of current calls

Set command AT+CLCCS[=<ccidx>]	Returns a list of current calls controlled by the MT for which the <ccidx> is allocated. Response [+CLCCS: <ccid1>,<dir>,<neg_status_present>,<neg_status>, <SDP_md>,<cs_mode>,<ccstatus>,<mpty>,[,<numbertype>, <ton>,<number>[,<priority_present>,<priority> [,<CLI_validity_present>,<CLI_validity>]]] [<CR><LF>+CLCCS: <ccid2>,<dir>,<neg_status_present>,<neg_status>, <SDP_md>,<cs_mode>,<ccstatus>,<mpty>,[,<numbertype>, <ton>,<number>[,<priority_present>,<priority> [,<CLI_validity_present>,<CLI_validity>]]] [...]]] If error is related to ME functionality: **+CME ERROR: <err>** Parameter See Chapter 8.3 “+CMCCSI Monitor of current calls”
Reference	Refer 27.007 section 8.74
Note	

8.4 ATD Originate a voice call

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Set command ATD<dial string>	Set command to originate a voice call. Response **OK** Parameter **<dial string>**: dialing number
Reference	
Note	

8.5 ATA Answer an incoming call

Set command ATA	Set command to answer an incoming call when the URC +CMCCSI:1,0,,,"",,5,,2,129,"12345678",0,0 indicates that there is an alerting MT call Response **OK** Parameter **NA**
Reference	
Note	

8.6 ATH Hang-up calls

Set command ATH	Set command to hang-up all calls Response **OK** Parameter **NA**
Reference	
Note	

8.7 AT+CHLD Hang-up a call or Hold/Resume a call

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Set command AT+CHLD=<command>	Set command to hold/resume a call, or hang-up a specific call. Response **OK** If error is related to ME functionality: **+CME ERROR: <err>** Parameter **<command>** 2 Hold/Resume a call, Swap two calls 71 Hang-up the first call 72 Hang-up the second call
Test command AT+CHLD =?	Response **+CHLD**: (list of supported **<command>**s) **OK** Parameter See set command
Reference	
Note	

8.8 AT+VTS DTMF tone

Set command AT+VTS=<command> [,<duration>]	Set command to send DTMF Response **OK** If error is related to ME functionality: **+CME ERROR: <err>** Parameter **<command>**: 0-9,*,#,A,B,C,D **<duration>**: 0-255, unit:100ms
Test command AT+VTS =?	Response **+VTS**: (list of supported **<command>**s) **OK**

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	Parameter See set command
Reference	
Note	

8.9 AT+VTD DTMF duration

Set command AT+VTD=<duration>	Set command to set DTMF duration Response **OK** If error is related to ME functionality: **+CME ERROR: <err>** Parameter **<duration>**: 0-255, unit:100ms
Test command AT+VTD =?	Response **+VTD**: (list of supported **<duration>**) **OK** Parameter See set command
Reference	
Note	

8.10 AT+CLCC List of current calls

Set command AT+CLCC	Returns a list of current calls Response **[+CLCC: <ccid1>,<dir>,<stat>,<mode>,<mpty>[,<number>,<type>[,<alpha>[,<priority>[,<CLI validity>]]]]** **[<CR><LF>** **+CLCC: <ccid2>,<dir>,<stat>,<mode>,<mpty>[,<number>,<type>[,<alpha>[,<priority>[,<CLI validity>]]]]** **[...]]]**

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	OK If error is related to ME functionality: **+CME ERROR: <err>** Parameter **<ccidx>**: integer type, value:1~ 6. Call identification number as described in 3GPP TS 22.030 [19] clause 6.5.5.1. This number can be used in +CHLD command operations. **<dir>:** integer type 0 mobile originated (MO) call 1 mobile terminated (MT) call **<stat>**: integer type (state of the call) 0 active 1 held 2 dialing (MO call) 3 alerting (MO call) 4 incoming (MT call) 5 waiting (MT call) **<mode>**: integer type (bearer/teleservice) 0 voice **<mpty>**: integer type 0 call is not one of multiparty (conference) call parties 1 call is one of multiparty (conference) call parties **<number>**: string type phone number in format specified by <type>. **<type>:** type of address octet in integer format (refer 3GPP TS 24.008 [8] clause 10.5.4.7)
Test command AT+CLCC=?	Response: **OK**
Reference	Refer 27.007 section 7.18
Note	

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8.11 AT+CVMOD Voice call mode

Set command AT+CVMOD=[<voice_mode>]	Set command selects the voice call mode for making a Mobile Originated voice call from the UE. Response **OK** Parameter **<voice_mode>**: integer type. The default value is manufacturer specific. 0 CS_ONLY 1 VOIP_ONLY 2 CS_PREFERED 3 VOIP_PREFERED
Read command AT+CVMOD?	Read command returns the current value of <voice_mode>. **+CVMOD**: <voice_mode> **OK** Parameter See Set command
Test command AT+CVMOD=?	Response +CVMOD: (list of supported **<voice_mode>**s) **OK**
Reference	3GPP TS 27.007, section 6.4
Note	Only VOIP_ONLY voice mode is available.

8.12 AT+CMCCS Monitor of current calls

Set command AT+CMCCS=[<n>]	This command activates or deactivates a call monitoring function in the ME. When this function is activated in the ME, the ME informs about events for calls with unsolicited result codes. Response **OK/ERROR/+CME ERROR** The unsolicited result code +CMCCSI is used for basic call information. +CMCCSI: <ccidx>,<dir>,<neg_status_present>,<neg_status>,<SDP_md>,<cs_mode>,<cc status>,<mpty>,<numbertype>,<ton>,<number>,<exittype>,<exitcause>

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eg: +CMCCSI:1,0,,,"",,2,,2,129,"12345678",0,0

Parameter
<n>: integer type

0 The call monitoring function is disabled.

2 The call monitoring function is enabled for basic call information (unsolicited result code +CMCCSI)

<ccidx>: integer type. Call identification number as described in 3GPP TS 22.030 [19] clause 6.5.5.1. This number can be used in +CHLD command operations. Value range is from 1 to N. N, the maximum number of simultaneous call control processes is implementation specific. The call identification number must not be reused until the unsolicited result code +CMCCSI has indicated that the <ccstatus>=1 (Idle)

<dir>: integer type

0 mobile originated (MO) call

1 mobile terminated (MT) call

<neg_status_present>:NA

<neg_status>:NA

<SDP_md>:NA

<cs_mode>:NA

<ccstatus>: integer type, indicating the state of the call.

1. Idle
2. Calling (MO); the call setup has been started
3. Connecting (MO); the call is in progress
4. Alerting (MO): an alert indication has been received
5. Alerting (MT); an alert indication has been sent
6. Active; the connection is established
7. Released; an outgoing (MO) call is released.
8. Released; an incoming (MT) call is released
9. User busy
10. User determined user busy
11. Call waiting (MO)
12. Call waiting (MT)
13. Call hold (MO)
14. Call hold (MT)

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	<mpty>:NA <numbertype> integer type. Indicating type of information in parameter <number>. 0 No valid information in parameter <number>. <number> shall then be set to empty string (""). 1 Number in <number> according to URI including the prefix specifying the URI type (see command +CDU). Parameter <ton> has no relevant information and is set to zero. 2 Number in <number> according to one of the formats supported by 3GPP TS 24.008 [8] clause 10.5.4.7) <ton>: type of number in integer format (refer 3GPP TS 24.008 [8] clause 10.5.4.7). The parameter is also set to zero when it has no meaningful content, e.g. when <numbertype>=1 <number>: string type phone number in format specified by <numbertype>. The used character set should be the one selected with command select TE character set +CSCS. When no number is available, <number> shall be set to empty string ("") <exittype>: integer type. Indicating type of information in parameter <exitcause>. 0 No valid information in parameter <exitcause> 1 Cause in <exitcause> according to 3GPP TS 24.008 [8], Annex H 2 Cause in <exitcause> according to 3GPP TS 24.229 [89], Annex A.2.1.4.1 <exitcause>: integer type. Additional information provided if relevant. If there is no relevant cause to report, or if<exittype>=0 the exit cause is set to 0 (<exit_cause>=0).
Read command AT+CMCCS?	Read command returns the current value of <n>. Response +CMCCS: <n> OK
Test command AT+CMCCS=?	Test command returns the values supported as a compound value. Response +CMCCS: (list of supported <n>s) OK
Reference	Refer 27.007 section 8.73

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note	

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9 SIM Control and Status AT Command

9.1 AT+CSIM Generic SIM access

Set command AT+CSIM=<length>,<command>	Set command: Set command transmits to the MT the <command> it then shall send as it is to the SIM/USIM. In the same manner the SIM/USIM <response> shall be sent back by the MT to the TA as it is. Response **+CSIM:<length>,<response>** **OK** If error is related to ME functionality: **+CME ERROR: <err>** Parameter **<length>:** integer type, the length of the characters that are sent to TE. **<command>:** command passed on by the MT to the SIM in the format as described in 3GPP TS 51.011 [28] **<response>:** response to the command passed by the SIM to the MT in the format as described in 3GPP TS 51.011 [28]
Test command AT+CSIM=?	Response **OK**
Reference	Refer 27.007 section 8.17
Note	The <command> can be a string type or a data type.

9.2 AT+CRSM Restricted SIM access

Set command AT+CRSM=<command> [,<fileid>[,<P1>,<P2>,<P3> [,<data>[,<pathid>]]]]	Set command transmits to the MT the SIM/USIM <command> and its required parameters. MT handles internally all SIM/USIM-MT interface locking and file selection routines. As response to the command, MT sends the actual SIM/USIM information parameters and response data. Response

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	+CRSM:<sw1>,<sw2>[,<response>] **OK** If error is related to ME functionality: **+CME ERROR: <err>** Parameter **<fileid>:** integer type, this is the identifier of a elementary datafile on SIM. Mandatory for every command except STATUS. NOTE: The range of valid file identifiers depends on the actual SIM and is defined in 3GPP 51.011[28] (SIM) or ETSI 102.221 (USIM). Optional files may not be present at all. **<command>:** string type (without double quotes), command passed on by the MT to the SIM/USIM in the format as described in 3GPP 51.011[28] (SIM) or ETSI 102.221 (USIM) 176 READ BINARY 178 READ RECORD 192 GET RESPONSE 214 UPDATE BINARY 220 UPDATE RECORD 242 STATUS **<p1>:** integer type, Parameters passed on by the MT to the SIM. These parameters are mandatory for every command, except GET RESPONSE and STATUS. The values are described in3GPP 51.011[28](SIM) or ETSI 102.221 (USIM). **<p2>:** integer type, same as <p1>. **<p3>:** integer type, same as <p1>. **<sw1>:** integer type; information from the SIM about the execution of the actual command. These parameters are delivered to the TE in both cases, on successful or failed execution of the command. **<sw2>:** integer type, same as <sw1>.

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	<response>: string type (without double quotes), Response of a successful completion of the command previously issued (hexadecimal character format; refer +CSCS). STATUS and GET RESPONSE return data, which gives information about the current elementary data field. This information includes the type of file and its size (refer to 3GPP 51.011[28](SIM) or ETSI 102.221 (USIM)). After READ BINARY or READ RECORD command the requested data will be returned. <response> is not returned after a successful UPDATE BINARY or UPDATE RECORD command)
Test command AT+CRSM=?	Response **OK**
Reference	Refer 27.007 section 8.18
Note	The <data> can be a string type or a data type.

9.3 AT+CPIN Enter PIN

Set command AT+CPIN=<pin>[,<newpin>]	Set command sends to the MT a password which is necessary before it can be operated (SIM PIN, SIM PUK, PH-SIM PIN, etc.). If the PIN is to be entered twice, the TA shall automatically repeat the PIN. If no PIN request is pending, no action is taken towards MT and an error message, +CME ERROR, is returned to TE. Refer subclause 9.2 for possible <err>. Response **OK** If error is related to ME functionality: **+CME ERROR: <err>** Parameter **<pin>:** string type, usually SIM PIN1or PIN2.If the requested password was a PUK, such as SIM PUK1 or SIM PUK2 or another password, then <newpin> is required. **<newpin>:** string type, same as <pin>. **<code>:** values reserved by the present document: READY MT is not pending for any password

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	SIM PIN MT is waiting SIM PIN to be given SIM PUK MT is waiting SIM PUK to be given PH-SIM PIN MT is waiting phone-to-SIM card password to be given PH-FSIM PIN MT is waiting phone-to-very first SIM card password to be given PH-FSIM PUK MT is waiting phone-to-very first SIM card unblocking password to be given SIM PIN2 MT is waiting SIM PIN2 to be given (this <code> is recommended to be returned only when the last executed command resulted in PIN2 authentication failure (i.e. +CME ERROR: 17); if PIN2 is not entered right after the failure, it is recommended that MT does not block its operation) SIM PUK2 MT is waiting SIM PUK2 to be given (this <code> is recommended to be returned only when the last executed command resulted in PUK2 authentication failure (i.e. +CME ERROR: 18); if PUK2 and new PIN2 are not entered right after the failure, it is recommended that MT does not block its operation). PH-NET PIN MT is waiting network personalization password to be given. PH-NET PUK MT is waiting network personalization unblocking password to be given. PH-NETSUB PIN MT is waiting network subset personalization password to be given. PH-NETSUB PUK MT is waiting network subset personalization unblocking password to be given. PH-SP PIN MT is waiting service provider personalization password to be given. PH-SP PUK MT is waiting service provider personalization

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	unblocking password to be given. PH-CORP PIN MT is waiting corporate personalization password to be given. PH-CORP PUK MT is waiting corporate personalization unblocking password to be given.
Read command AT+CPIN?	Read command returns an alphanumeric string indicating whether some password is required or not. Response **+CPIN:<code>** **OK**
Test command AT+CPIN=?	Response **OK**
Reference	3GPP TS 27.007, section 8.3
NOTE	

9.4 AT+CNUM Subscribe Number

Execute command AT+CNUM	Execute command returns the MSISDNs related to the subscriber (this information can be stored in the SIM/UICC or in the MT). When storing information in the SIM/UICC, if the currently selected card slot contains a SIM card or a UICC with an active GSM application, the information is stored in the EF_{MSISDN} under $DF_{Telecom}$. If the currently selected card slot contains a UICC with an active USIM application, the information is stored in the EF_{MSISDN} under ADF_{USIM}). If subscriber has different MSISDN for different services, each MSISDN is returned on a separate line. Refer subclause 9.2 for possible <err>. Response **+CNUM: [<alpha>],<number>,<type>** **OK** If error is related to ME functionality: **+CME ERROR: <err>** Parameter **<pin>:** string type, usually SIM PIN1or PIN2.If the requested

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	password was a PUK, such as SIM PUK1 or SIM PUK2 or another password, then <newpin> is required. **<alpha>:** optional alphanumeric string associated with <number>; used character set should be the one selected with command Select TE Character Set +CSCS **<number>:** string type phone number format specified by <type>. **<type>:** type of address octet in integer format (refer 3GPP TS 24.008 [8] subclause 10.5.4.7)
Test command AT+CNUM=?	Response **OK**
Reference	3GPP TS 27.007, section 7.1
NOTE	

9.5 AT+CCHO Open logical channel

Set command AT+CCHO=<dfname>	Set command causes the MT to return <sessionid> to allow the TE to identify a channel that is being allocated by the currently selected UICC, which is attached to ME. The currently selected UICC will open a new logical channel; select the application identified by the <dfname> received with this command and return a session Id as the response. The ME shall restrict the communication between the TE and the UICC to this logical channel. This <sessionid> is to be used when sending commands with Generic UICC Logical Channel access +CGLA commands. Response **<sessionid>** **OK** If error is related to ME functionality: **+CME ERROR: <err>** Parameter **<dfname>:** all selectable applications in the UICC are referenced by a DF name coded on 1 to 16 bytes; **<sessionid>:** integer type; a session Id to be used in order to target a specific application on the smart card (e.g.(U)SIM, WIM,

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	ISIM) using logical channels mechanism.
Read command AT+CCHO?	Read command returns the list of all <dfname>. Response +CCHO: <length>,<dfname> … **OK** Parameter <length>: the length of <dfname>. <dfname>: all selectable applications in the UICC are referenced by a DF name coded on 1 to 16 bytes.
Test command AT+CCHO=?	Response **+CME ERROR: <err>**
Reference	3GPP TS 27.007, section 8.45
NOTE	The <dfname> can be a string type or a data type.

9.6 AT+CCHC Close logical channel

Set command AT+CCHC=<sessionid>	Set command is used to close a communication session with the active UICC. The ME shall close the previously opened logical channel. The TE will no longer be able to send commands on this logical channel. The UICC will close the logical channel when receiving this command. Response **+CCHC** If error is related to ME functionality: **+CME ERROR: <err>** Parameter **<sessionid>:** integer type; a session Id to be used in order to target a specific application on the smart card (e.g.(U)SIM, WIM, ISIM) using logical channels mechanism.
Test command	Response

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AT+CCHC=?	**+CME ERROR: <err>**
Reference	3GPP TS 27.007, section 8.46
NOTE	

9.7 AT+CGLA Generic UICC logical channel access

Set command AT+CGLA=<sessionid>, <length>,<command>	Set command transmits to the MT the <command> it then shall send as it is to the selected UICC. In the same manner the UICC <response> shall be sent back by the MT to the TA as it is. Response +CGLA: <length>,<response> **OK** If error is related to ME functionality: **+CME ERROR: <err>** Parameter: **<sessionid>:** integer type; this is the identifier of the session to be used in order to send the APDU commands to the UICC. It is mandatory in order to send commands to the UICC when targeting applications on the smart card using a logical channel other than the default channel (channel "0"). **<length>:** integer type; length of the characters that are sent to TE in <command> or <response> (two times the actual length of the command or response) **<command>:** command passed on by the MT to the UICC in the format as described in 3GPP TS 31.101 [65] (hexadecimal character format; refer +CSCS) **<response>:** response to the command passed on by the UICC to the MT in the format as described in 3GPP TS 31.101 [65] (hexadecimal character format; refer +CSCS)
Test command	Response

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AT+CGLA=?	**+CME ERROR: <err>**
Reference	3GPP TS 27.007, section 8.43
NOTE	Compared to Restricted UICC Access command +CRLA, the definition of +CGLA allows TE to take more control over the UICC MT interface. The locking and unlocking of the interface may be done by a special <command> value or automatically by TA/MT (by interpreting <command> parameter). In case that TE application does not use the unlock command (or does not send a <command> causing automatic unlock) in a certain timeout value, MT may release the locking. The <command> can be a string type or a data type.

9.8 AT^DCID USIM Card Identification

Execute command AT^DCID	The command reads the ICCID of the USIM card. If no USIM card is present, or the USIM card is unreadable, no data is returned. Response 1) ^DCID: <ICCID> OK 2) ERROR 3) +CME ERROR: <err> Parameter <ICCID>: Integrate circuit card identity, a standard ICCID is a 20-digit serial number of the SIM card, it presents the publish state, network code, publish area, publish date, publish manufacture and press serial number of the SIM card.
Test command AT^DCID=?	Response **OK**
Reference	
NOTE	

9.9 AT^SIMWP SIM write protection

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Set command AT^SIMWP=<value>	This command is used to set SIM write protection configuration, each bit in <value> corresponds to a SIM write protection configuration for an EF file. Response **OK/ERROR/+CME ERROR** Parameter <value>: SIM write protection configuration. The default value is 0, indicating real-time writing EF files to SIM. Each bit in <value> corresponds to a SIM write protection configuration for an EF file. Bit0=0: write EFepsnsc to SIM when UE enters the deregistration state. Bit0=1: write EFepsnsc to SIM when UE switches off. Bit1=0: write EFepsloci to SIM when EPS location information updates. Bit1=1: write EFepsloci to SIM when UE switches off. Bit2=0: write EFloci to SIM when location information updates. Bit2=1: write EFloci to SIM when UE switches off.
Read command AT^SIMWP?	Read command returns the current value of SIM write protection configuration. Response ^SIMWP: <value> **OK** Parameter See Set command
Test command AT^SIMWP=?	Response ^SIMWP: (list of supported **<value>**s) **OK**
Reference	
NOTE	

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10 AIC defined AT commands for enhanced functions

Self-defined commands do not have to be implemented in accordance with the official syntax.

10.1 AT^DPIN2: Enter PIN2

Execute command AT^DPIN2=<pin2>	Execute command sends to the MT a pin2 code for verifying. NOTE1: MT is pending SIM PIN2(refer +CPIN). If SIM PIN2 has been verified successfully, MT shall enter to READY status (refer +CPIN); otherwise, status of MT shall be unchanged. NOTE2: If PIN2 is blocked and PUK2 required, the reported error cause is "operation not supported". Response: **OK** If error is related to ME functionality: **+CME ERROR: <err>** Parameter: <**pin2**> string type values
Test command AT^DPIN2=?	Response **OK**
Note	

10.2 AT^DESQ Signal quality indication in LTE

Set command AT^DESQ =[<mode>]	This command is used to enable/disable unsolicited result code "^**DESQU**: <**rsrp**>" when RSRP value changes. Response: **OK/ERROR/+CME ERROR** Parameter: <**mode**>: 0 disable intermediate result codes 1 enable intermediate result codes

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	<**rsrp**>: integer type; E-UTRAN (LTE-TDD RSRP) 0 < -140 dBm 1 -140 ≤ RSRP < -139 dBm 2 -139 ≤ RSRP < -138 dBm 3..94 -138≤ RSRP < -46 dBm 95 -46 ≤ RSRP < -45 dBm 96 -45 ≤ RSRP < -44 dBm 97 -44 ≤ RSRP dBm
Read command AT^DESQ?	Response **^DESQ: <mode>** **OK** Parameter See Set command
Test command AT^DESQ=?	Test command returns values supported as compound values. Response **^DESQ:** (list of supported <**mode**>s) **OK** Parameter See Set command
Reference	Used for EUTRAN RSCP changing report in LTE
Note	

10.3 AT^AQCI Query Cell Information

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Set Command AT^AQCI=<cell_type>[,<buffer_flag>[,<time_out>[,<search_prefer>]]]	This command is used to get the cell information, including service cell and neighbour cell. Response: Query service cell information: **^AQCI: "scell",<rat>,<is_tdd>,<roaming_flag>,<mcc>,<mnc>,<tac>,<cellid>,<freq_band_ind>,<ul_bandwidth>,<dl_bandwidth>,<earfcn>,<pcid>,<rsrp>,<rsrq>,<sinr>,<srxlev>,<rssi>** **OK** Query the information of neighbour cells: **^AQCI: "ncell",<intracell_flag>,<rat>,<mcc>,<mnc>,<tac>,<cellid>,<earfcn>,<pcid>,<rsrp>,<rsrq>,<sinr>,<srxlev>,<rssi>** **[...]** **OK** Parameter <cell_type>: Number format, indicating the cell type. 0 Service cell 1 Neighbour cell <buffer_flag>: Number format, indicating whether read neighbour cell information buffer directly. 0 Search current neighbour cell information. 1 Read neighbour cell information buffer directly. <time_out>: Number format, indicating the timeout value. 0 CP default value Others User specified timeout value, which is no more than 10. <search_prefer>: Number format, indicating search neighbour cell extended information priority.

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	0 User wishes to initiate neighbour cell search with low priority. 1 User wishes to initiate neighbour cell search with high priority. <rat>: string type, radio access technology. “LTE” UE is in LTE cell <is_tdd>: Number format, indicating the LTE mode 0 TDD 1 FDD <roaming_flag>: Number format, indicating whether UE is in roaming. 0 UE is not in roaming 1 UE is in roaming <intra_flag>: Number format, indicating the neighbour cell type. 0 Intra neighbour cell 1 Inter neighbour cell <mcc>: Number format. Mobile country code (first part of the PLMN code) <mnc>: Number format. Mobile network code (second part of the PLMN code) <cellid>: Cell ID. <pcid>: Physical cell ID <earfcn>: Number format. E-UTRA-ARFCN of the cell that was scanned. <freq_band_ind>: E-UTRA frequency band (see 3GPP 36.101). <ul_bandwidth>: Number format. UL bandwidth. 0 1.4MHz 1 3MHz 2 5MHz 3 10MHz 4 15MHz 5 20MHz <dl_bandwidth>: Number format. DL bandwidth. 0 1.4MHz 1 3MHz

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	2 5MHz 3 10MHz 4 15MHz 5 20MHz <tac>: Tracking area code. <rsrp>: Reference signal received power. <rsrq>: Reference signal received quality. <sinr>: Number format. Signal to interference plus noise ratio. Logarithmic value of SINR. <srxlev>: Number format. Select RX level value for base station in dB. <rssi>: Number format. Received signal strength indication.
Test Command AT^AQCI=?	Return list of supported <cell_type>s, <buffer_flag>s, <time_out>s and <search_prefer>s. Response **^AQCI: (0,1),(0,1),(0-10),(0,1)** **OK**
Note	

10.4 AT^SHUTDOWN system shutdown

Set command AT^SHUTDOWN	System will shutdown and save power off status. Response: **OK** If error: **ERROR**

10.5 AT^DPSHUTDW system deep shutdown

Set command AT^DPSHUTDW	System will deep shutdown and save power off status. This command supports if PMIC is gp202 so far.

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	Response: **OK** If error: **ERROR**

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11 AP Network Related AT commands

11.1 TP related AT commands

11.1.1 AT+FTPPORT

Set command AT+FTPPORT=<value>	Set control port for FTP. Response **OK** If error: **+CME ERROR: <err>** Parameters **<value>** FTP control socket port.Range 1~65535. Default port is 21.
Read command AT+FTPPORT?	Response **+FTPPORT: <value>** **OK**
Test command AT+FTPPORT=?	Response **+FTPPORT: (1~65535)** **OK**

11.1.2 AT+FTPMOD

Set command AT+FTPMOD=<value>	Set FTP working mode. Response **OK** If error: **+CME ERROR: <err>** Parameters **<value>** FTP mode.Range 0~1. 0 active mode. 1 passive mode.

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Read command AT+FTPMOD?	Response **+FTPMOD: <value>** **OK**
Test command AT+FTPMOD=?	Response **+FTPMOD: (0~1)** **OK**

11.1.3 AT+FTPTYPE

Set command AT+FTPTYPE=<value>	Set FTP transport type. Response **OK** If error: **+CME ERROR: <err>** Parameters **<value>** FTP transport type.Range 0~1. 0 Binary. 1 ASCII.
Read command AT+FTPTYPE?	Response **+FTPTYPE: <value>** **OK**
Test command AT+FTPTYPE=?	Response **+FTPTYPE: (0~1)** **OK**

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11.1.4 AT+FTPPUTOPT

Set command AT+FTPPUTOPT=<value>	Set FTP upload mode. Response **OK** If error: **+CME ERROR: <err>** Parameters **<value>** FTP upload mode.Range 1~3. 1 APPE, append data to the end of the file. 2 STOU, store unique. 3 STOR, store the data as a file at the server site.
Read command AT+FTPPUTOPT?	Response **+FTPPUTOPT: <value>** **OK**
Test command AT+FTPPUTOPT=?	Response **+FTPPUTOPT: (1~3)** **OK**

11.1.5 AT+FTPCID

Set command AT+FTPCID=<value>	Set PDP context ID used by FTP. Response **OK** If error: **+CME ERROR: <err>** Parameters **<value>** PDP context ID used by FTP.Range 1~16.

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Read command AT+FTPCID?	Response **+FTPCID: <value>** **OK**
Test command AT+FTPCID=?	Response **+FTPCID: (1~16)** **OK**

11.1.6 AT+FTPREST

Set command AT+FTPREST=<value>	Set the restart point.Then the FTP Client can be able to restart a file transfer at the point. Response **OK** If error: **+CME ERROR: <err>** Parameters **<value>** The restart point. Range 0~2^31-1.
Read command AT+FTPREST?	Response **+FTPREST: <value>** **OK**
Test command AT+FTPREST=?	Response **+FTPREST: (0~2^31-1)** **OK**

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11.1.7 AT+FTPSERV

Set command AT+FTPSERV=<host name>	Set the remote ftp server ip. Response **OK** If error: **+CME ERROR: <err>** Parameters **<hostname>** The IP address or domain name of the server.
Read command AT+FTPSERV?	Response **+FTPSERV: <hostname>** **OK**
Test command AT+FTPSERV=?	Response **+FTPSERV: <hostname>** **OK**

11.1.8 AT+FTPUN

Set command AT+FTPUN=<username>	Set the ftp authentication username. Response **OK** If error: **+CME ERROR: <err>** Parameters **<username>** The ftp authentication username.
Read command AT+FTPUN?	Response **+FTPUN: <username>** **OK**
Test command	Response

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AT+FTPUN=?	**+FTPUN: <username>** **OK**

11.1.9 AT+FTPPW

Set command AT+FTPPW=<password>	Set the ftp authentication password. Response **OK** If error: **+CME ERROR: <err>** Parameters **<password>** The ftp authentication password.
Read command AT+FTPPW?	Response **+FTPPW: <password>** **OK**
Test command AT+FTPPW=?	Response **+FTPPW: <password>** **OK**

11.1.10 AT+FTPGETNAME

Set command AT+FTPGETNAME=<file_name>	Set the name of the file to be downloaded. Response **OK** If error: **+CME ERROR: <err>** Parameters **<file_name>** The name of the file to be downloaded.
Read command AT+FTPGETNAME?	Response **+FTPGETNAME: <file_name>**

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	OK
Test command AT+FTPGETNAME=?	Response **+FTPGETNAME: <file_name>** **OK**

11.1.11 AT+FTPGETPATH

Set command AT+FTPGETPATH=<file_path>	Set the path of the file to be downloaded in ftp server. Response **OK** If error: **+CME ERROR: <err>** Parameters **<file_path>** The path of the file to be downloaded in ftp server.
Read command AT+FTPGETPATH?	Response **+FTPGETPATH: <file_path>** **OK**
Test command AT+FTPGETPATH=?	Response **+FTPGETPATH: <file_path>** **OK**

11.1.12 AT+FTPPUTNAME

Set command AT+FTPPUTNAME=<file_name>	Set the name of the file to be uploaded. Response **OK** If error: **+CME ERROR: <err>** Parameters **<file_name>** The name of the file to be uploaded.

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Read command AT+FTPPUTNAME?	Response **+FTPPUTNAME: <file_name>** **OK**
Test command AT+FTPPUTNAME=?	Response **+FTPPUTNAME: <file_name>** **OK**

11.1.13 AT+FTPPUTPATH

Set command AT+FTPPUTPATH=<file_path>	Set the path of the file that will be stored on the server. Response **OK** If error: **+CME ERROR: <err>** Parameters **<file_path>** The path of the file that will be stored on the server.
Read command AT+FTPPUTPATH?	Response **+FTPPUTPATH: <file_path>** **OK**
Test command AT+FTPPUTPATH=?	Response **+FTPPUTPATH: <file_path>** **OK**

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11.1.14 AT+FTPGET

Execute command AT+FTPGET=<mode> >	Start FTP downloading. Response **OK** If error: **+CME ERROR: <err>** Parameters **<mode>** Range 1~2. 1 Download file and save to local storage. 2.Directly transfer the downloaded data to the com port.
	Unsolicited Result Code: +FTPGET: 1,0,<transferlen> **<transferlen>**:Length of successfully downloaded data. Receiving this URC means that in mode 1, the file download was successful. +FTPGET: 1,<error> **<error>**: Error number.See section 13.3.22 for details. Receiving this URC means that in mode 1, the file download failed. +FTPGET: 2,1 +FTPGET: 2,<cnlength> …//output data +FTPGET: 2,<cnlength> …//output data +FTPGET: 2,0 **<cnlength>**: The length of the data read into the com port. Receiving this URC means that in mode 2, the file download was successful.

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	+FTPGET: 1,<error> **<error>**: Error number.See section 13.3.22 for details. Receiving this URC means that in mode 2, the file download failed.
Read command AT+FTPGET?	Response **+FTPGET: <mode>** **OK**
Test command AT+FTPGET=?	Response **+FTPGET: (1~2)** **OK**

11.1.15 AT+FTPPUT

Execute command AT+FTPPUT=<mode >	Start FTP uploading. Response **OK** If error: **+CME ERROR: <err>** Parameters **<mode>** Range 1~2. 1 Upload local file to ftp server. 2.Send the data from com port directly.

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	Unsolicited Result Code: +FTPPUT: 1,0,<transferlen> **<transferlen>**:Length of successfully upload data. Receiving this URC means that in mode 1, the file upload was successful. +FTPPUT: 1,<error> **<error>**: Error number.See section 13.3.22 for details. Receiving this URC means that in mode 1, the file upload failed. DOWNLOAD …//input data OK In mode 2, if the socket connection is successful, it will first response "DOWNLOAD", indicating that data can be written through the com port. Enter "+++" to end data entry. +FTPPUT: 2,0, <transferlen> Receiving this URC means that in mode 2, the file upload was successful. **<transferlen>**:Length of successfully upload data. + FTPPUT: 2,<error> **<error>**: Error number.See section 13.3.22 for details. Receiving this URC means that in mode 2, the file upload failed.
Read command AT+FTPPUT?	Response **+FTPPUT: <mode>** **OK**
Test command AT+FTPPUT=?	Response **+FTPPUT: (1~2)** **OK**

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11.1.16 AT+FTPMKD

Execute command AT+FTPMKD	Create a folder on the server. Response **OK** If error: **+CME ERROR: <err>** Parameters **N/A**
	Unsolicited Result Code: +FTPMKD: 1,0 Receiving this URC means that folder created successfully. +FTPMKD: 1,<error> **<error>**: Error number.See section 13.3.22 for details. Receiving this URC means that folder created unsuccessfully.
Read command AT+FTPMKD?	Read command is not supported.
Test command AT+FTPMKD=?	Response **+FTPMKD** **OK**
Note	The path of the folder to be created is defined by AT+FTPFGTPATH. The folder name is defined by AT+FTPGETNAME.

11.1.17 AT+FTPRMD

Execute command AT+FTPRMD	Delete a folder on the server. Response **OK** If error: **+CME ERROR: <err>** Parameters

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	N/A
	Unsolicited Result Code: +FTPRMD: 1,0 Receiving this URC means that folder deleted successfully. +FTPRMD: 1,<error> **<error>**: Error number.See section 13.3.22 for details. Receiving this URC means that folder deleted unsuccessfully.
Read command AT+FTPRMD?	Read command is not supported.
Test command AT+FTPRMD=?	Response **+FTPRMD** **OK**
Note	The path of the folder to be deleted is defined by AT+FTPGETPATH. The folder name is defined by AT+FTPGETNAME.

11.1.18 AT+FTPSIZE

Execute command AT+FTPSIZE	Get the file size on the server. Response **OK** If error: **+CME ERROR: <err>** Parameters **N/A**

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	Unsolicited Result Code: + FTPSIZE: 1,0,<size> **<size>**: file size. Receiving this URC means that getting file size successfully. + FTPSIZE: 1,<error>,0 **<error>**: Error number.See section 13.3.22 for details. Receiving this URC means that getting file size unsuccessfully.
Read command AT+FTPSIZE?	Read command is not supported.
Test command AT+FTPSIZE=?	Response **+ FTPSIZE** **OK**
Note	The path of the file is defined by AT+FTPGETPATH. The file name is defined by AT+FTPGETNAME.

11.1.19 AT+FTPLIST

Execute command AT+FTPLIST	Get the folder directory information. Response **OK** If error: **+CME ERROR: <err>** Parameters **N/A**
	Unsolicited Result Code: +FTPLIST: 1,1 ……list data…….. ……list data…….. OK Receiving this URC means that getting folder directory information successfully.

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	+FTPLIST: 1,<error> **<error>**: Error number.See section 13.3.22 for details. Receiving this URC means that getting folder directory information unsuccessfully.
Read command AT+FTPLIST?	Read command is not supported.
Test command AT+FTPLIST=?	Response **+FTPLIST** **OK**
Note	The path of the folder is defined by AT+FTPGETPATH.

11.1.20 AT+FTPQUIT

Execute command AT+FTPQUIT	Stop the current ftp session. Response **OK** If error: **+CME ERROR: <err>** Parameters **N/A**
Read command AT+FTPQUIT?	Read command is not supported.
Test command AT+FTPQUIT=?	Response **+FTPQUIT** **OK**

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11.1.21 AT+FTPOPEN

Execute command AT+FTPOPEN	Connect to remote ftp server. Response **OK** If error: **+CME ERROR: <err>** Parameters **N/A**
Read command AT+FTPOPEN?	Read command is not supported.
Test command AT+FTPOPEN=?	Response **+FTPOPEN** **OK**

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11.2 Network Share related AT commands

11.2.1 AT^PPPSHARE Set configurations for PPP dial

Set command AT^PPPSHARE=<cid>[,<shareMode>][,<pdp_type>][,<apn>]	Set configurations for PPP dial. Response: **OK** If error: **+CME ERROR: <err>** Parameters <**cid**>:integer type,the cid of PDP context,its range is 1~5. <**shareMode**>:integer type,the network sharing mode. 1 direct forwarding mode, default value. 2 routing mode. <**pdp_type**>:string type,the IP type of the PDP context. "IPV4V6" IPv4 and IPv6. "IP" IPv4 only. "IPV6" IPv6 only. <**apn**>:string type,the acess point name,depending on the operator registered.
Read command AT^PPPSHARE?	**^PPPSHARE: <cid>,<shareMode>[, <pdp_type>][,<apn>]** **OK**

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Test command AT^PPPSHARE=?	**^PPPSHARE: (list of supported cids),(list of supported share mode) ,(list of supported pdp types)** **OK**

11.2.2 ATD*99***cid# or ATDT*99***cid# Start PPP dial

Execute command ATD*99***cid# or ATDT*99***cid#	Start PPP dial. Response: **CONNECT** If error: **+CME ERROR: <err>** Parameters <**cid**>:integer type,the cid of PDP context,its range is 1~5.

11.2.3 AT^USBCFG Config mode of USB port

Execute command AT^USBCFG=<mode>	Set the work mode of USB port. Response: **OK** If error: **+CME ERROR: <err>** Parameters **<mode>** : integer type, the work mode of USB port. 0 the default mode,supports log printing,diag and at command

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	inputting. 1 supports RNDIS interface,log printing,diag and at command inputting. 2 supports ECM interfaces,log printing,diag and at command inputting. 3 supports msc storage,log printing,diag and at command inputting.
Read command AT^USBCFG?	**^USBCFG: <mode>** **OK**
Test command AT^USBCFG=?	**^USBCFG: (0-3)** **OK**

11.2.4 AT^USBTETHER Start or stop network share via USB interfaces

Execute command AT^USBTETHER=<state>,<cid>,<usbInterface>[,<shareMode>][,<apn>]	Start or stop network share via certain USB interfaces. Response: **OK** If error: **+CME ERROR: <err>** Parameterss <**state**>:integer type,to start or stop network share via certain USB interface. 1 Start network share via certain USB interface. 0 Stop network share via certain USB interface. <**cid**>:integer type,the cid of PDP context,its range is 1~5.

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	< **usbInterface**>:string type,the name of USB interface to transmit data packets,such as rndis0,ecm0,ecm1. <**shareMode**>:integer type, the network sharing mode. 1 direct forwarding mode. 2 routing mode,default value. <**apn**>:string type,the acess point name,depending on the operator registered.
Read command AT^USBTETHER?	**^USBTETHER: <state>,<cid>,<usbInterface>,<shareMode>,<apn>** **OK**

11.3 Dump PCAP log

11.3.1 AT^AEM Start or stop dump cp pcap log

Execute command AT^AEM=9,”cpcapon” or AT^AEM=9,”cpcapoff”	Start or stop dump cp pcap log. Response: **OK** If error: **+CME ERROR: <err>** Parameterss <9>:this is a fixed value. ”cpcapon”: start dump cp pcap log. ”cpcapoff”: stop dump cp pcap log. Note: if you need to set the packet capture length, set it as fllows AT^AEM=9,”cpcapon 1500”/ AT^AEM=9,”cpcapon 120”

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11.3.2 AT^CAPAPDATA Start or stop dump ap pcap log

Execute command AT^CAPAPDATA=”mobile”,”usb”,”ppp”,<len>	Start or stop dump ap pcap log. Response: **OK** If error: **+CME ERROR: <err>** Parameterss ”mobile”: start dump ap mobile interface pcap log. ”usb”: start dump ap usb interface pcap log. ”ppp”: start dump ap ppp interface pcap log. <len>: the packet capture length.

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12 File System related AT commands

The module supports the following physical storage media:

1) "/C": File storage on the Flash.
2) "/D": File storage on SD card.

The file system is used to store files in a hierarchical(tree) structure, and there are some definitions and conventions to use the AT command. Local storage space is mapped to "/C", "/D" for SD card.

NOTE: General rules for naming (both directory and files):

1) The length of actual fully qualified names of files can not exceed 128.
2) Directory and file names can not include the following characters: \ : * ? " < > |
3) Between directory name and file/directory name, use character "/" as list separator, so it can not appear in directory name or file name.
4) File name cannot begin with "." or "..".
5) If the last character of names is period "/" , will auto delete this character.

CME ERROR return value at **Summary of FS related CME ERRORS** chapter.

Command	Description
AT+GFCFG	Fs at common config.
AT+GFSINFO	Gets memory space information.
AT+GFLIST	List current directory file information.
AT+GFSIZE	Get file size.
AT+GFPUT	Upload a file to storage.
AT+GFGET	Download a file from storage.
AT+GFOPEN	Open file in storage.
AT+GFREAD	Read data from open file.
AT+GFWRITE	Write data to open file.
AT+GFSYNC	Sync open file data to storage.
AT+GFSEEK	Set open file offset.
AT+GFTRUNC	Truncate open file size.
AT+GFCLOSE	Close open file.
AT+GFDELETE	Delete files in storage
AT+GFMOVE	Move file.
AT+GFCHECK	Check file.
AT+GFMKDIR	Create directory.
AT+GFRMDIR	Remove directory
AT+GFCD	Set current work directory.
AT+GFRENAME	Rename file.
AT+GFCOPY	Copy file.
AT+GFFORMAT	Format volume.
AT+GFRWTEST	Get file information.

12.1 AT+GFCFG

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Fs at common config.

Test command **AT+GFCFG=?**	Response **+GFCFG: "timeout",(list of support<timeout>s)** **+GFCFG: "encoding",(list of support<input_format>s),(list of support <output_format>s)** **OK**
Set command **AT+ GFCFG="timeout"[,<timeout>]**	Input "timeout" only, response timeout set: **+GFCFG: "timeout",<timeout>** **OK** Set timeout, response: **OK** If there are any errors related to ME functionality: **+CME ERROR: <err>**
Set command **AT+ GFCFG="encoding"[,<input_format>[,<output_format>]]**	Input "encoding" only, response encoding set: **+GFCFG: "encoding",<input_format>,<output_format>** **OK** Set input_format and output_format, response: **OK** If there are any errors related to ME functionality: **+CME ERROR: <err>**
Defined Values	**"timeout":** Data input timeout config.
	<timeout>: Range:1~120, Unit:s, default 10s. Effect: AT+GFWRITE、AT+GFPUT
	"encoding": Input and output data encoding config.
	<input_format>: Input data format: 0: ASCII string, default 1: HEX string 2: ESCAPE string Effect: AT+GFWRITE、AT+GFPUT
	<output_format>: Output data format: 0: raw data 1: HEX string Effect: AT+GFREAD、AT+GFGET
	<err>: Error codes, please see Summary of FS related CME ERRORS.
Reference	
Note	

12.2 AT+GFSINFO

Gets memory space information.

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Execute command **AT+GFSINFO**	Get default volume size info, response: **+GFSINFO: <total_size>,<free_size>** **OK** If there are any errors related to ME functionality: **+CME ERROR: <err>**
Test command **AT+GFSINFO=<volume>**	Get input volume size info, response: **+GFSINFO: <total_size>,<free_size>** **OK** If there are any errors related to ME functionality: **+CME ERROR: <err>**
Defined Values	**<volume>:** Fs disk volume name, default is "/C" and "/D".
	<total_size>: Total size of <volume>. Unit:byte.
	<free_size>: Free size of <volume>. Unit:byte.
	<err>: Error codes, please see Summary of FS related CME ERRORS.
Reference	
Note	

12.3 AT+GFLIST

List current directory file information..

Set command **AT+GFLIST=<path>**	Get path file info, response: **[+GFLIST: <fname >,<size>]** **[…]** **OK** If there are any errors related to ME functionality: **+CME ERROR:<err>**
Execute command **AT+GFLIST**	Get current path file info, response: **[+GFLIST: <fname >,<size>]** **[…]** **OK** If there are any errors related to ME functionality: **+CME ERROR: <err>**
Defined Values	**<path>:** File or directory path. If input path is "/C/**" or "/D/**", will get all directory and file informat. Path max length is 128bytes.
	<fname>: File name.
	<size>: File size. Unit:byte.
	<err>: Error codes, please see Summary of FS related CME ERRORS.
Reference	
Note	

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12.4 AT+GFSIZE

Get file size.

Set command **AT+GFSIZE=<fname>**	Get file size, response: **+GFSIZE: <size>** **OK** If there are any errors related to ME functionality: **+CME ERROR: <err>**
Defined Values	**<fname>**: File name.
	<size>: File size. Unit:byte.
	<err>: Error codes, please see Summary of FS related CME ERRORS.
Reference	
Note	

12.5 AT+GFPUT

This command is used to upload a file to storage.

After executing the setup command and returning >, the module switches to data mode. When the uploaded file size reaches <fsize>, or <timeout> no data input, the module will automatically exit the data mode. Data can be transmitted by the user by performing +++.

Set command **AT+GFPUT=<fname>[,<fsize>[,<data>]]**	If input data, response: **+GFPUT: <upload_size>** **OK** If not input data, response: > TA switches to data mode (that is, pass-through mode) to receive binary file data. When the total size of the input data reaches <fsize> (in bytes), TA will switch back to command mode. **+GFPUT: <upload_size>** **OK** If there are any errors related to ME functionality: **+CME ERROR: <err>**
Defined Values	**<fname>:** The path name of the upload file. The maximum length is 128 bytes.
	<fsize>: Estimated file size to be uploaded. If fsize not set, will be continue receive data. Unit:byte.
	<data>: Input data.
	<upload_size>: The actual uploaded file size, Unit: bytes.

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	<err>: Error codes, please see Summary of FS related CME ERRORS.
Reference	
Note	1. DOS 8.3 <filename> filename format is recommended. 2. The +++ sequence causes TA to terminate the command and switch to command mode. Previously uploaded data will remain in the file. 3. When executing the command, must return > before enter data.

12.6 AT+GFGET

This command is used to download a file from storage. If the download file not in storage, an error will be reported.

Set command **AT+GFGET=<fname>**	Response **+GFGET: <fsize>,<data>** **OK** If there are any errors related to ME functionality: **+CME ERROR: <err>**
Defined Values	**<fname>:** The path name of the download file. The maximum length is 128 bytes.
	<data>: The actual downloaded file data.
	<err>: Error codes, please see Summary of FS related CME ERRORS.
Reference	
Note	1. DOS 8.3 <filename> filename format is recommended.

12.7 AT+GFOPEN

This command is used to open a file in storage.

Set command **AT+GFOPEN=<fname>[,<mode>]**	Response **+GFOPEN: <fhandle>** **OK** If there are any errors related to ME functionality: **+CME ERROR: <err>**
Execute command **AT+GFOPEN?**	Response **[+GFOPEN: <fname>,<fhandle>,<mode>]** **[...]** **OK**

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Defined Values	**<fname>:** The path name of the download file. The maximum length is 128 bytes.
	<fhandle>: The open file handle. Data type is int. Range:0~65535.
	<mode>: Open file mode: **0:** Read only. Return error if file does not exist. **1:** Write only. Truncate it if file exist, otherwise create it. **2:** Write only. Append data if file exist, otherwise create it. **3:** Read and Write. Truncate it if file exist, otherwise create it. **4:** Read and Write. Append data if file exist, otherwise create it.
	<err>: Error codes related to ME functions.
Reference	
Note	

12.8 AT+GFREAD

This command is used to read a file from storage.

Set command **AT+GFREAD=<fhandle>[,<length>]**	Response **+GFREAD: <fhandle>,<total>,<accum>,<cur_length>,<data>** OK If there are any errors related to ME functionality: **+CME ERROR: <err>**
Defined Values	**<fhandle>:** The file operation handle. Data type is int. Range:0~65535.
	<length>: The expected length of the file to read, which is the file length by default.
	< total >: The total length of the file actually read.
	<accum>: Accumulated read packet length.
	<cur_length>: Current packet data size.
	<data>: Read data.
	<err>: Error codes, please see Summary of FS related CME ERRORS.
Reference	
Note	

12.9 AT+GFWRITE

This command is used to write data to open file in storage.

Set command **AT+GFWRITE=<fhandle>[,<length>[,<data>]]**	If input data, response: **+GFWRITE: <fhandle>,<write_length>** **OK** If not input data, response: **>** TA switches to data mode (that is, pass-through mode) to receive binary file data. When the total size of the input data reaches <length>

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	(in bytes), TA will switch back to command mode. **+GFWRITE: <fhandle>,<write_length>** **OK** If there are any errors related to ME functionality: **+CME ERROR: <err>**
Defined Values	**<fhandle>:** The file operation handle. Data type is int. Range:0~65535.
	<length>: Estimated file size to be uploaded. If length not set, will be continue receive data. Unit:byte.
	<write_length>: The length of the file actually write.
	<data>: Input file data.
	<err>: Error codes, please see Summary of FS related CME ERRORS.
Reference	
Note	

12.10 AT+GFSYNC

This command is used to sync open file data to storage.

Set command **AT+GFSYNC=<fhandle>**	Response: **OK** If there are any errors related to ME functionality: **+CME ERROR: <err>**
Defined Values	**<fhandle>:** The file operation handle. Data type is int. Range:0~65535.
	<err>: Error codes, please see Summary of FS related CME ERRORS.
Reference	
Note	

12.11 AT+GFSEEK

This command is used to set file offset.

Set command **AT+GFSEEK=<fhandle>[,<offset>,<fromwhere>]**	If only set <fhandle>, response: **+GFSEEK: <fhandle>,<current_offset>** Set all parameter, response: **OK** If there are any errors related to ME functionality: **+CME ERROR: <err>**
Defined Values	**<fhandle>:** The file operation handle. Data type is int. Range:0~65535.
	<current_offset>: Current file offset.

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	<offset>: Number of bytes moved by file pointer. Data type is int. Unit:byte.
	<fromwhere>: File pointer movement mode: **0** Start file pointer **1** Current file pointer **2** End file pointer
	<err>: Error codes, please see Summary of FS related CME ERRORS.
Reference	
Note	

12.12 AT+GFTRUNC

This command is used to truncate file.

Set command **AT+GFTRUNC=<fhandle>,<length>**	Response: **OK** If there are any errors related to ME functionality: **+CME ERROR: <err>**
Defined Values	**<fhandle>:** The file operation handle. Data type is int. Range:0~65535.
	<length>: Truncate file length.
	<err>: Error codes, please see Summary of FS related CME ERRORS.
Reference	
Note	

12.13 AT+GFCLOSE

This command is used to close the open file.

Set command **AT+GFCLOSE=<fhandle>**	Response **OK** If there are any errors related to ME functionality: **+CME ERROR:<err>**
Defined Values	**<fhandle>:** The file operation handle. Data type is int. Range:0~65535.
	<err>: Error codes, please see Summary of FS related CME ERRORS.
Reference	
Note	

12.14 AT+GFDELETE

This command is used to delete the file.

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Set command **AT+GFDELETE=<fname>**	Response **OK** If there are any errors related to ME functionality: **+CME ERROR:<err>**
Defined Values	**<fname>:** The path name of the download file. The maximum length is 128 bytes. If the parameter ends with "/", delete the folder and all subfolders and subfiles.
	<err>: Error codes, please see Summary of FS related CME ERRORS.
Reference	
Note	

12.15 AT+GFMOVE

This command is used to move file in storage.

Set command **AT+GFMOVE=<old_fname>,<new_fname>**	Response **OK** If there are any errors related to ME functionality: **+CME ERROR: <err>**
Defined Values	**<old_fname>:** The path name of the move file. The maximum length is 128 bytes.
	<new_fname>: The file new name. The maximum length is 128 bytes.
	<err>: Error codes, please see Summary of FS related CME ERRORS.
Reference	
Note	

12.16 AT+GFCHECK

This command is used to check file in storage.

Set command **AT+GFCHECK=<fname>[,<verify_alg>]**	Response **+MFCHECK: <check_code>** **OK** If there are any errors related to ME functionality: **+CME ERROR: <err>**
Defined Values	**<fname>:** The path name of the file. The maximum length is 128 bytes.
	<verify_alg>: Algorithms for verify file: 0: MD5, default 1: SHA 2: SHA256 3: CRC
	<check_code>: Verification result. Type is string.
	<err>: Error codes, please see Summary of FS related CME ERRORS.

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Reference	
Note	

12.17 AT+GFMKDIR

This command is used to create directory.

Test command **AT+GFMKDIR=<dir>**	Response **OK** If there are any errors related to ME functionality: **+CME ERROR: <err>**
Defined Values	**<dir>:** Create directory name. The maximum length is 128 bytes.
	<err>: Error codes, please see Summary of FS related CME ERRORS.
Reference	
Note	

12.18 AT+GFRMDIR

This command is used to remove directory.

Set command **AT+GFRMDIR=<dir>**	Response **OK** If there are any errors related to ME functionality: **+CME ERROR: <err>**
Defined Values	**<dir>:** Remove directory name. The maximum length is 128 bytes.
	<err>: Error codes, please see Summary of FS related CME ERRORS.
Reference	
Note	

12.19 AT+GFCD

This command is used to set current work directory.

Set command **AT+GFCD=<curr_dir>**	Response **OK** If there are any errors related to ME functionality: **+CME ERROR: <err>**
Defined Values	**<curr_dir>:** Set current work directory. The maximum length is 128 bytes.
	<err>: Error codes, please see Summary of FS related CME ERRORS.
Reference	
Note	

12.20 AT+GFRENAME

This command is used to rename file in storage.

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Set command **AT+GFRENAME=<old_fname>,<new_fname>**	Response **OK** If there are any errors related to ME functionality: **+CME ERROR: <err>**
Defined Values	**<old_fname>:** The path name of the rename file. The maximum length is 128 bytes.
	<new_fname>: The file new name. The maximum length is 128 bytes.
	<err>: Error codes, please see Summary of FS related CME ERRORS.
Reference	
Note	

12.21 AT+GFCOPY

This command is used to copy source file path to destination path.

Set command **AT+GFCOPY=<old_fname>,<new_fname>**	Response **OK** If there are any errors related to ME functionality: **+CME ERROR: <err>**
Defined Values	**<old_name>:** The path of the copy file. The maximum length is 128 bytes.
	<new_name>: The file new name. The maximum length is 128 bytes.
	<err>: Error codes, please see Summary of FS related CME ERRORS.
Reference	
Note	

12.22 AT+GFFORMAT

This command is used to format file system.

Execute command **AT+GFFORMAT**	Format default “/C” volume, response **OK** If there are any errors related to ME functionality: **+CME ERROR: <err>**
Set command **AT+GFFORMAT=<volume>**	Response **OK** If there are any errors related to ME functionality: **+CME ERROR: <err>**
Defined Values	**<volume>:** Fs volume name: "/C"、"/D"
	<err>: Error codes, please see Summary of FS related CME

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	ERRORS.
Reference	
Note	

12.23 AT+GFRWTEST

This command is used to test file system.

Execute command **AT+GFRWTEST**	Test default "/C" volume, Response: **OK** If there are any errors related to ME functionality: **+CME ERROR: <err>**
Set command **AT+GFRWTEST =<start>[,< volume >[,<interval>[,<length>]]]**	Response: **OK** If there are any errors related to ME functionality: **+CME ERROR: <err>**
Defined Values	**<start>:** 0: stop fs read and write test; 1.start fs read and write test.
	<volume>: Fs volume name: "/C"、"/D".
	< interval>: Interval between rounds.
	<length>: Read or write file length.
	<err>: Error codes, please see Summary of FS related CME ERRORS.
Reference	
Note	

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13 Camera related AT commands

13.1 AT+ QCAMOPEN Open camera

Set command AT+QCAMOPEN=width,height	+QCAMOPEN: open camera at width,height Response: **OK** If error: **+QCAMOPEN ERROR: 49** Parameters <**image_width, image_heigh**>:image_width=width, image_heigh=height.
Test command AT+QCAMCLOSE=?	**OK**

13.2 AT+QCAMCLOSE Close camera

AT+QCAMCLOSE	+QCAMCLOSE: close camera Response: **OK**
Test command AT+QCAMCLOSE=?	**OK**

13.3 AT+QCAMTEST Test camera

AT+QCAMTEST	+QCAMTEST: test camera and then enter preview Response: If support jpeg, send the jpeg image to pc. QCAMTEST result addr: xxxx, size: xxxx **OK**

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	If error: **+QCAMTEST ERROR: 49**
Test command	**NULL**

13.4 AT+QCAMTESTBLT Test camera with jpeg support

AT+QCAMTESTBLT	+QCAMTESTBLT: test camera and not support if no jpeg Response: send the jpeg image to pc. QCAMTEST result addr: xxxx, size: xxxx **OK** If error: **+QCAMTEST ERROR: 49**
Test command	**NULL**

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14 FOTA related AT commands

14.1 AT+FOTACRT Set HTTPS certification for FOTA

Execute command AT+FOTACRT=<size>	Set HTTPS certification for FOTA. Response: **FOTACRT START** Then starting send certification. **OK** If received <size> bytes, return OK. If error: **+CME ERROR: <err>** Parameterss <**size**>: size of certification of HTTPS.
Test command AT+FOTACRT=?	Response: **AT+FOTACRT=<size>**

14.2 AT+FOTA Start FOTA downloading

Execute command AT+FOTA=<url>,<port>,<user_name>,<pwd>,<reserve>,<dl_step>,<upd_step>	Start fota firmware downloading from network. Response: OK +IND:"FOTA","HTTPSTART",0 +IND:"FOTA","DOWNLOADING",<recv_size>,<percent> …… +IND:"FOTA","DOWNLOADING",<recv_size>,<percent> +IND:"FOTA","HTTPEND",0,0 +IND:"FOTA","BASIC","UPDATING",<percent> ……

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	+IND:"FOTA","BASIC","UPDATING",<percent> +IND:"FOTA","HTTPSTART",1 +IND:"FOTA","DOWNLOADING",<recv_size>,<percent> …… +IND:"FOTA","DOWNLOADING",<recv_size>,<percent> +IND:"FOTA","HTTPEND",0,1 +IND:"FOTA","EXT","UPDATING",<percent> …… +IND:"FOTA","EXT","UPDATING",<percent> If error: **+CME ERROR: <err>** Parameterss <**url**>: URL of FOTA firmware. <**port**>: HTTP port. <**user_name**>: User name of login. <**pwd**>: Password of login. <**reserve**>: reserve. <**dl_step**>: Download notification maximum count，5-100; If 0, do not send download URC. <**upd_step**>: Upgrade notification maximum count，5-100; If 0, do not send upgrade URC.
Test command AT+FOTA=?	Response: **+FOTA=<url>,<port>,<user_name>,<pwd>,(0,1),(0,5-100),(0,5-100)**

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15 Wifi Scan related AT commands

15.1 AT+WIFISCAN

Execute command AT+WIFISCAN[=<on_off>[,<ap_num>[,<timeout>[,<channel_ids>[,<mac>[,<ssid>]]]]]]	Execution command causes the TA to return one or more lines of information about wifi aps which are scanned. Response **OK/ERROR/CME ERROR** Parameter < **on_off** >: integer type. trigger start or stop wifi scan 0: stop wifi scan; 1 start wifi scan; default: start wifi scan < **ap_num** >: integer type. the max ap number which expect scanned default: 10 < **timeout** >: integer type. the time out of the whole wifi scan. default: 5 seconds < **channel_ids** >: integer type. (1~13) the channel id which expect scanned. 0: scan all channels. 10: scan single channel 10. (1,6,11): scan channels 1, 6, 11. default: scan all channels < **mac** >: string type. filter wifi aps which is same with <mac> Default: no filter < **ssid** >: string type. filter wifi aps which is same with <ssid> Default: no filter
	Unsolicited Result Code: Whole scan report: +WIFISCAN: 0, <**ap_num_scanned**>,< **ap_num_expexted**> <**ap_num_scanned**>: the ap number which has scanned < **ap_num_expexted**>: the max ap number which expect scanned Wifi ap information report:

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	+WIFISCAN: 1, <**channel_ids**>,<**mac**>,<**rssi**>,<**ssid**>,<**security**>
Read command AT+WIFISCAN?	Read command returns current wifi scan params. Response +WIFISCAN: <**ap_num**>,<**timeout**>,<**channel_ids**>,<**mac**>,<**ssid**> OK
Test command AT+WIFISCAN=?	+WIFISCAN: (ap_num_range),(timeout_range),(channel_id_range),-,- Response +WIFISCAN: (1-100),(2000-10000),(1-13),-,- OK
Example	AT+WIFISCAN=1,20,5000,(1,6,11) OK +WIFISCAN: 0,4,20 +WIFISCAN: 1,1,1C:20:DB:8D:D7:80,-30,"happy family",- +WIFISCAN: 1,6,1C:20:DB:8D:D7:80,-30,-,- +WIFISCAN: 1,6,1C:20:DB:8D:D7:90,-30,"family",- +WIFISCAN: 1,11,1C:20:DB:8D:D7:80,-30,-,- AT+WIFISCAN=? +WIFISCAN: (1-100),(2000-10000),(1-13),-,- OK AT+WIFISCAN? +WIFISCAN: 5,2000,6,-,- OK
Note	Not support mac and rssi filter and security type currently

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16 Product Number

16.1 AT^IMEI

This command is used to read and write imei number.

Test command **AT^IMEI=?**	Response **^IMEI: (index),(imei)** **OK**
Query command **AT^IMEI =<index>**	Response **^IMEI = <index>,<imei>** **OK** If there are any errors related to ME functionality: **+CME ERROR:<err>**
Set command **AT^IMEI =<index>,<imei>**	Response **OK** If there are any errors related to ME functionality: **+CME ERROR:<err>**
Defined Values	**<index >:** imei index, support 0 and 1.
	< imei >: If code definded IMEI_AT_CHECKDIGIT, must be input 14byte length, device will be calculation the check digit. otherwise, must be input 15bytes.
	<err>: Error codes, please see Summary of FS related CME ERRORS.
Reference	
Note	

16.2 AT^SN

This command is used to read and write SN number.

Test command **AT^SN=?**	Response **^SN: (0,1),()** **OK**
Query command **AT^SN =<index>**	Response **^SN = <index>,<sn>** **OK** If there are any errors related to ME functionality: **+CME ERROR:<err>**

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Set command **AT^SN =<index>,<sn>**	Response **OK** If there are any errors related to ME functionality: **+CME ERROR:<err>**
Defined Values	**<index >:** sn index, support 0 and 1.
	<sn>: max support 64 byte length.
	<err>: Error codes, please see Summary of FS related CME ERRORS.
Reference	
Note	

16.3 AT^MAC

This command is used to read and write MAC address.

Test command **AT^MAC=?**	Response **^MAC: (mac)** **OK**
Query command **AT^MAC?**	Response **^MAC = <mac>** **OK** If there are any errors related to ME functionality: **+CME ERROR:<err>**
Set command **AT^MAC =<mac>**	Response **OK** If there are any errors related to ME functionality: **+CME ERROR:<err>**
Defined Values	**<mac>:** must be 12byte length.
	<err>: Error codes, please see Summary of FS related CME ERRORS.
Reference	
Note	

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17 Station

17.1 AT^STATION

This command is used to read and write station information.

Test command **AT^STATION=?**	Response **^STATION: <name>,<flag>** **OK**
Query all station command **AT^STATION?**	Response **^STATION = <index>,<name>,<flag>** **…** **OK** If there are any errors related to ME functionality: **+CME ERROR:<err>**
Query station flag command **AT^STATION=<name>**	Response **^STATION = <name>,<flag>** **OK** If there are any errors related to ME functionality: **+CME ERROR:<err>**
Set command **AT^ STATION =<name>,<flag>**	Response **OK** If there are any errors related to ME functionality: **+CME ERROR:<err>**
Defined Values	**<name>:** station name.
	<flag>: station flag. 0: not do 1: fail 3: pass
	<index>: station index.
	<err>: Error codes, please see Summary of FS related CME ERRORS.
Reference	
Note	

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18 Calibration

18.1 AT+AFC

This command is used to query afc calibration information.

Query afc command **AT+AFC?**	Response **+AFC = <flag>,<freq>,<ct>,<cal>,<hpm>,<lpm>** **OK** If there are any errors related to ME functionality: **+CME ERROR:<err>**
Defined Values	**<flag>:** calibration flag.
	<freq>: calibration frequency.
	<ct>: calibration afc coarse tuning.
	<cal>: calibration afc calibration value.
	<hpm>: calibration high power mode 32k value.
	<lpm>: calibration low power mode 32k value.
	<err>: Error codes, please see Summary of FS related CME ERRORS.
Reference	
Note	

18.2 AT+CALIBAND

This command is used to query calibration band information.

Test command **AT+CALIBAND=?**	Response **+CALIBAND: <band_num>** **OK**
Query calibration band command **AT+CALIBAND?**	Response **+CALIBAND= <time>,<version>,<band_num>** **OK** If there are any errors related to ME functionality: **+CME ERROR:<err>**

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Query bandnum flag command **AT+CALIBAND =<band_num>**	Response **+CALIBAND = <band_num>,<flag>** **OK** If there are any errors related to ME functionality: **+CME ERROR:<err>**
Defined Values	**< band_num>:** calibration band number.
	<time>: calibration time.
	<version>: calibration tool version.
	<flag>: calibration band flag.
	<err>: Error codes, please see Summary of FS related CME ERRORS.
Reference	
Note	

18.3 AT+NSTBAND

This command is used to query nst band information.

Query nst band command **AT+ NSTBAND?**	Response **+NSTBAND = <band_num>** **OK** If there are any errors related to ME functionality: **+CME ERROR:<err>**
Defined Values	**<band_num>:** calibration band number.
	<err>: Error codes, please see Summary of FS related CME ERRORS.
Reference	
Note	

18.4 AT+RFID

This command is used to query rfid information.

Query rfid command **AT+RFID?**	Response **+RFID = <hw_name>[,<rfid_name>]** **OK** If there are any errors related to ME functionality: **+CME ERROR:<err>**
Defined Values	**< hw_name >:** hardware name.

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	<rfid_name>: rfid name.
	<err>: Error codes, please see Summary of FS related CME ERRORS.
Reference	
Note	

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19 Appendix

19.1 Summary of ERRORS and Messages

This clause gives the descriptions for ERROR or CME ERROR:<err> or CEER or other related error report modes.

19.1.1 Summary of AT general CME ERRORS

Code of <err>	Meaning
0	phone failure
1	no connection to phone
2	phone-adaptor link reserved
3	operation not allowed
4	operation not supported
5	PH-SIM PIN required
6	PH-FSIM PIN required
7	PH-FSIM PUK required
10	SIM not inserted
11	SIM PIN required
12	SIM PUK required
13	SIM failure
14	SIM busy
15	SIM wrong
16	incorrect password
17	SIM PIN2 required
18	SIM PUK2 required

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20	memory full
21	invalid index
22	not found
23	memory failure
24	text string too long
25	invalid characters in text string
26	dial string too long
27	invalid characters in dial string
30	no network service
31	network timeout
32	network not allowed – emergency calls only
40	network personalization PIN required
41	network personalization PUK required
42	network subset personalization PIN required
43	network subset personalization PUK required
44	service provider personalization PIN required
45	service provider personalization PUK required
46	corporate personalization PIN required
47	corporate personalization PUK required
48	hidden key required (NOTE: This key is required when accessing hidden phonebook entries.)
49	some parameters of the at command are invalid
50	at command too long
51	Invalid at command
52	at command handled timeout
53	not support the parameter value
54	fail to handle the AT command
55	modem not alive
56	no idle link to use between AP and CP
57	fail to write AT commands to CP via IPC link
100	Unknown

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19.1.2 Summary of AT SMS CMS ERRORS0

Code of <err>	Meaning
0…127	3GPP TS 24.011 [6] clause E.2 values
128...255	3GPP TS 23.040 [3] clause 9.2.3.22 values.
300	ME failure
301	SMS service of ME reserved
302	operation not allowed
303	operation not supported
304	invalid PDU mode parameter
305	invalid text mode parameter
310	(U)SIM not inserted
311	(U)SIM PIN required
312	PH-(U)SIM PIN required
313	(U)SIM failure
314	(U)SIM busy
315	(U)SIM wrong
316	(U)SIM PUK required
317	(U)SIM PIN2 required
318	(U)SIM PUK2 required

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320	memory failure
321	invalid memory index
322	memory full
330	SMSC address unknown
331	no network service
332	network timeout
340	no +CNMA acknowledgement expected
500	unknown error
...511	other values in range 256...511 are reserved
512...	manufacturer specific

19.1.3 Summary of GPRS-related CME ERRORS

Code of <err>	Meaning
Errors related to a failure to perform an Attach	
103	Illegal MS
106	Illegal ME
107	GPRS services not allowed
111	PLMN not allowed
112	Location area not allowed
113	Roaming not allowed in this location area
Errors related to a failure to Activate a Context	
132	service option not supported
133	requested service option not subscribed
134	service option temporarily out of order
149	PDP authentication failure
Other GPRS errors	
150	invalid mobile class
148	unspecified GPRS error

19.1.4 Summary of enhancement related CME ERRORS

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Code of <err>	Meaning
600	at command aborted
601	inter-RAT changing
603	number of password attempts reached
700	invalid storage device
701	invalid phonebook type
702	power off
703	first input PIN/PIN2/PUK/PUK2 is invalid value
704	second input PIN/PIN2/PUK/PUK2 is invalid value
705	operation condition not satisfied PIN/PIN2 disable or enable

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706	operation condition not satisfied PIN/PIN2 block and enable
707	operation condition not satisfied PIN/PIN2 block and disable
708	operation condition not satisfied PUK/PUK2 block
709	SIM type incompatible(SIM inserted in UTRAN mode)
800	The maximal length of numbers exceed 20
801	The maximal length of numbers exceed 40
802	The maximal length of numbers exceed 60
803	The maximal length of numbers exceed 80
804	Subaddress memory capacity exceeded
805	The maximal length of subaddress exceed 22
806	The maximal length of subaddress exceed 44

19.1.5 Summary of enhancement related CMS ERRORS

Code of <err>	Meaning
600	at command aborted
601	short message service parameter(SMSP) EF is not present in SIM
602	cell broadcast message indentifier selection (CBMI) EF is not present in SIM
603	cell broadcast message indentifier range selection (CBMIR) EF is not present in SIM
604	SMSP and CBMi EFs are not present in SIM
605	CBMI and CBMIr EFs are not present in SIM
606	SMSP and CBMIr EFs are not present in SIM

19.1.6 Summary of FTP related CME ERRORS

Code of <err>	Meaning
1060	AT command parameter parsing fail
1061	FTP thread creating fail
1062	FTP openning fail
1063	Get IP address fail
1064	Socket disconnecting fail

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1065	Deleting FTP Client fail
1066	Getting FTP Server file list fail
1067	Deleting FTP packet pool fail
1068	Openning local file fail
1069	FTP evevt notification creating fail
1070	FTP active/passive mode setting fail
1071	Packet allocating fail
1072	Reading data from packet fail
1073	Working path setting fail
1074	Creating folder fail
1075	Deleting folder fail
1076	Setting download offset value fail
1077	Downloading file and saving fail
1078	Downloading and transferring file by com port fail
1079	Uploading local file fail
1080	Uploading data by com port fail
1081	Getting file size fail

19.1.7 Summary of Network Share related CME ERRORS

Code of <err>	**Meaning**
1100	the network interface enaged
1101	the pdp enaged
1102	the usb interface not up
1103	fail to activate the pdp
1104	certain internal errors

19.1.8 Summary of FS related CME ERRORS

Code of <err>	**Meaning**
50	Invalid argument
1050	File system error
1051	File not exists
1052	File exists
1053	File too large
1054	File table overflow

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1055	File open fail
1056	File opened
1057	File write fail
1058	File read fail
1059	File Read only
1060	Bad file description
1061	File delete fail
1062	File list fail
1063	No space left on fs
1064	Operation file timeout occurred

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